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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 315 — Past Questions & Materials
by Renance

Course: PHY 315
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 300

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Paper 1 of 3 — PHY 315 2016 2017

Level 300 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria
College of Science
Department of Physics
Second Semester Examination

Academic Session:	2016/2017	Time Allowed:	2 hrs
Course Title:	Energy and Environment	Course Code:	PHY 315
Course Unit:	2 units	Date:	3/4/2017
Instructions:	ANSWER Any Three QUESTIONS		

Q1 A) Environmental Physics is an interdisciplinary subject? Discuss.
 B) i) Discuss Energy and metabolism as it affects the human body. ii) The human body obey the laws of thermodynamics. If that the first law is given as: $dQ = dU + pdV$, show how the human body obeys the Gibbs free energy; $dG = dH - T.dS$
 C) i) Discuss the various mechanism of Energy transfer as applicable to the human body. ii) Discuss how the human body can survive in hot and cold climate and list the various stages of hyperthermia and hypothermia respectively.

Q2 A) i) Discuss noise as a source of Environmental pollution
 ii) Explain what you understand by threshold of hearing and pain
 B) State and discuss the maximum permissible noise level for i) residential environment ii) commercial environment iii) industrial environment iv) sleeping environment
 C) Discuss the structure and composition of the atmosphere

Q3 A) Write short note on the following
 i) Residence Time ii) Ozone and Ozone depletion iii) Greenhouse gas and greenhouse Effect iv) Atmospheric aerosol
 B) What is "Global warming".
 C) i) Explain what you understand by hydrological cycle.
 ii) Discuss briefly the Physics of the Cloud

Q4 A) In a well tabulated table, discuss the seven categorizes of wind in terms of their Effects and Speed
 B) i) Discuss the Physics of wind creation ii) Global convection
 C) Write short note on the following four forces acting on parcel of air in the atmosphere
 i) Gravitation ii) Pressure Gradient iii) Coriolis frictional force iv) Frictional force

Q5 A) i) Discuss the concept Energy as a non-homogeneous Commodity.
 ii) State and explain five forms of energy you know.
 B) i) Distinguish between Renewable and Non-Renewable energy.
 iii) Fossil fuel is a non-renewable energy resource, Discuss.
 C) i) What is Soil to Environmental Physicist, State the particle dimension of the fraction of soil.
 ii) State the expression for the fraction of soil volume occupied by solid and porosity.

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Paper 2 of 3 — Phy 315 Exam PQ

Level 300 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria

College of Science
Department of Physics

First Semester Examination

Academic Session: 2019/2020 Time Allowed: 2 hrs
 Course Title: Energy and Environment
 Course Code: PHY 315
 Course Unit: 2 units
 Instructions: ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS

Q1

- A. i) Discuss the concept Earth Energy Budget with the aid of a diagram
 ii) Discuss the Energy as a non-homogeneous Commodity.
- B. i) State and explain five forms of energy.
 ii) Fossil fuel is a non-renewable energy resource, Discuss.
- C. i) Discuss briefly the following sources of energy; Solar Cell or photovoltaic energy, Biomass energy and Geothermal energy
 ii) Distinguish between Renewable and Non-Renewable energy.

Q2

- A i) What is Environment to the Physicist
 ii) Environmental Physics is an interdisciplinary subject? Discuss.
- B. i) Discuss Energy and metabolism as it affects the human body.
 ii) Discuss the various mechanism of Energy transfer as applicable to the human body.
- C. Discuss how the human body can survive in cold climate and list the various stages of hypothermia.

Q3

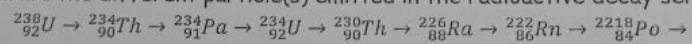
- A. Discuss the structure and composition of the atmosphere
- B. i) Discuss noise as a source of Environmental pollution
 ii) Explain what you understand by threshold of hearing and pain
- C. State and discuss the maximum permissible noise level for i) residential environment ii) commercial environment iii) industrial environment iv) sleeping environment

Q4

- A. Write short note on the following
- Residence Time
 - Ozone and Ozone depletion
 - Greenhouse gases
 - Greenhouse Effect
- B. What is "Global warming".
- C. i) Explain what hydrological cycle.
ii) Discuss briefly the Physics of the Cloud.

Q5

- A. In a well tabulated table, discuss the seven categorizes of wind in terms of their Effects and Speed
- B. Discuss briefly the following i) hydrosphere ii) Lithosphere iii) Biosphere
- C. i) What is radioactive decay (series)
ii) State the different particle(s) emitted in the radioactive decay series below



Paper 3 of 3 — Phy 315 Materials (ENERGY__ENVIRONMENT)

Level 300 • First Semester • Handout

energy, the
environment,

CHAPTER 3

and health

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In this chapter, the principal environ-

ABSTRACT mental and health impacts of energy
are discussed according to the scale at which
they occur. About half of the world's households
use solid fuels (biomass and coal) for cooking
and heating in simple devices that produce
large amounts of air pollution—pollution that is
probably responsible for 4–5 percent of the
global burden of disease. The chief ecosystem
impacts relate to charcoal production and
fuelwood harvesting.

At the workplace scale, solid-fuel fuel cycles
create significant risks for workers and have
the largest impacts on populations among
energy systems. In communities, fuel use is
the main cause of urban air pollution, though
there is substantial variation among cities in
the relative contributions of vehicles and
stationary sources. Diesel-fuelled vehicles,
which are more prominent in developing
countries, pose a growing challenge for urban
health. The chief ecosystem impacts result
from large-scale hydropower projects in
forests, although surface mining causes signif-
icant damage in some areas.

At the regional scale, fine particles and
ozone are the most widespread health-

damaging pollutants from energy use, and can extend hundreds of kilometres from their sources. Similarly, nitrogen and sulphur emissions lead to acid deposition far from their sources. Such deposition is associated with damage to forests, soils, and lakes in various parts of the world. At the global scale, energy systems account for two-thirds of human-generated greenhouse gas increases. Thus energy use is the human activity most closely linked to potential climate change. Climate change is feared to have significant direct impacts on human health and on ecosystems.

There are important opportunities for 'no regrets' strategies that achieve benefits at more than one scale. For example, if greenhouse gas controls are targeted to reduce solid fuel use in households and other energy systems with large health impacts (such as vehicle fleets), significant improvements can occur at the local, community, regional, and global scales. ■

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Chapter 3: Energy, the Environment, and Health

Because of their ubiquity and the harvesting, processing, size, energy systems influence every category of environmental. Nearly every category of insult and impact. Indeed, large distribution, and use of fuels and other sources of energy have major environmental insult multiple-volume treatises have been environmental implications. Insults include devoted to discussing the environmental major land-use changes due to fuel cycles and impact. problem of just part of the energy system in such as coal, biomass, and hydropower, which single countries (as with U.S. electric power have implications for the natural as well as human production in ORNL and RFF, 1992-98). A detailed environment.¹ Perhaps the most important insult from energy review of the environmental connections of energy systems systems is the routine and accidental release of pollutants. Human world-wide is beyond the scope of this volume. Indeed, simply catastrophic activities disperse a wide variety of biologically and climatologically logging the routes of insults and types of impacts of energy systems active elements and compounds into the atmosphere, surface waters, world-wide would take substantially more space than is available

and soil at rates far beyond the natural flows of these substances. The here, even if accompanied by little comment.

results of these alterations include a 10-fold increase in the acidity of rain In addition, for three other reasons reproducing catalogues

and snow over millions of square kilometres and significant changes involving simple listings of insults and impacts for each of the many

in the global composition of the stratosphere (upper atmosphere) types of energy systems would not serve the interests of readers.

and troposphere (lower atmosphere). First, many detailed studies in recent years have done this job much

The rough proportions of various pollutants released into the better than we could here. Thus we will cite a range of such material

environment by human activities are shown in table 3.1. Note the to enable interested readers to expand their understanding. In addition,

importance of energy supply systems, both industrial and traditional, there is a substantial amount of such information in other chapters,

for example, on the environmental and health impacts of renewable

in the mobilisation of such toxic substances as sulphur oxides and

energy systems in chapter 7 and of fossil and nuclear power systems

particles as well as in the release of carbon dioxide, the principal

in chapter 8. Chapter 8 also addresses the technological implications

greenhouse gas. Also shown is the human disruption index for each

of reducing urban pollution according to changes in local willingness

substance, which is the ratio of the amount released by human activities

to pay for health improvements. Chapter 1 discusses some of the

to natural releases. This indicates that together with other human

relationships between environment and energy development, and

activities, energy systems are significantly affecting the cycling of chapter 9 has much discussion of the implications of various future

important chemical species at the global scale. Although by energy scenarios for greenhouse gas emissions.

themselves these indexes do not demonstrate that these insults are The second reason relates to our desire to help readers understand

translated into negative impacts, their magnitudes provide warning the relative importance of the problems. The significance of known

that such impacts could be considerable. environmental impacts from energy systems varies by orders of magnitude,

In the past hundred years most of these phenomena have grown from the measurable but minuscule to the planet-threatening. Just

from local perturbations to global disruptions. The environmental as the other chapters in this volume must focus on just a few of the

transition of the 20th century—driven by more than 20-fold growth most important energy system issues for the next half-century, we

in the use of fossil fuels and augmented by a tripling in the use of must do so for environmental impacts.

traditional energy forms such as biomass—has amounted to no Finally, we feel that it is as important to give readers a framework

less than the emergence of civilisation as a global ecological and for thinking about environmental impacts as it is to document

geochemical force. current knowledge about individual problems. Thus we have devoted

The impacts from energy systems, however, occur from the household much of our effort to laying out the problems in a systematic manner

to the global scale. Indeed, at every scale the environmental impacts using scale as the organising principle. Near the end of the chapter

of human energy production and use account for a significant we also discuss two of the most common analytical frameworks for

portion of human impacts on the environment. making aggregate comparisons involving a range of environmental

This chapter examines the insults and impacts of energy systems impacts from energy systems: economic valuation and comparative

according to the scale at which the principal dynamics occur— risk assessment using fuel-cycle analysis.

meaning the scale at which it makes the most sense to monitor, Given space limitations and the reasons summarised above, we

evaluate, and control the insults that lead to environmental impacts. focus below on the two or three most important environmental

In addition, some cross-scale problems are considered to illustrate insults and impacts at each scale. This approach brings what may

the need to control insults occurring at one scale because of the seem to be a geographic bias as well—examples at each scale tend

impacts they have at other scales. Impacts are divided into two to be focused not only on the most important problems but also on

broad categories: those directly affecting human health (environ- the places in the world where the problems are most severe. We

mental health impacts) and those indirectly affecting human welfare recognise that there are innumerable other impacts and places that

through impacts on the natural environment (ecosystem impacts). could be mentioned as well, but we offer this set as candidates for

Because of their ubiquity and size, energy systems influence nearly those that ought to have the highest priority in the next few decades.

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Chapter 3: Energy, the Environment, and Health

Indeed, if these environmental problems were brought under report's goal of exploring the sustainability of current practices. In

control, the world would have moved most of the way towards a later chapters, as part of efforts to examine the feasibility of

sustainable energy future from an environmental standpoint. advanced energy conversion technologies, new sources of energy,

This chapter focuses almost entirely on the environmental insults and enhanced end-use efficiencies, the potential environmental and impacts associated with today's energy systems, in line with this impacts of future energy systems are explored.

TABLE 3.1. ENVIRONMENTAL INSULTS DUE TO HUMAN ACTIVITIES BY SECTOR, MID-1990S

Natural Human Share of human disruption caused by			
Insult	baseline	disruption	Commercial Traditional Agriculture Manufacturing, (tonnes a year) indexa energy supply energy supply other
Lead emissions to atmosphere	12,000	18	41% (fossil fuel Negligible Negligible 59% (metal burning, including processing, additives) manufacturing, refuse burning)
Oil added to oceans	200,000	10	44% (petroleum Negligible Negligible 56% (disposal harvesting, of oil wastes, processing, including motor and transport) oil changes)
Cadmium emissions to atmosphere	1,400	5.4	13% (fossil 5% (traditional 12% (agricultural 70% (metals emissions to fuel burning) fuel burning) burning) processing, manufacturing, refuse burning)
Sulphur emissions to atmosphere	31 million	2.7	85% (fossil 0.5% (traditional 1% (agricultural 13% (smelting, to atmosphere (sulphur) fuel burning) fuel burning) burning) refuse burning)
Methane flow to atmosphere	160 million	2.3	18% (fossil 5% (traditional 65% (rice paddies, 12% (landfills) fuel harvesting fuel burning) domestic animals, and processing) land clearing)
Nitrogen fixation (as nitrogen oxide (nitrogen) and ammonium)c	140 million	1.5	30% (fossil 2% (traditional 67% (fertiliser, 1% (refuse fuel burning) fuel burning) agricultural burning) burning)
Mercury emissions to atmosphere	2,500	1.4	20% (fossil 1% (traditional 2% (agricultural 77% (metals to atmosphere fuel burning) fuel burning) burning) processing, manufacturing, refuse burning)
Nitrous oxide flows to atmosphere	33 million	0.5	12% (fossil 8% (traditional 80% (fertiliser, Negligible fuel burning) fuel burning) land clearing, aquifer disruption)
Particulate emissions to atmosphere	3,100 million	0.12	35% (fossil 10% (traditional 40% (agricultural 15% (smelting, non-emissions to fuel burning) fuel burning) burning) agricultural land atmosphere clearing, refuse)
Non-methane	1,000 million	0.12	35% (fossil 5% (traditional 40% (agricultural 20% (non-

hydrocarbon fuel processing fuel burning) burning) agricultural land
emissions to and burning) clearing, refuse
atmosphere burning)

Carbon dioxide 150 billion 0.05e 75% (fossil fuel 3% (net 15% (net 7% (net
flows to (carbon) burning) deforestation for deforestation for deforestation for
atmosphere fuelwood) land clearing) lumber, cement
manufacturing)

Note: The magnitude of the insult is only one factor determining the size of the actual
environmental impact. a. The human disruption index is the ratio
of human-generated flow to the natural (baseline) flow. b. The automotive portion of anthropogenic
lead emissions in the mid-1990s is assumed to
be 50 percent of global automotive emissions in the early 1990s. c. Calculated from total nitrogen
fixation minus that from nitrous oxide. d. Dry mass.

e. Although seemingly small, because of the long atmospheric lifetime and other characteristics of
carbon dioxide, this slight imbalance in natural flows

is causing a 0.4 percent annual increase in the global atmospheric concentration of carbon dioxide.

Source: Updated from Holdren, 1992 using Houghton and others, 1994; IPCC, 1996b; Johnson and
Derwent, 1996;

Lelieveld and others, 1997; Nriagu, 1989, 1990; Smithsonian Institution, 1996; Smith and Flegal,
1995; and WRI, 1998.

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Chapter 3: Energy, the Environment, and Health

Household scale moving up the ladder (OTA, 1992).

The oldest human energy technology, the home cooking fire, Although there are local exceptions,
history has generally shown

persists as the most prevalent fuel-using technology in the world. that when alternatives are
affordable and available, populations tend

For much of the world's population, the home cooking fire accounts to move up the ladder to
higher-quality fuel-stove combinations.

for most direct energy demand. Household fuel demand accounts Although all of humanity had its
start a quarter of a million years ago

for more than half of energy demand in most countries with per at the top of the energy ladder in
those times (wood), only about

capita incomes under \$1,000 (see figure 2.1). half has moved up to higher-quality rungs (figure
3.1). The remaining

The 'energy ladder' is a useful framework for examining trends half is either still using wood or has
been forced by local wood shortages

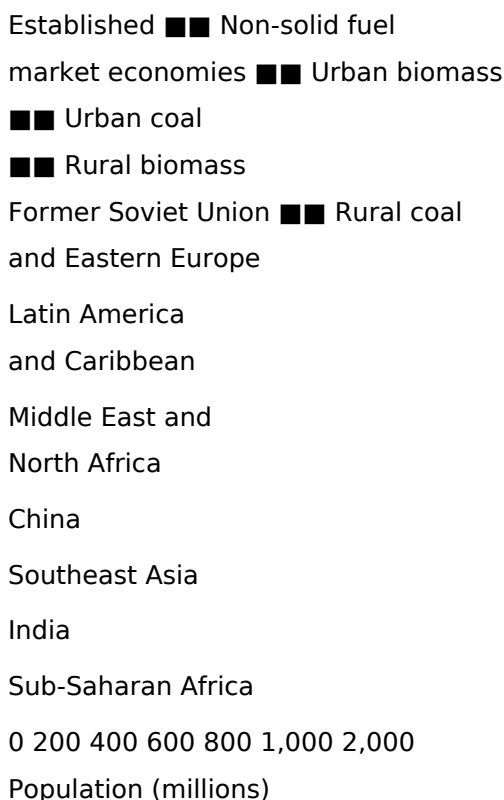
and impacts of household fuel use (see figure 10.1). The ladder down the ladder to crop residues,
dung, or, in some severe situations,

ranks household fuels along a spectrum running from simple to the poorest-quality fuels such as
shrubs and grass.

biomass fuels (dung, crop residues, wood) through fossil fuels Throughout history in places where
coal is easily available, local

(kerosene and gas) to the most modern form (electricity). The wood shortages have led some populations to move to coal for fuel-stove combinations that represent rungs in the ladder toward household use. This shift occurred about a thousand years ago in to become cleaner, more efficient, more storable, and more so in the United Kingdom, for example, although it is relatively uncommon controllable in moving up the ladder.² But capital costs and there today (Brimblecome, 1987). In the past 150 years such transitions dependence on centralised fuel cycles also tend to increase in occurred in Eastern Europe and China, where coal use still persists

FIGURE 3.1. POPULATION AND HOUSEHOLD FUEL USE BY REGION, 1995



Source: Updated from Reddy, Williams, and Johansson, 1997.

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Chapter 3: Energy, the Environment, and Health

Physical form and contaminant

in millions of households (see figure content are the two characteristics change, intensification of grazing,

3.1). In terms of the energy ladder, of fuels that most affect land-use shifts, and fuel harvesting. coal represents an upward movement their pollutant emissions Nevertheless, as with deforestation, there

in terms of efficiency and storability. are some poor areas where harvesting of

Because of these characteristics and its when burned. wood and brush plays an important role.

higher energy densities, it is possible to ship Although the link between fuelwood

coal economically over longer distances than wood harvesting and deforestation is far from universal, and to efficiently supply urban markets. In these senses, coal there are localised cases in which fuelwood demand seems is like other household fossil fuels. Unlike kerosene and gas, to contribute significantly to forest depletion. Most prominent however, coal is often a dirtier fuel than wood. among these are places, mainly in Sub-Saharan Africa, where commercial charcoal production is practised. In these areas temporary kilns (legal or illegal) in forested areas are used until local wood resources are depleted, then moved or rebuilt elsewhere. The chief environmental impacts of household fuel cycles relate to wood resources are depleted, then moved or rebuilt elsewhere. harvesting and combustion. In the 1970s books and newspapers Charcoal, being a relatively high-quality and high-density fuel, can called attention to the 'other' energy crisis, referring to the growing need to transport economically across long distances to urban markets. and alarming shortage of woodfuel affecting a large fraction of the world population that depended on it. Thus large charcoal-using cities can have 'wood sheds' extending hundreds of kilometres along roadways, though there is evidence desertification were often also occurring in such places, it was that significant regrowth often occurs over long enough periods. In perhaps a logical conclusion that fuel demand was to blame. It is some arid and semiarid areas, harvesting by woodfuel traders to still common today to read that deforestation is caused by fuel meet urban demand seems to contribute to forest depletion, gathering in rural areas of developing countries. Detailed studies in although, again, regrowth is often occurring. The quality of many areas around the world, however, have rarely documented regrowth in terms of biodiversity and other ecosystem parameters is cases in which fuel demand is a significant cause of deforestation. not well documented, however. The most important cause by far seems to be expansion of agricultural The harvesting of dung and crop residues as fuel does not have lands, followed by lumbering and road building. Indeed, the causation much direct environmental impact. But in some areas it may deprive is more often the reverse—that is, the shortage of fuelwood is due local soils of needed nutrients and other conditioners. Indeed, in to deforestation, rather than the other way around. most rural areas the use of dung as fuel rather than fertiliser is Part of the misunderstanding stems from the assumption that probably a sign of poverty and lack of fuel alternatives. Crop rural households gather woodfuel from forests. In many areas residues, on the other hand, consist of a wide variety of materials, villagers gather significant amounts of fuelwood from what Gerald many of which do not have much value as fertiliser or soil conditioner.

Leach has called ‘invisible trees’—the trees around fields, next to Indeed, in some cases disposal becomes a serious problem if these houses, along roads, and so on that do not show up in most satellite residues are not gathered for fuel. In these cases the usual practice remote sensing surveys or national forest statistics. Thus when is to burn the residues in place on the fields, with consequent estimates of local fuelwood demand appear to exceed growth rates pollution implications (although sometimes with benefits in terms in local forests, it does not necessarily imply that deforestation is of pest control). Consequently, regardless of development level, air taking place. Conversely, if deforestation is known to be occurring, pollution from post-harvest burning of farmland is a significant it does not mean that fuel demand is necessarily the reason. seasonal source of air pollution in many agricultural areas around

Similarly, desertification in the Sahel and elsewhere in Sub- the world (see chapter 10). Harvesting and preparation of household Saharan Africa has links to fuel demand. But it has been difficult to biomass fuels also have occupational health impacts on women and separate out the influence of all the relevant factors, including climate children, due, for example, to heavy loads and burns (see the section on workplace scale, below).

BOX 3.1. HEALTH-DAMAGING POLLUTANTS IN SOLID Combustion

FUEL SMOKE FROM HOUSEHOLD STOVES IN INDIA Physical form and contaminant content are the two characteristics

of fuels that most affect their pollutant emissions when burned. It is

Biomass smoke generally difficult to pre-mix solid fuels sufficiently with air to assure

- Small particles, carbon monoxide, nitrogen dioxide.
- Formaldehyde, acrolein, benzene, 1,3-butadiene, toluene, good combustion in simple small-scale devices such as household

styrene, and so on. stoves. Consequently, even though most biomass fuels contain few

- Polyaromatic hydrocarbons such as benzo(a)pyrene.

noxious contaminants, they are usually burned incompletely in

Coal smoke household stoves and so produce a wide range of health-damaging

- All the above plus, depending on coal quality, sulphur oxides

and such toxic elements as arsenic, lead, fluorine, and mercury. pollutants (box 3.1). Wood and other biomass fuels would produce

little other than non-toxic products, carbon dioxide, and water when

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FIGURE 3.2A. ENERGY FLOWS IN A TYPICAL WOOD-FIRED COOKING STOVE

1 kilogram of wood

15.33 megajoules

Into pot In products of incomplete combustion Waste heat
2.76 megajoules (18%) 1.23 megajoules (8%) 11.34 megajoules (74%)

Source: Smith and others, 2000 a.

FIGURE 3.2B. INDOOR CONCENTRATIONS OF HEALTH-DAMAGING POLLUTANTS FROM A TYPICAL WOOD-FIRED COOKING STOVE

1 kilogram of wood per hour in
15 ACH 40 cubic metre kitchen

Carbon monoxide Particles Benzene 1,3-Butadiene Formaldehyde
150 mg/m³ 3.3 mg/m³ 0.8 mg/m³ 0.15 mg/m³ 0.7 mg/m³
(10 mg/m³) (0.1 mg/m³) (0.002 mg/m³) (0.0003 mg/m³) (0.1 mg/m³)

Note: Dozens of other health-damaging pollutants are known to be in woodsmoke. Mg/m³ stands for milligrams per cubic metre. Numbers in parentheses are typical standards set to protect health. Source: Smith and others, 2000 a.

FIGURE 3.2C. GREENHOUSE GAS EMISSIONS FROM A TYPICAL BIOMASS COOKSTOVE

1 kilogram of wood
454 grams carbon

Other greenhouse gas carbon

Carbon dioxide Methane carbon Nitrous oxide

Carbon monoxide: 37.5 grams (131 grams)
403 grams 3.8 grams (86 grams) 0.018 grams (4.7 grams)
Hydrocarbons: 6.3 grams (69 grams)

Note: Numbers in parentheses are carbon dioxide equivalents of non-carbon dioxide gases. Source: Smith and others, 2000 a.

combusted completely. But in practice sometimes as much as one-fifth because it is solid, but also often contains significant intrinsic

of the fuel carbon is diverted to products of incomplete combustion, contaminants that add to its emissions of health-damaging pollutants.

many of which are important health-damaging pollutants. Most prominent among such emissions are sulphur oxides (see box

Coal, on the other hand, is not only difficult to burn completely 3.1). But in many areas coal also contains arsenic, fluorine, and

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other toxic elements that can lead to serious health-damaging To estimate the health damage from pollution, it is necessary to

pollutants. Tens of millions of people in China, for example, are exposed take into account the amount of pollution released. Equally important,

to such pollutants from household coal use. however, is the behaviour of the population at risk. Even a large

Petroleum-based liquid and gaseous fuels, such as kerosene amount of pollution will not have much health impact if little of it

and liquefied petroleum gas, can also contain sulphur and other reaches people. But a relatively small amount of pollution can have

contaminants, though in much smaller amounts than in many coals. a big health impact if it is released at the times and places where

In addition, their physical forms allow much better pre-mixing with people are present, such that a large fraction is breathed in. Thus it

air in simple devices, assuring much higher combustion efficiencies is necessary to look not only at where the pollution is but also at

and lower emissions of health-damaging pollutants in the form of where the people are.

products of incomplete combustion. Furthermore, stoves for petroleum- Unfortunately, pollution from household stoves is released right

based liquid and gaseous fuels are much more energy efficient than at the times and places where people are present—that is, in every

those for coal. As a result emissions of health-damaging pollutants household every day. This is the formula for high pollution exposures:

per meal from these fuels are at least an order of magnitude less significant amounts of pollution often released in poorly ventilated

than those from solid fuels (Smith and others, 2000 a).

spaces at just the times when people are present. Moreover, because

Not only do solid-fuel stoves produce substantial emissions of

of their nearly universal responsibility for cooking, women and their

health-damaging pollutants per meal, but a large fraction do not

have chimneys for removing the emissions from the home. youngest children are generally the most exposed.

Consequently, indoor concentrations of health-damaging pollutants Thus, although the total amount of health-damaging pollution

can reach very high levels. Figure 3.2a shows the energy flows of a released from stoves

world-wide is not high relative to that from

typical wood-fired cooking stove, in which a large fraction of the fuel large-scale use of fossil fuels, human exposures to a number of

energy is lost because of low combustion efficiency or low transfer important pollutants are much larger than those created by outdoor

of the heat to the pot. Figure 3.2b shows the excessive pollutant lev- pollution. As a result the health effects can be expected to be

els commonly reached in these circumstances, well beyond World higher as well.

Health Organization guidelines. Even in households with chimneys, In many ways the harvesting impacts and air pollution from use

however, heavily polluting solid-fuel stoves can produce significant of biomass fuels are the result of fuel shortages, particularly where

local outdoor pollution. This is particularly true in dense urban inferior forms (dung and crop residues) are in use. Thus these can

slums, where such ‘neighbourhood’ pollution can be much higher be considered part of the health effects of too little energy, along

than average urban pollution levels. with lower nutrition and chilling (box 3.2).

BOX 3.2. HEALTH EFFECTS OF TOO LITTLE WOODFUEL

Lack of sufficient fuel for heating and cooking has several negative health impacts. First, in Household coping strategies for fuelwood shortages many places women and children must walk further and work harder to gather fuel, using more energy and time and placing themselves at increased risk of assault and natural hazards such as leeches and snakes. In time, addition, nutrition can be negatively affected and injury if families have to walk long distances to gather cooking fuel. When seasonal changes natural hazards result in longer fuel collection times, families commercial of special foods for are unable to compensate by reducing the time spent on agricultural activities. Instead fuels children, pregnant Inferior biomass fuels or lactating women, the time is subtracted from resting and food preparation. on fuelwood Inferior fuels, such as twigs and grass, that are used as substitutes in times of shortage require more attention from women during cooking, keeping them from other allocation tasks. These fuels also produce more health-damaging smoke and are inadequate for processing more nutritious foods such as cereals and beans (since they have long cooking times). The figure at right outlines some coping strategies adopted by households to deal with fuel shortages and their health consequences. Source: Agarwal,1985; Brouwer,1994.

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BOX 3.3. NATIONAL BURDEN OF DISEASE FROM HOUSEHOLD SOLID FUEL USE IN INDIA

National surveys, including the 1991 national census, show that nearly 80% of Indian household solid fuel use in a national survey Tuberculosis has been associated with because households rarely use coal. Cardiovascular (heart) disease. Although

households use biomass as their primary in India involving nearly 90,000 households there are apparently no studies in biomass-using cooking fuel. As a result, a large portion of the as well as in smaller studies. Although this households, studies of urban air pollution suggest Indian population is potentially exposed to relationship is not yet established with that in India 50,000–190,000 women under 30 indoor and outdoor levels of pollution produced complete certainty, it would be highly suffer from pollution-related heart disease. by cooking stoves. Based on risks derived significant because tuberculosis is on the Adverse pregnancy outcomes. Stillbirth solely from studies of the health effects of rise in many developing countries due to and low birthweight have been associated individual diseases occurring in biomass-using HIV infection and the increase in drug- with solid fuel use by pregnant women in households in developing countries, many in resistant strains. In India 50,000–130,000 Latin America and India. Low birthweight is a India itself, it is possible to estimate the total cases of tuberculosis in women under 15 big problem in developing countries because national burden of disease in India from use are associated with solid fuel use. it is a risk factor for a range of health problems. of these fuels: Chronic respiratory disease, such as In India, however, there are too few studies to Acute respiratory infection. More than a chronic bronchitis, is almost entirely due to calculate the impacts of solid fuel use on dozen studies around the world have found smoking in the industrialised world. But adverse pregnancy outcomes. that household use of solid fuels is associated studies in Asia and Latin America have found Total. Because there is more uncertainty with acute respiratory infection in young children the chronic respiratory disease develops in in the estimates for tuberculosis and heart (although, as with all the diseases discussed women after long years of cooking with solid disease, only the low ends of the estimated here, there are other important risk factors, fuels. In India 19,000–34,000 women under ranges are used. In India 410,000–570,000 including malnutrition and crowding). Acute 45 suffer from chronic respiratory disease premature deaths a year in women and children, respiratory infection is the leading cause of linked to solid fuel use. of 5.8 million total, seem to be due to biomass death of the world's children and the largest Lung cancer, which is also dominated fuel use. Given the age distribution of these category of ill health in the world in terms of by smoking in industrialised countries, has deaths and the associated days of illness disease burden. Almost 9 percent of the global been found to result from long-term exposure involved, 5–6 percent of the national burden burden of ill health and 12 percent of India's to cooking with coal in more than 20 studies of disease in women and young children can

is due to acute respiratory infection. Acute in China. No such effect has been shown be attributed to biomass fuel use in households.

respiratory infection linked to solid fuel use is for biomass fuels, however. In India 400–800 For comparison, about 10 percent of the Indian

estimated to cause 290,000–440,000 premature women under 45 suffer from lung cancer national burden of disease is attributed to lack

deaths a year in Indian children under 5. linked to solid fuel use; the number is small of clean water and sanitation.

Source: Smith, 2000; Smith and others, 2000; Murray and Lopez, 1996.

Estimated health effects tobacco, AIDS, heart disease, or cancer (Murray and Lopez, 1996).

Considering the sizes of the relevant populations and the exposures Given that India contains about one-quarter of the world's solid-fuel

to health-damaging pollutants, there has been relatively little scientific cooking stoves, the global impact could be expected to be about

investigation of the health effects of indoor air pollution in developing four times larger, or about 2 million deaths a year in women and

countries relative to studies of outdoor air pollution in cities. children. This is roughly compatible with World Health Organization

Nevertheless, enough has been done to enable rough estimates of estimates of about 2.5 million, estimates that were generated by

the total impact of air pollution, at least for women and young extrapolating studies from industrialised country cities to developing

children (who suffer the highest exposures). country conditions (WHO, 1997). The global burden of disease from

Four main types of health effects are thought to occur, based on major risk factors, including indoor air pollution, is shown in figure 3.3.

studies in households that use solid fuels and corroborated by studies

of active and passive smoking and outdoor air pollution (Smith, 1998):

Greenhouse gases

■ Infectious respiratory diseases such as acute respiratory infections

The same incomplete combustion processes that produce emissions and tuberculosis.

■ Chronic respiratory diseases such as chronic bronchitis and

of health-damaging pollutants from household solid-fuel stoves also

lung cancer. produce greenhouse gas emissions. (Greenhouse gas emissions and

■ Adverse pregnancy outcomes such as stillbirth and low birth- their global impacts are described below, in the section on the global

weight in babies born to women exposed during pregnancy. scale.) A large amount of fuel carbon is typically diverted to gaseous

■ Blindness, asthma, and heart disease (less evidence to date). products of incomplete combustion, all of which cause greater

The best estimates of such effects for developing countries have global warming per carbon atom than would be the case if complete

been done for India (box 3.3). These indicate that household solid combustion occurred and all the carbon was released as carbon fuel use causes about 500,000 premature deaths a year in women dioxide (see figure 3.2c). The most powerful of these is methane, and children under 5. This is 5–6 percent of the national burden which over a 20-year period causes more than 20 times the global of ill health, or 6–9 percent of the burden for these two warming from the same amount of carbon as carbon dioxide population groups.³ This is comparable to, though somewhat (equivalent to a discount rate of about 4 percent). less than, the estimated national health impacts of poor water and Greenhouse gas emissions from several of the most important sanitation at the household level—and more than the national household fuels in developing countries (as measured in India) are burdens of such major health hazards as malaria, tuberculosis, shown in figure 3.4. Because of significant emissions of non-carbon

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gas emissions over the entire fuel cycle in each case, for example,

FIGURE 3.3. GLOBAL BURDEN OF DISEASE

FROM SELECTED MAJOR RISK FACTORS, 1995 including methane leaking from biogas digesters and releases from

oil refineries making kerosene. Nevertheless, the extremely low greenhouse gas emissions from biogas stoves illustrate the potentially Indoor air pollution great advantage for greenhouse gas emissions of processed biomass fuels such as biogas. Such fuels can be both renewably harvested as Outdoor air well as burned as liquids or gases with high combustion efficiency. pollution

(The section on cross-scale impacts, below, discusses some of the Occupational potential opportunities for reducing impacts at the household and hazards

global scales through improvements in household cooking.)

Poor water and Reducing the human health and global warming impacts of sanitation

household stoves will require better stoves with higher efficiency, Physical inactivity lower emissions, and cleaner fuels. These issues are discussed in chapter 10. Of course, the largest greenhouse gas emissions are from energy Hypertension systems used in industrialised countries, as discussed in later sections.

Unsafe sex or

unwanted pregnancy Workplace scale

The extraction, transport, use, and waste management of energy

Illicit drug use

sources involve important health hazards related to the work and

workplaces involved in these activities. Many of the jobs involved,

Tobacco use

such as forestry and mining for solid fuels, are particularly dangerous.

Many workers are engaged in these jobs, particularly in countries

Alcohol use

that are rapidly developing their industries and the energy sources

Malnutrition

that the industries require. In addition, much of the work needed

for household energy supply in developing countries is carried out

0 5 10 15 20 as a household task that does not figure in national statistics as an

Percentage of total 'occupational issue'.

This section analyses these health issues based on the type of energy

Note: Burden of indoor air pollution extrapolated from data for India.

source and give examples of how the effects have been documented

Source: Smith, 2000; Murray and Lopez, 1996.

in different countries. The fourth edition of the International Labour

dioxide greenhouse gases, solid biomass fuels, even though renewable, Organisation's

Encyclopaedia of Occupational Safety and Health

can have a larger greenhouse gas commitment per meal than fossil (Stellman, 1998) provides additional detail about energy jobs and

fuels, kerosene, and liquefied petroleum gas. These relationships their health hazards.

have several important policy implications:

■ Even if renewably harvested, many biomass fuel cycles are not Biomass

greenhouse gas neutral because of their substantial production As noted, wood, crop residues, dung, and the like are common energy

of products of incomplete combustion. sources for poor households in developing countries. Wood is also

■ In some situations, therefore, substitution of fossil fuels for still widely used in industrialised countries, in some cases promoted

renewable biomass might be recommended to reduce greenhouse in the interest of reducing greenhouse gas emissions. Wood and

gas emissions. agricultural waste are often collected by women and children (Sims,

■ To be greenhouse gas neutral, biomass fuel cycles must be based on 1994). Such collection is part of daily survival activities, which also

renewable harvesting and must have close to 100 percent combust- include water hauling, food processing, and cooking (see chapters

tion efficiency (which most do not in their current configurations). 2 and 10). An analysis in four developing countries found that

■ Improved biomass stoves should be designed to increase overall women spend 9–12 hours a week on these activities, while men

efficiency and to reduce combustion inefficiency, which is the spend 5–8 hours (Reddy, Williams, and Johansson, 1997). Women's

cause of greenhouse gas and health-damaging pollutants. role in firewood collection is most prominent in Nepal (2.4 hours a

Stoves using biogas, which is made in household or village anaerobic digester (0.8 hours for women and 0.8 hours for men).

digesters from dung (see chapter 10), have by far the least greenhouse gas emissions. Firewood collection may be combined with harvesting of wood

household gas emissions per meal—only about 10 percent of those for local use in construction and small-scale cottage industry

liquefied petroleum gas and a factor of 80 less than the average stove manufacturing. This subsistence work is often seasonal, unpaid,

burning dung directly (see figure 3.4). A complete comparison of and not recorded in national economic accounts. Globally about 16

these fuel-stove combinations would require evaluating greenhouse gas emissions. million people are involved in forestry (Poschen, 1998), more than

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Pollution from household stoves

14 million of them in developing countries is released right at the times and the health of the workers. About 1

countries and 12.8 million in sub-Saharan Africa—places where people are present—percent of the global workforce is

subsistence forestry. that is, in every household engaged in mining, but these workers

A number of health hazards are associated with the basic conditions of the forest. every day. occupational accidents each year. Armstrong

Forest workers have a high risk of insect bites, and Menon (1998) offer a detailed review of stings from poisonous plants, leech infestation, cuts, occupational health and safety issues in coal mining

falls, and drowning. In tropical countries the heat and humidity and other mining.

humidity put great strain on the body, while in temperate countries Underground coal miners are exposed to the hazards of excavating

the effects of cold are a potential hazard. The work is outside, and transporting materials underground. These hazards include

in sunny countries ultraviolet radiation can be another health hazard, injuries from falling rocks and falls into mine shafts, as well as

increasing the risk of skin cancer and eye cataracts (WHO, 1994). injuries from machinery used in the mine. There are no reliable

All forestry work is hard physical labour, with a risk of ergonomic data on injuries of this type from developing countries (Jennings,

damage such as painful backs and joints as well as fatigue, which 1998), but in industrialised countries miners have some of the highest

increases the risk of injuries. Heavy loads of firewood contribute to rates of compensation for injuries—and the situation is likely to be

ergonomic damage (Poschen, 1998). Women carrying heavy loads worse in developing countries. In addition, much of the excavation of firewood are a common sight in areas with subsistence forestry involves drilling into silica-based rock, creating high levels of silica (Sims, 1994). Falling trees, sharp tools, dangerous machinery, and dust inside the mine. Pneumoconiosis silicosis is therefore a falls from heights are the main causes of injuries. In addition, the common health effect in coal miners (Jennings, 1998).

living conditions of forestry workers are often poor, and workers may be spending long periods in simple huts in the forest with limited protection against the weather and poor sanitary facilities. FIGURE 3.4. GREENHOUSE GAS EMISSIONS

FROM HOUSEHOLD FUELS

Urbanisation leads to the development of a commercial market for firewood and larger-scale production of firewood from logs or from smaller waste material left over after logs have been harvested. Energy forestry then becomes more mechanised, exposing (57%) workers to additional hazards (Poschen, 1998). Motorised hand tools (such as chain saws) become more common, resulting in high risks of injuries, noise-induced hearing losses, and 'white finger' disease caused by vibration of the hands. In addition, fertilisers and pesticides become part of the production system, with the potential for poisoning those who spray pesticides. As forestry develops further, Kerosene (50%) more logging becomes mechanised, with very large machinery reducing the direct contact between workers and materials. Workers in highly mechanised forestry have only 15 percent of the injury risk of highly skilled forestry workers using chain saws (Poschen, 1998). Still, firewood production remains an operation that requires manual handling of the product at some stage and so tends to remain hazardous. Another health aspect of wood-based energy is the risk of burning wood that has been treated against insect damage with copper-arsenic compounds or that has been painted with lead paint. Such wood residues may be harder to sell and so may be used to a greater extent by firewood production workers in stoves and open fires. When burned, poisonous arsenic and lead compounds will be emitted with the smoke. These compounds are health hazards when inhaled. (9%)

1 10 100 1,000
Coal

1 10 100 1,000
Coal

1 10 100 1,000
Coal

1 10 100 1,000
Coal

1 10 100 1,000
Coal

Grams of carbon per megajoule delivered

Coal is a major global energy source, accounting for 23 percent of (carbon dioxide equivalent) total energy consumption. It was the primary energy source from 1900 until 1960, when it was overtaken by oil (WHO, 1997). Coal can be Note: Includes warming from all greenhouse gases emitted: carbon dioxide, methane, carbon monoxide, non-methane hydrocarbons, and nitrous oxide. Weighted by stove distribution in India. Numbers in parentheses

Like mining in general, both operations are inherently dangerous to are average stove energy efficiency. Source: Smith and others, 2000 b.

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Solid biomass fuels, even though

Other health hazards specific to renewable, can have a larger greenhouse to elemental iron.) Additional

underground coal mining include gas commitment per meal than health hazards are created for workers

coal dust, which can cause 'coal worker's fossil fuels, kerosene, and when the large amounts of ash produced

pneumoconiosis' or anthracosis, often by power stations and industry need to combined with silicosis. Coal dust is explosive, liquefied petroleum gas. be transported and deposited. Crane (1998)

and explosions in underground coal mines are reviews the health hazards faced by power a constant danger for coal miners. Coal inherently generation workers.

burns, and fires in coal mines are not uncommon. Once such

a fire has started, it may be almost impossible to extinguish. Apart Oil and gas

from the danger of burns, the production of smoke and toxic fumes Oil and gas exploration, drilling, extraction, processing, and transport

create great health risks for the miners. involve a number of the same hazards as mining: heavy workload,

Even without fires, the coal material produces toxic gases when ergonomic hazards, injury risk noise, vibration, and chemical expo-

it is disturbed: carbon monoxide, carbon dioxide, and methane sures (Kraus, 1998). This work is often carried out in isolated areas

(Weeks, 1998). Carbon monoxide is extremely toxic because it with inclement weather conditions. Long-distance commuting may

binds to haemoglobin in the blood, blocking oxygen transport and also be involved, causing fatigue, stress, and traffic accident risks.

creating chemical suffocation (Bascom and others, 1996). Carbon The ergonomic hazards lead to risks of back pain and joint

monoxide is a colourless and odourless gas and so gives no warning pain. Injury hazards include burns and explosions. Skin damage

before the symptoms of drowsiness, dizziness, headache, and from exposure to oil and to chemicals used in drilling creates a

unconsciousness occur. Carbon dioxide, also colourless and need for well-designed protective clothing. In addition, many oil

odourless, displaces oxygen in underground air and can also cause and gas installations have used asbestos to insulate pipes and

suffocation. Another health hazard in mining is exhaust from the equipment. Inhalation of asbestos dust in the installation and repair

diesel engines used in underground machinery and transport vehicles. of such equipment creates a risk of lung cancer, asbestosis, and

This exhaust contains very fine particles, nitrogen oxides, and carbon mesothelioma (WHO, 1998a). monoxide, all of which pose serious health hazards (Bascom and A lot of exploration and drilling for oil and gas occur offshore.

others, 1996). Exposure to fine particles in diesel exhaust increases This involves underwater diving, which is dangerous. In addition,

the risk of lung cancer (Holgate and others, 1999). weather-related exposures can be extreme, particularly since the

Surface coal mining avoids some of the hazards of working work often requires round-the-clock operations (Kraus, 1998).

underground. Still, it involves the risk of injuries from machinery,

falls, and falling rocks. In addition, coal mining is energy-intensive Hydropower and other renewables

work, and heat, humidity, and other weather factors can affect Major hazards occur when a hydroelectric power station is built,

workers' health. The machinery used is also noisy, and hearing loss because this usually requires constructing a large dam, excavating

is a common among miners. Another health hazard is the often underground water channels, and building large structures to house

squalid conditions under which many coal workers and their families the generator. McManus (1998) lists 28 occupational hazards

in developing countries live, creating particular risk for the diseases potentially involved in the construction and operation of hydroelec-

of poverty. In addition, such workers are likely to receive part of tric power stations. These include asbestos exposure, diesel and

their compensation in the form of coal for use as household fuel, welding fumes, work in confined spaces or awkward positions,

with consequent indoor and neighbourhood pollution. drowning, electrocution, noise, heat, electromagnetic fields, vibration,

After extraction, coal needs to be processed and transported to weather-related problems, and chemical exposures from paints,

residential areas, power stations, and industries. This creates other oils, and PCBs (polychlorinated biphenyls). As in any industry,

types of occupational hazards (Armstrong and Menon, 1998). For however, proper attention to health and safety can keep the risks to

instance, coal for residential use is often ground and formed into acceptable levels.

briquettes. This work involves high levels of coal dust as well as The manufacture of wind and solar power equipment involves the

noise hazards. Loading, transportation, and off-loading of large typical hazards in manufacturing: injuries, noise, chemical exposures,

amounts of coal involves ergonomic, noise, and injury hazards. and so on. In addition, the technologies for solar electricity generation

The large-scale use of coal in power stations and industry creates involve new chemical compounds, some based on rare metals with

yet more hazards. One is the conversion of coal to coke in steel poorly known toxic properties (Crane, 1998).

production. This process distils a large number of volatile polycyclic

aromatic hydrocarbons in coal, the so-called coal tar pitch volatiles Nuclear danger

(Moffit, 1998). Exposure to these hydrocarbons puts coke oven Nuclear power generation has its own hazards due to the radiation

workers at twice the lung cancer risk of the general population danger involved in mining, processing, and transporting uranium,

(IARC, 1984). (This process is not entirely associated with energy as well as the radiation in nuclear power stations. In addition,

supply, as an important aim is to provide carbon to reduce iron oxides occupational hazards will develop as countries start to deal with the

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backlog of radioactive waste. Due to the major potential risk to the 40–60 percent of the population over 15. Thus miners may account

general public from a malfunctioning nuclear power station, the for 1 percent of the population over 15, or about 30 million people

safety of stations is always paramount. This has contributed to a low world-wide. If half of these miners are coal miners, the number of

average occupational health risk for workers in the stations miners killed each year in accidents would be about the same as for

(Morison, 1998). forestry workers (30,000–150,000). Another approach to this

The mining of uranium has been an important occupational calculation is through total coal production. If applied to the world's

health hazard in nuclear power generation, as underground mining coal production today, about 70 percent of which is in developing

for uranium often entails high exposure to radon, a radioactive countries, the mean death rate in U.S. mines from 1890–1939 of 3.1

gas emitted from uranium. Radon exposure leads to an increased deaths per million tons produced would predict 16,000 coal

risk of lung cancer. In addition, the same occupational hazards in mining deaths a year world-wide (ORNL and RFF, 1994a). This may

mining noted above occur, although the relatively high energy be low, however, because China alone has about 6,500 coal mining

content of uranium ore means that there are fewer health effects per deaths a year according to official statistics, which tend to be incomplete

unit of electricity produced. (Horii, 1999). The estimate of 6,500 of 16,000 deaths, on the other

Until the Chernobyl accident, relatively few nuclear power station hand, is roughly consistent with China's 30 percent share of global workers had been affected by radiation exposure. In that accident, production (BP, 1998). however, 40 workers lost their lives in the fire or due to acute For energy production and distribution as a whole, occupational radiation exposure. The long-term impact on workers exposed mortality may sum to 70,000–350,000 a year. These numbers are during the accident in the form of cancer and other radiation- likely to exclude many cases of occupational disease (such as cancers related effects is not yet known, however. The clean-up after the caused by asbestos or radiation) and deaths among the many workers accident may eventually create substantially more effects. As many in informal workplaces. The upper limit of the numbers, however, as 900,000 army, police, and other workers were called on to take may also be inflated by the crude estimates of mortality rates and part. Many workers were needed because they were only allowed to number of workers. Occupational mortality rates in energy jobs in work for a short time, until they had reached the maximum industrialised countries are generally 10–30 times lower than in allowable radiation dose. In some cases this dose was reached in a developing countries (Kjellstrom, 1994; ILO, 1998), indicating that few minutes. Studies are now being undertaken to establish the more effective prevention programs could eliminate more than 90 exposure of each clean-up worker and the long-term health impacts percent of the deaths referred to above. Still, energy-related jobs (WHO, 1996). have inherent health risks that need to be considered when assessing the full impact of energy production and distribution. Number of workers and quantitative Although too often ignored in discussions of environmental health effects estimates health risks, the burden of occupational disease and injury is It is difficult to estimate the number of workers involved in meeting substantial on a global scale. It is conservatively estimated that with the energy requirements of communities. As noted, in poor well over 1 million deaths a year, nearly 3 percent of the global communities much of this work is carried out by family members, burden of ill health is directly attributable to occupational conditions particularly women, who are not formally employed. In addition, (Leigh and others, 1996). This is substantial, accounting for more much of this work is carried out by small industries that are not than motor vehicles, malaria, or HIV and about equal to tuberculosis always recorded in national employment statistics. or stroke. Although the fraction due directly to supplying energy As noted, an estimated 16 million people are involved in forestry, is unclear, energy systems employ many millions of people world- most of them in developing countries. In industrialised countries wide in jobs substantially riskier than average—particularly in jobs

with reliable statistics, the occupational mortality rate for agricultural producing solid fuels. workers is 5–10 times the average for all workers (Kjellstrom, 1994). Because of the additional risks in forestry, mortality rates for Community scale these workers are possibly twice as high again, or 32,000–160,000 Energy systems are associated with a vast array of insults and at a global level. Not all of this activity is directly related to fuel impacts (see table 3.1). Many of these are expressed at the demand, however. community scale, including problems associated with coal and uranium As noted, miners are a large occupational group in international mining, petroleum and gas extraction, water use and contamination statistics (UN Demographic Yearbooks). They represent up to 2 percent by power plants, thermal pollution, and noise from wind farms. of the economically active population in certain developing countries. Here we can only focus on the largest of these impacts world-wide. Mining is an extremely dangerous occupation. Recent data show Urban air pollution is the chief environmental impact of energy that occupational mortality rates for miners are up to 20 times the systems at the community level. Although there are industrial and average for all occupations (ILO, 1998). The range of mortality other sources of some pollutants, the vast bulk—whether measured rates may be as wide as that for forestry (2–10 per 1,000 workers by mass or by hazard—is generated by fuel combustion or, as in the per year). In most countries the economically active population is case of photochemical smog, is created in the urban atmosphere by

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About 1 percent of the global precursor chemicals largely released workforce is engaged in mining, (National Research Council, 1998).⁵ in the course of fuel use. From the but these workers account for In the late 1990s the mean annual 1930s to the 1950s a number of urban 8 percent of the 15,000 fatal cocentration of PM10 in North American, air pollution episodes in the industri- occupational accidents Western European, and Japanese cities ranged alised world brought air pollution to the from 30–45 micrograms per cubic metre attention of the public. The first major each year. (figure 3.5). (The U.S. standard is 50 micrograms improvements came by banning the burning of per cubic metre.) In the 1960s particulate levels refuse and coal within city limits. By the early 1970s the were probably two to four times higher. (Small particles infamous London smogs (and their parallels in other cities), caused were not measured routinely until the mid-1980s, so previous by coal combustion, were memories. Two other community-level levels have to be inferred from measurements of total particles.)

impacts are also discussed in this section: those due to large hydro- Still, industrialised countries face a number of energy-related air electric dams and to nuclear power. pollution challenges. Nitrogen dioxide and ozone levels exceed standards in many cities, particularly in sunny cities with large auto Fuel-derived air pollution fleets such as Los Angeles (California) and Athens (Greece). The recent in cities of industrialised countries evidence suggesting that small particles (less than 2.5 micrograms)

During the past 25 years the cities of the industrialised world have may be even better indicators of ill health than PM10 has led the generally brought energy-derived urban air pollutants under even United States to propose new regulations aimed at PM2.5, potentially greater control. In the United States, for example, emissions per unit putting a number of cities out of compliance. European countries of useful energy from power plants and automobiles—the two are also considering such regulations. Since long-term data are not widely largest urban energy polluters—have fallen 65 percent and 50 percent available, it is not clear how much PM2.5 levels have decreased in in health hazard (weighted by the relative standards for each pollutant).⁴ recent decades, partly because such particles are transported over Japan and Western Europe have achieved similar results. much larger areas than larger particles.

In the power sector these achievements have mostly come about This focus on even smaller particles has brought diesel exhaust by relying more on nuclear power and natural gas and by requiring smokestack controls for particles and nitrogen and sulphur oxides particles under more scrutiny. Unlike gasoline, diesel produces a at coal-fired power plants. In addition, thermal power plants have significant amount of emissions of particles that are not only smaller become more efficient, and more improvements are expected, but may have chemical properties that make them more dangerous.

particularly for those using gas (see chapter 8). For vehicles, the This feature raises questions about the future of diesel-fuelled vehicles, reductions have come from a mix of improvements in engine even though such vehicles can be slightly more fuel-efficient and combustion, increases in fuel efficiency (in North America), and the cost-effective than gasoline-fuelled vehicles. The tendency for many nearly universal requirement of catalytic converters (devices to help countries to keep diesel prices low relative to gasoline—as a means control pollutant emissions). Thus, despite significant increases in of assisting farming, fishing, and other industries—can artificially power production and vehicle use since 1975, overall emissions of promote diesel passenger vehicles. (See the section on cross-scale most pollutants are now lower. impacts, below, for a discussion of the economic implications of As a result of these emission reductions, urban air quality has diesel particle health effects.)

generally improved throughout the industrialised world. Although Since the 1980s studies have seemed to indicate that there is no fuel combustion produces a number of health-damaging pollutants, threshold for the health effects of particle pollution. In other words, as explained above, small particles are probably the best single there no longer seems to be an identifiable level that can be termed indicator. Suspended small particles are a mix of primary combustion safe. All that can be said is that the effect is lower at lower levels, particles—carbonaceous materials and associated trace elements— but does not seem to disappear even at the lowest (background) and secondary conversion products such as sulphate and nitrate concentrations. Indeed, in the late 1990s European and global aerosols. In many parts of the world, windblown and urban dust can offices of the World Health Organization revised their particle guidelines also be significant contributors to suspended particles. to reflect the absence of thresholds (figure 3.6).

Small particles are deposited deep in the lungs, where their Because it is rarely (if ever) practical to set a standard of zero for clearance is slow and their ability to cause damage is enhanced. pollutants with significant natural sources, standard setting is much Small particles also carry adsorbed trace metals and carcinogenic harder for pollutants with no threshold for significant effects. hydrocarbons into the lungs, intensifying the potential for health Policy-makers must determine that the benefits of fuel combustion damage. Assessments of the human health effects of air pollutants outweigh the extra mortality produced by the resulting pollution— increasingly focus on these small particles. Still, there are few for example, that the 5 percent increase in mortality ‘allowed’ by a measurements of these particles in most cities, although more cities PM10 standard of 50 micrograms per cubic metre above background are measuring PM10 (particles less than 10 micrograms), which (see figure 3.6) is acceptable given the societal advantages of fuel is considered a better indicator than simple total particulate levels use. This is a difficult determination, and much more politically

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FIGURE 3.5. GLOBAL DISTRIBUTION OF URBAN PM 10 CONCENTRATIONS

Cumulative percentage of urban population

0% 20% 40% 60% 80% 100%

Sweden

Norway ■■ Urban population

Belarus ■■ Annual concentration

Iceland

France

Switzerland
Slovak Rep.
New Zealand
Canada
Australia
Netherlands
Ireland
United States
Germany
United Kingdom
Finland
Croatia
Venezuela
Czech Rep.
Greece
Singapore
Denmark
South Africa
Turkey
Austria
Kenya
Belgium
Argentina
Spain
Egypt
Japan
Portugal
Hungary
Italy
Malaysia
Russian Federation
Latvia
Hong Kong (China)
Korea, Rep. of
Ecuador
Romania
Brazil
Nicaragua
Colombia
Costa Rica
Ukraine

Panama
 Lithuania
 Mexico
 Ghana
 Honduras
 Guatemala
 Chile
 Poland, Rep.
 Nepal
 Philippines
 Thailand
 Bulgaria
 Iran, Islamic Rep.
 China
 India
 Indonesia
 Nigeria
 Pakistan

0 100 200 300

Population-weighted PM10 concentration (micrograms per cubic metre)

Note: In many cases, PM10 levels have been entirely estimated from measurements of total particles. Source: WRI, 1998; WHO, 1998b.

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set. The likely result will be continuous pressure to tighten particle

FIGURE 3.6. PROVISIONAL GLOBAL

AIR QUALITY GUIDELINES FOR PARTICLES standards, with stronger incentives for lower particle emissions

from vehicles, power plants, and other fuel-using sources. Indeed,

Change in PM10 (micrograms per cubic meters) as discussed in later chapters, emission reductions are the driving

0 50 100 150 200 forces for new power and transport technologies.

0 As recently as the early 1990s, the main source of lead emissions throughout the world was tetra-ethyl-lead used as an additive to raise octane in gasoline. But nearly every country now has a plan to Percent mortality increase

5 remove lead from gasoline (box 3.4). Still, significant numbers of children in many industrialised and developing countries have blood lead levels above those thought to affect cognitive development 10

(intelligence). These levels will decline as lead is removed from the

rest of the world's gasoline, although industrial and other sources must be controlled as well.

15

Fuel-derived air pollution
in cities of developing countries

Developing country cities have much higher mean pollutant
20 concentrations than industrialised country cities (see figure 3.5).

Change in daily mortality
with change in daily PM10 level In cities in China and India, averages seem to be nearly 200
micrograms per cubic metre of PM10, though there is much
Source: WHO, 1999. variation by season and city. Such concentrations must be causing
significant premature deaths—perhaps 15 percent or more above
difficult than endorsing a standard that has some scientific validity background levels. Indeed,
estimates for premature mortality from
of being below a 'no effects' level, which is how most standards are urban air pollution range from
170,000–290,000 a year in China

BOX 3.4. GETTING THE LEAD OUT: A SLOW SUCCESS STORY

"The current consensus is that no amount of of leaded gasoline fell to a low 0.15 grams a the
Middle East, are dominated by powerful

Pb [lead] in the environment can be considered litre. In 1991 unleaded petrol was introduced oil
companies and state-owned refineries.

safe" (Schwela and Zali, 1999). Although this and taxes were changed to make it cheaper Although
in 1994 two-thirds of global

was not the first reason to remove lead from than leaded fuel. By the end of 1997 unleaded
gasoline sales were unleaded, additional

gasoline, it soon became the driving force. fuel accounted for about 75 percent of gasoline efforts
are needed. Several mechanisms can

The introduction of catalytic converters sales. In addition, more stringent exhaust encourage the
reduction of lead in gasoline.

spawned the need for unleaded gasoline to emission standards were implemented for The most
promising is to set fuel taxes so that

protect the devices. Shortly after, concerns gasoline-fuelled vehicles, promoting an unleaded
unleaded gasoline is cheaper than leaded

about the health effects of lead emissions led fuel market. Finally, oil companies agreed to fuel.
Fuel filler inlets should be required in

to an increase in sales of unleaded gas and a phase out leaded gasoline by July 1998. automobiles
to allow only the narrower nozzles

reduction in the lead content of leaded fuel. Mexico has also taken steps to reduce the of unleaded
fuel pumps to be used. Requiring

Since leaded gasoline has been responsible lead content of fuel, though it still has far to catalytic
converters in vehicles would further

for about 90 percent of lead emissions, it was go. Since 1994 the lead content of leaded fuel
decrease the use of leaded fuel. Emphasising

the most logical target for reduction (Lovei, 1998). has been cut to 0.15 grams a litre. But it other benefits of using unleaded gasoline, such

Many nations have taken action to phase appears that the Mexican National Petroleum as lower exhaust emissions of hydrocarbons,

out lead from fuel. Canada, Japan, and the Co. has recently raised lead levels. No also promotes the reduction of lead in

United States have completely phased out government agency has the authority to gasoline (Schwela and Zali, 1999). Finally, as

leaded gasoline (Lovei, 1998; Schwela and ensure fuel quality, making enforcement of shown in Brazil, it is possible to substitute

Zali, 1999). Los Angeles (California) saw a 98 low lead levels a challenge. Unleaded gasoline 5–10 percent ethanol for lead as an octane

percent reduction in the lead exposure of accounts for 46 percent of sales in Mexico booster, thereby promoting renewable fuels.

commuters between 1979 and 1989. In Western City and 84 percent in Monterrey (which is An enduring urban myth exists that some

Europe leaded gasoline has a very low lead wealthier and closer to the U.S. border). older cars need lead to operate well. As long

content, and unleaded fuel has a large market Leaded fuel is still cheaper, however. Mexico as the fuel has the correct octane level, no

share in most countries. In addition, a few is implementing new standards requiring engine needs lead. Indeed, in many cases

developing nations have lowered or even catalytic converters and so unleaded gas. the removal of lead will have direct benefits

banned lead in gasoline (Lovei, 1998). But large problems remain in many developing in the form of less engine maintenance. The

Over the past 20 years Singapore has countries. The biggest lead problems are in persistence of this myth slows the introduction

taken significant steps to phase out lead in Africa and in petroleum-exporting nations. These of low-lead fuels despite technical evidence

fuel. Between 1980 and 1987 the lead content countries, including Venezuela and those in to the contrary.

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Dams affect Earth

(World Bank, 1997; Florig, 1997) at scales rivalling other major projected increase in population, and 90,000–200,000 in India human activities, such as urbanisation, economic activity, and

(Murray and Lopez, 1996; Saksena urbanisation and energy use, it will be necessary to

and Dayal, 1997). develop and deploy new, even cleaner and

The causes of air pollution in developing road building. more efficient energy-using technologies. A country cities are much more varied than in number of advanced power plant technologies industrialised countries. Although automobile own- potentially offer such performance (see chapter 8).

ership rates are much lower, there tend to be many other In addition, some near-commercial vehicle technologies

types of vehicles as well, including two- and three-wheelers using may allow vehicle densities in developing country cities to grow for

highly polluting two-stroke engines. There also tend to be larger several decades and still meet air quality goals (box 3.5). Strong

portions of light-duty and heavy-duty vehicles using diesel rather pollution controls will be needed to bring these technologies into

than gasoline. In addition to power plants and large industries wide use, however.

with limited pollution controls, developing country cities tend to In addition to technical changes in vehicles of all types (not just

have large numbers of small boilers, engines, and open fires in private cars), a range of other improvements will be needed if the

the commercial and light-industry sectors, as well as in informal world's cities are to accommodate the greater demand for transport

sectors such as street food. These enterprises tend to rely on the that increases in population and income will bring. These include

most inexpensive and thus dirty fuels in inefficient applications improvements that result in significant and sustained enhancement

without pollution controls—and so have high emissions per unit of in the attractiveness of public transport, land-use planning to

useful energy. reduce the need for intraurban trips, and implementation of policy

Furthermore, such cities often do not have adequate refuse and tools such as time-of-day, congestion, and central-zone pricing. In

garbage collection, leading to street-side trash burning, a highly addition, significant switches to public transport might occur

polluting activity. Even when collected, trash often burns uncontrollably through such means as including the cost of vehicle insurance in the

in landfills in or near cities, wasting potential energy and producing price of fuel and taxing employer-provided parking as income (see

clouds of noxious emissions. Another major non-energy source of chapter 11).

particle pollution in many cities is dust from construction sites and

unmanaged empty land. Finally, unlike in industrialised countries, a Hydroelectric dams 6

large fraction of urban households in developing countries still use Dams, large and small, have brought tremendous benefits to many

solid fuels for cooking and space heating in inefficient stoves with regions, including important contributions to development in

no pollution controls (see figure 3.1). Although individually industrialised countries. It is important not to deny these benefits

small, their large number means that these stoves can contribute to developing countries. But such dams need to be designed and

significantly to urban pollution. constructed with care. Although dams frequently serve many

In addition to dealing with trash, dust, and other non-energy purposes—including flood control, irrigation, navigation, and

issues, the most pressing need for pollution control in developing recreation—major dams (those over 150 metres high, with 25 country cities is to reduce and eventually eliminate small- and cubic kilometres of storage, or 1,000 megawatts of electricity) tend medium-scale combustion of dirty fuels. For stationary sources, this to have hydropower as one of the their main objectives. Such dams means shifting away from solid fuels (coal and biomass) and often have big impacts on the environment. There are more than high-sulphur fuels of any kind. For mobile sources, it means 300 major dams world-wide, and nearly all have hydropower as a dealing soon with, in order of priority, two-stroke engines, poorly major component of their function. The environmental impact per operating diesel engines, and gasoline-fuelled vehicles without unit of electricity production, however, can often be smaller for catalytic converters. In addition, as is happening in Bangkok large than for small dams. The type rather than the size can be the (Thailand) and New Delhi (India), there is great advantage in most important factor (Gleick, 1992). shifting vehicle fleets (taxis, buses, government vehicles) to clean- With a total capacity of about 640,000 megawatts of electricity, burning gaseous fuels such as compressed natural gas or liquefied hydropower provides about one-fifth of the world's electricity petroleum gas (Mage and Zali, 1992). (Gleick, 1992). In Central and South America hydropower provides about 60 percent of electricity; in Asia this figure is about 15 per- Urban pollution control in the longer run cent. Itaipu, on the border of Brazil and Paraguay, is the most pow- Because the best commercial technology in terms of energy erful hydropower dam built to date, with a capacity of 12,600 efficiency and emissions has not been deployed completely in megawatts of electricity. It cost \$20 billion to build. When finished, industrialised countries and has been used little in developing countries, China's Three Gorges Dam will produce about 18,200 megawatts of much improvement is possible in the next 20 years without switching electricity and may cost as much as \$75 billion (The Economist, to advanced technologies. In the longer term, however, if air pollution 1999). Thus hydroelectric dams are the most expensive energy levels are to be brought down to and kept at low levels given the projects in the world.

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No major river in the world is without existing or planned hydro- other major human activities, such as urbanisation and road building.

electric dams. Nearly four-fifths of the discharge of the largest rivers Direct human impacts. During the 20th century 30–60 million

in Canada, Europe, the former Soviet Union, and the United States people were flooded off their lands by dams (Dunn, 1998). The are strongly or moderately affected by flow regulation, diversions, World Bank, using Chinese government figures, estimates that 10.2 and the fragmentation of river channels by dams (Dynesius and million people were displaced by reservoirs in China between 1950 Nilsson, 1994). More than 500,000 square kilometres—the area of and 1989 (World Bank, 1993). Given that a number of major dams Spain—have been inundated by dam reservoirs world-wide, though are under construction or planned in developing countries, there not all for hydropower (Collier, Webb, and Schmidt, 1996). will be no slackening in the pace of population displacement. (Indeed, many hydropower plants have no reservoirs.) Globally, China's Three Gorges Dam, for example, is expected to displace about 200 cubic kilometres of water a year—about 7 percent of the more than 1 million people, and the proposed Pa Mong Dam freshwater consumed by human activities—are evaporated from the between Lao PDR and Thailand is expected to displace more than surface of reservoirs due to their increased exposed surface area 500,000 (Gleick, 1998). (Shiklomanov, 1998). Thus dams affect Earth at scales rivalling Large population resettlements can have a number of direct

BOX 3.5. ALTERNATIVE VEHICLES

With growing energy and environmental more efficient, and less polluting, and generates into water and electricity without using com- concerns surrounding today's conventional only the power required for most operations. bustion. The hydrogen gas can come from a vehicles, a great deal of research is going Hybrid electric vehicles have several advantages number of sources, including multiple forms into alternative vehicles. Four main types of over conventional vehicles and fewer disavan- of pure hydrogen and a variety of hydrocarbon alternative vehicles have the potential to reduce tages than electric vehicles. Hybrid electric fuels. Fuel-cell vehicles have many advantages the environmental and efficiency deficits vehicles have higher fuel economy and lower over conventional vehicles. Fuel cells have of conventional vehicles and to become emissions than vehicles with internal combustion a much greater engine efficiency and fuel commercially available in the near future. engines, and better range and more rapid economy, drastically reduce pollution emissions Electric vehicles are powered by rechargeable refuelling than electric vehicles. Hybrid electric (including greenhouse gas emissions), and batteries and have no internal combustion vehicles also reduce petroleum consumption can use a wide variety of fuels, promoting engines. The battery, which can be made of and increase energy diversity by using alternative energy diversity.

lead-acid, nickel-metal hydride, and lithium-ion engines, which can use a range of fuels. But In addition, they are quieter and smoother in operation, can be recharged at home or, in the hybrid electric vehicles are still expensive and have tested at high performance levels, have long driving ranges, and have future, at recharging stations. Electric vehicles not yet fully developed. Programs are in place have several environmental benefits relative to develop and improve hybrid electric vehicles, about the same operating costs as conventional to conventional vehicles, including no tailpipe and several automobile manufacturers are or automobiles. Still, there are several draw-emissions and lower hydrocarbon, carbon will soon be marketing models. backs to fuel-cell vehicles, including the monoxide, and nitrogen oxide emissions Compressed natural gas vehicles are lack of infrastructure to distribute hydrogen (including emissions from the production of powered by an abundant, inexpensive fuel or another fuel (unless gasoline is used), electricity). Other advantages include lower composed largely of methane. Natural gas difficult storage of pure hydrogen, and maintenance costs and the elimination of is a clean-burning fuel with lower carbon possible safety concerns. Major automobile the need for complicated tailpipe emission controls. dioxide, carbon monoxide, hydrocarbon, and companies are planning to have commercially But electric vehicles also have several disadvantages, such as the environmental concerns partly due to the lower carbon content per currently demonstrating prototypes and of an increase in electricity use, increasing unit of energy in natural gas relative to other improving on them. Large cost reductions need emissions of sulphur oxides, and possible fossil fuels. In addition to its environmental to occur, however, and fuel infrastructure contamination from the disposal and recycling benefits, natural gas vehicles are cheaper to of batteries. There are also disadvantages in maintain, requiring service less frequently than issues must be resolved before fuel-cell terms of convenience and cost, such as conventional vehicles as well as having a vehicles are ready for the marketplace. lengthy recharging and lack of infrastructure lower cost of refuelling. Converting vehicle Of these four alternatives to conventional for recharging stations, short driving ranges fleets such as taxis, three-wheelers, and vehicles, electric vehicles have the fewest (though electric vehicles are good for local buses to natural gas is an important interim barriers to market entry. But they probably trips and commuting for two-car households), way to improve air quality in developing have the least consumer appeal in terms of

an inability to maintain high speeds, and high country cities. Conversion costs are relatively environmental improvements and convenience.

battery costs. Today electric vehicles cost small, although baggage space is reduced Fuel-cell vehicles will probably be found to

about \$30,000, which is too expensive for because of the need to add pressurised be the most environmentally friendly, but they

most markets. tanks. It is hard to use compressed natural are the furthest from commercial development.

Hybrid electric vehicles combine the battery gas for private vehicles because of the need Hybrid electric vehicles also offer a good

and electric motor of an electric vehicle with to create many fuelling stations. Urban vehicle option in the near future, with convenience

the internal combustion engine and fuel tank fleets, on the other hand, can operate with and environmental benefits. All these cars

of a conventional vehicle, to obtain the benefits relatively few centralised fuelling stations. will likely begin to enter the market in the

from both technologies. The engine, which Fuel-cell vehicles operate by combining next 5–10 years, and infrastructure will have

is much smaller than that in a conventional vehicle, hydrogen and oxygen gases into an electro- to be built to accommodate all of them as

operates at a constant power load and so is chemical device, a cell, that converts them well as today's automobiles.

Source: American Honda Motor Company, 1999; California Energy Commission, 1998; California Environmental Protection Agency, 1999; Ford Motor Company, 1998, 2000; General Motors Corporation, 1999; Global Toyota, 1999; Gould and Golob, 1997; Hanisch, 1999; Krebs, 1999; Kremer, 2000; Mark, Ohi, and Hudson, 1994; Matsumoto, Inaba, and Yanagisawa, 1997; Mendler, 1997; National Fuel Cell Research Center, 1999; Natural Gas Vehicle Coalition. 2000a, b; Neil, 1999; Steinbugler and Williams, 1998; USDOE, 1995; USEPA, 1994, 1998b.

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TABLE 3.2. ECOLOGICAL INSULTS AND IMPACTS OF LARGE DAMS

Insult caused by dam	Impacts seen	Severity of impact	Example of impact
Changes in the Deterioration of downstream	Depends on the sensitivity	Enhanced algae growth in	
the reservoir consumes the			
chemical properties ecosystem caused by inability of the affected ecosystem	oxygen in the epilimnion and, as it decays, the mass		
of release water to process the increased dis-	(tropical ecosystems are sinks to the already		
oxygen-deficient hypolimnion, where			
solved minerals especially sensitive) decay processes reduce the oxygen concentration even			
further, resulting in acid conditions at lower levels and			
the dissolution of minerals from the reservoir bed.			
Changes in the Thermal pollution often results	Diversity, biomass, distri-	Productivity levels in the	
surface waters of new reservoirs			

thermal properties in species diversity reduction, bution, and density of fish often increase before long-term declines occur (Horne, of release water species extinction, and produc- stocks can be affected, 1994). China's Three Gorges Dam may be the final critical tivity changes in the reservoir disrupting breeding cycles factor for driving to extinction the Yangtze River dolphin.

Changes in the flow Erosion increases downstream Erosion of natural Changes in the downstream river morphology and rate and timing of of dam. Settling of sediments riverbeds can disturb the ecosystem productivity. release water in the reservoir causes high nurseries and spawning sediment loads to be picked of many aquatic organ- up in the area immediately isms, disturbing their below the dam breeding cycles

Changes in the High trap efficiencies of Effects often noticed Before the Aswan High Dam was constructed, the Nile sediment load of dams prevent the natural most severely in high- carried about 124 million tonnes of sediment to the sea the river processes of sediments productivity areas each year, depositing nearly 10 million tonnes on the and associated nutrients downstream from the floodplain and the delta. Today 98 percent of the sediment refreshing downstream soils dam that no longer remains behind the dam, resulting in a drop in soil receive annual fertilisation productivity and depth, among other serious changes to Egypt's floodplain agriculture (Pottinger, 1997).

Changes in Increased likelihood of lower Reduced access to Within nine years of the closure opening of the Hoover the dynamics water tables, which can create potable water is a Dam, 110 million cubic metres of material had been of downstream problems in areas near the huge problem in many washed away from the first 145 kilometres of riverbed riverbeds dam where groundwater is developing countries below the dam (McCully, 1996). a major source

Changes in The loss of sediment in the Financially expensive for Over the past 80 years dams have reduced by four-fifths the coastal area rivers flowing through deltas many areas where there is the sediment reaching the coasts of southern California. morphology and into the sea often results a large population living near This has reduced the beach cover at the base of cliffs in a gradual process of delta the coastal zone. along these shorelines, causing cliffs to collapse and coastal degradation (Jenkins and others, 1988).

social and health impacts. The social and cultural stress, loss of Ecosystem impacts. An internal survey of World Bank hydroelectric

income, disruption of traditional support services, and other problems dam projects found that 58 percent were planned and built without facing displaced populations often lead to lowered health status. any consideration of downstream impacts—even when these Even when efforts are made to resettle people in new areas, it is impacts could be predicted to cause coastal erosion, pollution, and difficult to locate land of similar productivity because other groups other problems (Dixon, 1989). The main ecological insults and are likely to already occupy the best areas. Some 13,500 people impacts of large dams (not just those producing hydropower) are have been swept to their deaths by the 200 or so dams (outside summarised in table 3.2. China) that have collapsed or been overtopped in the 20th century. Dams and greenhouse gases. The work assessing the impacts of In 1975 in Henan, China, about 230,000 people died from a series dams on greenhouse gas emissions is incomplete, but some estimates of dam bursts (Gleick, 1998). have been made. The most immediate changes are in the carbon Disease can spread from vectors that thrive in secondary dam flow between the flooded vegetation and the atmosphere. The systems, such as irrigation canals and even dam reservoirs. decomposition of plants and soils causes the gradual release of their Mosquitoes carrying malaria, for example, have thrived in conditions stored carbon (Rudd and others, 1993). created by dams. The parasitic disease schistosomiasis has also From a greenhouse gas standpoint, it might be thought that become more prevalent through the creation of habitats for snails vegetation decaying in a reservoir would be no worse than the same that act as the disease vector. Nearby populations, for example, amount of deforestation. Because of the low-oxygen conditions near suffered nearly universal infection after several large African dams and in the bottoms of many reservoirs, however, relative to deforestation were filled, including Aswan (Egypt), Akosombo (Ghana), and Sennar a larger fraction of the biomass carbon is likely to be released as (Sudan) (Nash, 1993). methane rather than as carbon dioxide. Since methane is a much

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more powerful greenhouse gas than carbon dioxide, the global than those from new natural gas power systems—with the impor- warming impacts are greater than the same amount of carbon tant exception of carbon dioxide, which is not produced by nuclear released as carbon dioxide. power. If public concerns about reactor safety, proliferation, and The peak greenhouse gas emissions, however, are unlikely to waste disposal can be satisfied, nuclear power may be able to play rival those of a similarly sized fossil power plant, emissions from a significant role in de-carbonising the world energy system in the

which would not decrease with age like those from a reservoir. In next 50 years (see chapter 8). addition, it is difficult to determine the baseline in tropical forests—that is, how much methane and other non-carbon dioxide greenhouse gases are released in natural conditions. In colder climates Nested between local-scale issues—such as the health effects of reservoirs apparently emit greenhouse gases at much lower rates urban pollution—and global-scale issues—such as climate change—are a number of regional-scale problems that affect human health and ecosystems over areas the size of countries and Nuclear power continents. The most important regional-scale issues are acid There are two main environmental concerns about nuclear power, deposition, tropospheric ozone, and suspended fine particles. both mostly with regard to its potential impacts on human health. Matched with the regional spatial scale is a temporal scale that One involves the highly radioactive products produced by nuclear requires air pollutants to remain aloft for periods ranging from fission inside power reactors. Such products require careful days to weeks and thereby be transported by prevailing winds and management at the reactor and during and after disposal. The other transformed by chemical reactions. Gases and fine particles meet concern revolves around the weapons-grade plutonium or uranium this criterion; larger particles (greater than 1 micron or so in diameter) that might be clandestinely derived from the nuclear power fuel tend to settle out quickly and are considered contributors to local, cycle to make bombs or other weapons of mass destruction by rather than regional, impacts. Fine particles may be solid (such as nations or subnational groups (see chapter 8). elemental ‘black’ carbon) or liquid (such as aerosol droplets). The routine (non-accidental) emissions of pollutants from the Contributing to regional pollution are a number of precursor harvesting, processing, and conversion of nuclear fuels are not negligible species, most of which are generated by the use of fossil fuels and negligible. And more than many technologies, they are vulnerable to biofuels. Prominent among them are sulphur dioxide (SO₂) and being enhanced by mismanagement. Still, the impacts of these emissions—nitrogen oxides (NO_x). Sulphur dioxide is released during the sions are generally substantially less than those involved with combustion of the sulphur found naturally in fossil fuels, while producing power with current coal technologies, the chief competitor nitrogen oxides originate either as fuel nitrogen or as atmospheric in many areas. Although involving different pollutants, routine emissions—nitrogen oxidised during combustion. Other species of importance are sions from nuclear power systems are probably no more dangerous particulate matter (PM), carbon monoxide (CO), methane (CH₄), and

TABLE 3.3. ANTHROPOGENIC EMISSIONS OF IMPORTANT SPECIES BY REGION, 1990 (MILLIONS OF TONNES)

	Sulphur dioxide	Nitrogen oxides	Carbon monoxide	Non-methane volatile organic compounds	Methane
Region (as sulphur)	(as nitrogen)	monoxide	organic compounds	Energy- Non- related	Energy- Non- related
Western Europe	8.8	2.5	3.6	0.4	45
Eastern Europe and former Soviet Union	13.5	3.4	3.5	0.6	47
North America	11.6	0.7	7.6	0.3	82
Asia	17.9	3.0	5.6	1.9	165
Rest of world	8.8	4.1	3.2	4.4	105
Total	60.6	13.6	23.5	7.6	444
	74.2	31.1	975	177.5	320.4

Note: These numbers are slightly different from those in table 3.1 because of different assumptions and methods. Energy-related sources include the combustion, extraction, processing, and distribution of fossil fuels and biofuels. Non-energy-related sources include industrial processes, deforestation, savannah burning, agricultural waste burning, and uncontrolled waste burning. Source: Olivier and others, 1996.

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Energy activities account

non-methane volatile organic compounds (NMVOC), released during emissions of sulphur dioxide (see above, in the section on green-incomplete combustion and other and 76 percent for house gases) on a 20-year time horizon activities. Ammonia (NH₃) is a significant of about 4.5 due to its influence on the regional pollutant, but fuel combustion is nitrogen oxides. atmospheric lifetime of methane (IPCC, not its primary source. 1990). In addition, carbon monoxide is toxic to

When emissions of these primary species are humans and is a critical component of many photo-released into the atmosphere, they form a complex, reactive chemical reactions in the atmosphere. It is a scavenger of

‘soup,’ the chemical and physical behaviour of which is determined hydroxyl radicals and so influences the production of ozone. There

by such factors as temperature, humidity, and sunlight. The primary are many relatively easy ways to reduce carbon monoxide emis-

species are transported and deposited, influencing the health of sions—catalytic converters for automobiles, improved household

humans and of natural ecosystems. But these primary species are stoves, and reuse of carbon monoxide gas in industry.

also transformed into secondary species—such as sulphate, nitrate, Non-methane volatile organic compounds consist of a variety

acids, ozone, and miscellaneous organic compounds—that can have of chemical species. In China, for example, the mix of organic

effects even more damaging than their precursors and in areas far compounds is 46 percent paraffins, 32 percent olefins, 21 percent

removed from the primary sources. This can lead to transboundary aromatics, and 1 percent aldehydes (Piccot, Watson, and Jones, 1992).

problems, where a country or region has little control over the emis- These compounds are important in the chemistry of the atmosphere

sions that damage its environment. because of their influence on the formation and destruction of ozone and methane. Non-methane volatile organic compounds are

Emissions and energy a product of the incomplete combustion of fossil fuels, biofuels, and

A snapshot of global and regional anthropogenic (human-caused) other carbonaceous materials.

They are also emitted during the

emissions in 1990 is provided in table 3.3. The emissions are extraction, processing, and handling of gaseous and liquid fossil

partitioned into those derived from energy-related activities (including fuels. And they are released through the evaporation of miscellaneous

combustion, extraction, processing, and distribution of fossil fuels organic products in industry and households.

and biofuels) and those derived from non-energy activities (which have Ammonia is a significant component of regional emissions. Being

a wide variety of sources, including industrial processes, deforestation, an alkaline substance, it neutralises acids in the atmosphere. But

savannah burning, agricultural waste burning, and uncontrolled once it is deposited on land, it can be converted to acid through

waste burning). Non-anthropogenic sources (volcanoes, soils) are biochemical processes in the soil. Ammonia emissions are largely

not included. derived from animal waste, fertiliser application, and fuel combustion.

Energy activities account for 82 percent of anthropogenic emissions In 1990 energy-related activities accounted for just 5 percent of

of sulphur dioxide and 76 percent for nitrogen oxides. Energy activities global ammonia emissions—2.7 of 52.0 teragrams (Olivier and

play a less dominant role for the three other species—56 percent others, 1998). Most ammonia emissions are from Asia and other

for non-methane volatile organic compounds, 46 percent for carbon developing countries, due to the rural nature of these countries, the

monoxide, and 34 percent for methane. The smaller role of energy intensive use of fertiliser for food production, and the heavy use of

in emissions of these three species reflects the important contributions fossil fuels. In 1990 ammonia emissions in Asia were 22.5 teragrams,

of deforestation, savannah burning, and agricultural waste burning compared with 3.5 teragrams in Western Europe and 4.1 teragrams in the generation of products of incomplete combustion, coupled in North America. with rice cultivation and enteric fermentation in the case of methane. Nevertheless, table 3.3 demonstrates the critical contribution of Future emissions energy to emissions of regional-scale pollutants. It also highlights Sulphur dioxide. The latest energy projections indicate that global the importance of the developing world in current patterns of sulphur dioxide emissions will likely stay roughly constant between regional emissions. 1990 and 2020, at about 59 teragrams of sulphur (Nakićenović, Grübler, Sulphur dioxide and nitrogen oxides play a role in the formation and McDonald, 1998). This 'middle-course' scenario incorporates of acid deposition, because they can be transformed to acids in modest economic growth, continued reliance on fossil fuels, and the atmosphere. The transformation products are fine particles, the elimination of trade barriers. At the regional level, however, a solid or aerosol, in the form of sulphates and nitrates. In addition, distinctive pattern emerges for all the important species. Emissions will nitrogen oxides are a major precursor to the formation of regional decline in the industrialised regions of the Northern hemisphere— tropospheric ozone. Finally, sulphates and nitrates have the ability to Europe, the former Soviet Union, North America—and increase scatter and absorb radiation and so contribute to global and regional sharply the developing regions of the Southern hemisphere and the climate change, probably with a net cooling effect. Far East—Latin America, Africa, Asia (figure 3.7). Carbon monoxide is an important regional atmospheric pollutant In Western Europe strong national environmental policies, from several perspectives. It acts as an indirect greenhouse gas with changes in national energy policies, and implementation of the

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FIGURE 3.7. SULPHUR DIOXIDE EMISSIONS BY REGION, 1990 AND 2020 (PROJECTED)

■ 1990

■ 2020

■ China/2020

30

Teragrams of sulphur

20

10

0

Eastern NFC ACT BAT

Europe/Former Western North

Soviet Union Europe America Asia Rest of the world

Source: Nakićenović, Grübler, and McDonald, 1998; Foell and others, 1995.

1985 Helsinki Protocol and 1994 Oslo Protocol (under the 1979 Europe). A mid-range estimate is 20 teragrams of sulphur, to be achieved

Convention on Long-range Transboundary Air Pollution) have driven if all large, new facilities are fitted with flue-gas desulphurisation

down sulphur dioxide emissions. As a result the region could see a units and other fossil-fuel users switch to low-sulphur fuels.

60 percent drop in sulphur dioxide emissions between 1990 and China continues to be the largest contributor to Asia's sulphur

2020. Similarly, in North America the adoption by the United States dioxide emissions, emitting about half of the continental total. But

of the 1990 amendments to the Clean Air Act has reduced sulphur the establishment in 1997 of China's Two Control Zone policy for

dioxide emissions. North America's sulphur dioxide emissions in sulphur dioxide emissions has generated optimism that emissions

2020 are expected to be about 35 percent below 1990 levels. In will not grow as fast as once thought. Emissions of 11.9 teragrams

Central and Eastern Europe and the former Soviet Union, a 50 percent of sulphur in 1995 are used as a baseline, and the plan is to

reduction is anticipated. limit national emissions to 12.3 teragrams of sulphur in 2000 by

The problem of sulphur dioxide emissions has shifted to the capping emissions in certain provinces at their 1995 levels. While

developing world, with emissions in Latin America, Africa, and the implementation and enforcement questions linger, there is a

Middle East expected to increase by about 30 percent between 1990 commitment at the highest level in China to holding down sulphur

and 2020. The main problem region is Asia, where emissions are dioxide emissions. The official (but undocumented) estimate for

already high—with 17 teragrams of sulphur emissions in 1990— China's emissions in 2020 is 19.5 teragrams of sulphur.

and could double by 2020. If that happens, Asia will account for 58 The message from figure 3.7 is one of opportunity. With rapid

percent of global emissions, much of them from China. growth in Asia, many of the coal-fired plants projected to be needed

Three emission scenarios from the RAINS-ASIA model are also after 2000 have yet to be built. Thus the opportunity exists to fit

shown in figure 3.7 (Foell and others, 1995; Arndt and others, these plants with emission controls or lower-emission technology at

1997; Streets and others, 1999). Driven by a similar energy forecast, the time of construction. The incremental cost of emission reduction

the model projects that Asia's sulphur dioxide emissions in 2020 is then the only hurdle to be overcome—though it is a high hurdle

will be bounded by an upper value of 40 teragrams of sulphur (under (\$25 billion a year for the ACT scenario, rising to \$65 billion a year

the assumption of no further control policies beyond those in place for the BAT scenario). Substitution of natural gas for coal is an in 1990—the NFC scenario) and a lower value of 6 teragrams of sulphur attractive interim measure, and any possibilities for increasing the (with the assumption of very tight controls, similar to those in Western efficiency of energy use and moving towards renewable energy

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would reduce emissions of sulphur, nitrogen, and other species. The Acid deposition ecologically driven scenario (chapter 9), for example, would lower Acid deposition—or acid precipitation in its ‘wet’ form—is perhaps global 2020 emissions from 59 teragrams of sulphur to 34 teragrams. the most important regional-scale manifestation of energy generation Nitrogen oxides. The situation for nitrogen oxides emissions is through fuel combustion. Acid deposition occurs when sulphur even more challenging, because of the added emissions from dioxide and nitrogen oxides are oxidised in the atmosphere to—transportation. Though nitrogen emissions were not estimated by sulphuric acid (H_2SO_4) and nitric acid (HNO_3), respectively, and Nakićenović, Grübler, and McDonald (1998), other analyses suggest dissolved in rainwater. Clean rainwater is naturally acidic, with a pH a regional pattern similar to that of sulphur dioxide. An earlier of about 5.6. In the industrialised regions of Europe, North America, study, Energy for Tomorrow’s World (WEC, 1993), which was more and Asia, rainfall pH values of 4.0-6.0 are common—and values as optimistic about economic growth, forecast a 13 percent increase in low as 3.0 have been measured in isolated events. global emissions of nitrogen oxide between 1990 and 2020 (from Acid deposition is a problem because it causes damage to natural 24 teragrams of nitrogen to 27 teragrams) under case B assumptions. and human-made surfaces with which it comes into contact. If soils The increase in Asia was 70 percent (from 6.8 teragrams of nitrogen contain insufficient alkali to neutralise the acid, damage can be caused to 11.5 teragrams). Use of the RAINS-ASIA model, with its daunting to vegetation, particularly sensitive tree species and agricultural view of the growth of fossil-fuel-based energy systems in Asia, yields crops. Lakes can become acidified, leading to the demise of fish an estimated increase of more than 300 percent in this period (van populations. Over time the entire natural structure and function of Aardenne and others, 1999). ecosystems can change. Manufactured materials can be attacked: Carbon monoxide and non-methane volatile organic compounds. Though there are no published projections for emissions of carbon metal surfaces rust, and alkaline materials like concrete, limestone,

monoxide and non-methane volatile organic compounds, carbon and marble are eroded (box 3.6). monoxide emissions are unlikely to increase in Asia, because In Europe forest damage has long been attributed to acid deposition.

inefficient combustion of biofuels will fall and inefficient vehicles Despite emission reductions, the health of European forests still

will be replaced. On the other hand, emissions of non-methane seems to be deteriorating (UNECE, 1996). In a 1995 survey of 117,000

volatile organic compounds may grow rapidly as expanding trees in 30 countries, more than 25 percent showed signs of significant

industrial production calls for greatly increased solvent use, defoliation, and more than 10 percent showed significant leaf

increased vehicle use generates more hydrocarbons, and rising discoloration. Both direct and indirect effects of air pollution, of

living standards increase the demand for domestic and commercial which acid deposition is but one part, are considered the cause.

paints and solvents. Together, the expected rise in emissions of Surveys of forest soils show that, while sulphur deposition has

nitrogen oxides non-methane volatile organic compounds bodes dropped drastically since the 1970s, nitrogen deposition is still

ill for the formation of regional ozone in the developing world. high, impairing soil chemistry and nutrient status. For acidification

BOX 3.6. ENVIRONMENTAL IMPACTS OF ACID DEPOSITION

In general, the exposure-response relationships Although the impacts of acid rain on sensitive populations, such as children and

between acid deposition and impacts on terrestrial systems are known with less asthmatics, as in Hong Kong, China (Hong

ecosystems, materials, visibility, and human certainty, several aspects are likely outcomes Kong Municipal Government, 1999). Another

health are complex. Some are reasonably well of acid deposition. Effects on soil include potential human health problem comes from

understood, but others involve poorly known reducing the availability of nutrients and increased levels of toxic metals leached from

relationships involving climate, geography, enhancing the solubility of metals. But soil, especially aluminium, into drinking water

other chemicals, and time. Much research has nitrogen deposition into the soil can enhance in rural areas (Environment Canada, 1999).

been devoted to studies in North America and its nutrient content, and some soils are fairly Acid precipitation is also known to have

Western Europe, while relatively little has been resistant to damage. Acid deposition can negative non-ecological consequences. It

done in Asia—where most of the growth in cause damage to foliage of trees and crops, causes the erosion of materials and structures,

acid-depositing emissions is expected over however (Curren and Nixon, 1992). Forests, leading to aesthetic and functional damage as

the next few decades. especially those at high elevations, are also well as increased maintenance costs. This damage to structures includes those that have Acid deposition has harmful effects on affected by acid deposition directly through a great deal of historical significance and are many lakes and rivers, hurting aquatic life. In increased susceptibility to natural stresses considered highly valuable. Another impact of affected regions such as eastern Canada, and indirectly through a loss of nutrients acid deposition is haze, or a lessening of visibility, lakes have acid levels that are unsafe for fish obtained from soil (USEPA, 1999). Considerable largely an aesthetic problem (USEPA, 1999). and other aquatic life. While species of fish uncertainty relates to long-term impacts that The largest documented economic vary in their sensitivities to acidification, those may not yet have been observed (NAPAP, 1998). disruptions have been to fishery, forestry, with low tolerance decline in population, at Several human health problems are and agricultural industries. The damage times to the point of extinction. This not only linked to acid deposition. For example, many occurring to their products is causing a loss affects the species directly harmed, but loss respiratory diseases, including bronchitis and of productivity and jobs (Environment Canada, of species diversity damages the ecosystem emphysema, are likely caused or aggravated 1999). Furthermore, recreational use of as a whole due to the interdependence by sulphur particulates and nitrogen oxides. aquatic regions and forests has diminished, among species (Curren and Nixon, 1992). Respiratory problems are particularly noted in causing a loss in revenue (NAPAP, 1998).

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of surface waters, there appears to be an overall improvement (with was observed. It appears that neutralisation of acid inputs by deep higher pH, for example), probably as a result of reductions in acid soils and underlying bedrock may be averting ecosystem damage. deposition (UNECE, 1997). With projected reductions in sulphur Because of the heterogeneity of Chinese soils, however, local and nitrogen emissions through 2020, continued progress is acidification and damage may be occurring in sensitive areas that expected towards healthier ecosystems in Europe. have not been studied. A more recent survey of acidification in China North America has seen significant reductions in the sulphate by the Norwegian Institute for Water Research (Lydersen and others, concentration and pH of precipitation as a result of the 1990 1997) reported severe effects of acid deposition on soils, water

amendments to the Clean Air Act. Reductions in nitrate concentration bodies with high loadings showing typical signs of acidification, and have not been observed, however, because requirements for lower observed effects on surface water organisms.

nitrogen oxide emissions did not go into effect until 1996 (NAPAP, Zhao and Xiong (1988, p. 342) describe some of the severe 1998). On the whole, it is too early to tell if there has been significant materials damage observed in Chongqing: "Metal structures are cant improvement in the health of ecosystems. There is evidence of scraped of rust and painted once every 1–3 years. Shells of buses recovery in some New England lakes, but the U.S. Environmental are generally replaced every 1–2 years. Structural components Protection Agency has reported that additional reductions in sulphur made of stainless steel become rusty after a few years. Some con- and nitrogen deposition will be needed to fully restore the health of crete works built in the 1950s have corroded in such a manner that sensitive Adirondack lakes (USEPA, 1999). High-elevation spruce fir the gravel is exposed. It is estimated that the corrosion depth reaches forests in the eastern United States continue to show signs of damage. 0.5 cm in less than 30 years".

But, as in Europe, there is reason to hope for improvement. In northern China, by contrast, rainwater pH values are typically Asia is the region of greatest concern. Acid deposition is being 6.0–7.0. Although emissions are high in many parts of northern reported throughout Asia (Wang and Wang, 1995), with many areas China, meteorological conditions are more conducive to pollutant receiving levels that exceed the carrying capacity of their soils. Long- dispersion, and windblown dust from central Asian deserts tends to range transport is receiving scientific and political attention as neutralise the acidity. The line delineating acid precipitation in countries receive increasing pollution from neighbouring and even China (pH of 5.6) extends just west of Beijing and along the eastern distant countries (Huang and others, 1995; Ichikawa and Fujita, edge of the Greater Khingan mountain range. Since 1982 the area 1995; Streets and others, 1999). By far the worst episodes of acid receiving acid deposition may have expanded by 600,000–700,000 deposition occur in southwestern China (Zhao and Xiong, 1988). square kilometres (Wang and Wang, 1995).

Average rainwater pH values of 4.0–5.0 are observed in the Sichuan Acidification is responsible for much of the air pollution-related Basin, and values below 3.0 have been recorded in individual damage in China, though the relative roles of acid rain, dry deposition episodes. Atmospheric conditions in Sichuan and Guizhou of sulphur dioxide, nitrates, particulates, ozone, and other factors

provinces, with weak winds and frequent temperature inversions, have not been determined. Areas with lower rain acidity see much

are conducive to high pollutant concentrations. Emissions are also less damage than Chongqing and neighbouring cities of southwestern

high there because of the widespread burning of high-sulphur coal China. Acid rain damage to crops, forests, materials, and human

in small stoves and medium-size industrial boilers. health in China in 1995 is estimated to total more than \$13 billion

Southwestern China has seen damage from acid deposition. (China Daily, 9 March 1998).

Sulphur deposition levels are more than 10 grams of sulphur per In Asia there is also considerable concern about the fate of

square metre per year, making the situation comparable to the cultural materials as pollution levels rise. Concerns about the dete-

worst parts of the former Czechoslovakia in the 1960s and 1970s. rioration of the Taj Mahal were raised as far back as 1981 (Lal,

Zhao and Xiong (1988) report the following effects in the vicinity of Gauri, and Holdren, 1981).

Throughout Asia, cultural buildings and

Chongqing and the provinces of Sichuan and Guizhou: monuments made of alkaline-based materials are vulnerable to attack.

■ A 50 percent dieback of pine forests on Nanshan Mountain, Glass, paper, textiles, and archives are also subject to accelerated

about 20 kilometres from Chongqing, attributed to acid deposi- deterioration in the warm, moist, polluted atmospheres of Asia.

tion and air pollution. These problems are greatly under-appreciated and should be given

■ A more than 50 percent reduction in biomass production in high priority in future research before rich areas of cultural heritage

commercial spruce forests in areas experiencing rain with a pH are destroyed.

of less than 4.5. Finally, although not yet major emitters, Sub-Saharan Africa and

■ A yellowing of rice in large areas near Chongqing after rainfalls Latin American have the potential for significant sulphur emissions

with a pH of less than 4.5. as fossil fuel use increases.

Seip and others (1995) sampled soil water and stream water in

a 7-hectare catchment near Guiyang in Guizhou Province, about 350 Tropospheric ozone

kilometres south of Chongqing. Sulphate concentrations were very Ozone is an important air pollutant that can cause damage to crops,

high, pH values were as low as 4.3, and aluminium concentrations trees, and human health. It is a major component of the harmful

were elevated. Despite these factors, no apparent damage to vegetation smog that forms in urban areas during periods of high temperature,

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Acid deposition is being

intense solar radiation, low wind reported throughout Asia, with Alkaline dust is also important

speed, and an absence of precipitation. many areas receiving levels that in Asia because of its ability to

In the polluted air mass, ozone is exceed the carrying capacity neutralise the acidity of precipitation produced by a complex set of chemical and deposition. In the spring (March, reactions involving nitrogen oxides and non- of their soils. April, May) large dust storms build in the methane volatile organic compounds. North Taklamakan and Gobi deserts and the loess America and Europe are developing coordinated plateau areas of China and Mongolia. These storms

strategies to reduce emissions of ozone precursors and are associated with strong cold fronts and prevailing westerly

thereby reduce some of the health and ecosystem damage attributed winds. Dust particles are lifted as high as 6 kilometres into the

to it. Although there is still progress to be made in these regions, it atmosphere and transported over long distances to eastern China,

is again in Asia that concern is greatest. the Republic of Korea, Japan, the Pacific Ocean, and even North

Episodes of high ozone concentrations are now common in the America. The dust contains high concentrations of calcium, which

megacities (cities containing more than 10 million people) of neutralises part of the acidity in rainfall. Thus, while sulphate levels

southern Asia that have industrial emissions (producing volatile in northeast Asian deposition are high and similar to those in North

organic compounds), transportation (producing nitrogen oxides), America and Europe, pH values are less acid (typically 5.3–7.0+).

and conducive climates—Bangkok (Thailand), Hong Kong (China), Although large amounts of carbonaceous particles are emitted

Mumbai (India), and Shanghai (China), to name a few. In addition, from the burning of coal, most of the larger particles fall to ground

the formation and transport of regional ozone have been observed quickly and are not part of the regional emissions picture. Similarly,

in measurement campaigns such as PEM-West (Akimoto and others, a large portion of particles is collected, for even in the most polluted

1996; Jaffe and others, 1996). Ozone concentrations were observed regions some form of particulate collection is usually employed.

to be regionally enhanced by photochemical activity in continental Nevertheless, a certain portion of fine particles from fuel combustion

air masses passing through areas with high nitrogen oxides emissions. is carried aloft and transported over long distances. These particles

The potential effects of elevated ozone concentrations on human are usually less than 1 micron in diameter and consist of carbonaceous

health and crop production in Asia are just beginning to be explored solids—so-called black carbon—and organic compounds in

(Chameides and others, 1994). Studies in the West have established aerosol form. These particles can participate in chemical reactions,

that crop yields are depressed by repeated exposures to ozone levels contribute to reduced visibility, and lead to soiling of surfaces. They

above 50–70 parts per billion; these concentrations are exceeded in scatter and absorb solar radiation and hence play a role in global

fall and winter throughout large areas of southern China. There is warming. They also affect cloud albedo (ability to reflect sunlight),

concern that damage to winter wheat and other crops in the Yangtze because their hydrophilic qualities increase the number of cloud

Delta may endanger China's ability to meet increasing food demands. condensation nuclei. On balance, black carbon is thought to

These analyses are still in their infancy, however, and much more contribute a net warming of about 0.5 degrees Celsius (C) globally

work is needed on meteorological analysis, the gathering of (Penner, Ghan, and Walton, 1991). monitoring data, field studies on crop responses to elevated The combustion of biofuels and coal in rural households and

concentrations, and regional assessments of economic impact. Until diesel fuel in vehicles is a prime contributor to these fine particles.

more of this work is done in Asia, a definitive statement cannot There is an urgent need to better characterise the anthropogenic

be made about the relationship between regional emissions of non- emissions of primary particles from Asian sources, both by size and

methane volatile organic compounds and nitrogen oxides and chemical and physical characteristics. Diesel vehicles that are poorly

impacts on human health and vegetation. designed, operated, and maintained emit large quantities of fine

particles in much of the developing world.

Suspended fine particles Forest fires are a large source of particle emissions in all size

Particulate emissions are relatively well controlled in the ranges. Some of these fires are of natural origin (caused by lightning

industrialised world. Control systems on stationary and mobile strikes), while others are caused by human activities such as forest

sources are effective in limiting the release of primary particles, and clearing. The fires in Indonesia in the summer of 1997 caused a

secondary fine particles (such as aerosols) are being checked months-long regional air pollution episode in Indonesia, Malaysia,

by reductions in emissions of their precursors. In the outdoor Singapore, and parts of Thailand and the Philippines. The health of

environments of many Asian cities, however, concentrations of fine tens of millions of people was affected. Increases in acute respiratory

particles are very high, exacerbated by domestic solid-fuel combustion, infections, asthma, and conjunctivitis were noted in Kuala Lumpur

small-scale industrial activities, and inefficient transportation (Malaysia), Sarawak (Malaysia), and Singapore. Tests on school

systems (see above). In many parts of the world the build-up of children in Malaysia noted significant decreases in lung function, the

secondary fine particles over large regional areas during hot, dry chronic effects of which will not be known for a long time (Brauer and spells leads to regional haze, impaired visibility, inhalation health Hisham-Hashim, 1998). Fine particles from such fires can be transported effects, and related ecosystem problems. thousands of kilometres if atmospheric conditions are conducive.

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Regional climate change the industrial revolution kept the mean global surface air temperature

In the early 1990s it was recognised that sulphate aerosols can about 33 degrees Celsius warmer than it would have been in absence

influence the global climate by scattering and absorbing incoming of an atmosphere with such natural levels of greenhouse gases.

solar radiation (Charlson and others, 1992) and hence exerting a (This natural 'greenhouse effect' is highly beneficial to life on Earth,

cooling effect. The role of sulphate aerosols has now been clarified since without it the average temperature would be far below freezing.⁸)

(IPCC, 1996a). Indeed, sulphate aerosols contribute negative radia- Although water vapour contributes the largest part of the natural

tive forcing (of about -0.4 watts per square metre) that offsets the greenhouse effect, its concentration in the atmosphere globally—

positive forcing of carbon dioxide and other greenhouse gases. on which the size of the water-vapour contribution to the green-

Hence a reduction in sulphur dioxide or nitrogen oxide emissions house effect depends—is not significantly affected by emissions of

would be expected to reduce sulphate aerosol concentrations and water vapour from human activities. The most important anthropogenic

increase the potential for global warming. The radiative forcing is greenhouse gas emissions are those of carbon dioxide (CO_2), which

spatially inhomogeneous, with values as large as -11 watts per arise mainly from combustion of fossil and biomass fuels and from

square metre over heavily polluted areas such as Central Europe deforestation (see below).^{9, 10} An important indirect effect of human

and eastern China. activities on the atmospheric concentration of water vapour results

Lal and others (1995) have suggested that sulphate aerosols can from increased evaporation of water from the surface of Earth

also interfere with local climates. The cooling effect of the aerosol because of the warming caused by increasing concentrations of

haze reduces the difference between land and sea temperatures and anthropogenic greenhouse gases in the atmosphere. The resulting

weakens the monsoon. In addition, the cooler land surface reduces increase in atmospheric water-vapour content further warms Earth's

evaporation and lowers the amount of water vapour in the atmosphere. surface—a significant 'positive feedback' in the anthropogenic

The authors estimate that sulphate aerosols in the Asian atmosphere greenhouse effect.¹¹ will reduce monsoon rainfall over India and parts of China by the Concerns developed many decades ago that human-caused middle of this century. The calculated reduction of 7–14 percent increases in the carbon dioxide content of the atmosphere might over the fertile north and central Indian plains would be a serious accentuate the natural greenhouse effect enough to disturb the global threat to agricultural production. It also appears that large-scale climatic patterns to which human habitation, agriculture, forestry, forest fires can reduce rainfall regionally. and fisheries had become accustomed. As a result, in 1958 scientists began to take direct measurements of the atmospheric concentration

Global scale: climate change of carbon dioxide at locations far from its main human sources.¹² from greenhouse gases The continuous record of such measurements, at various remote The two most important human-caused problems associated with locations on land and from ships and aircraft, has revealed a steady environmental processes operating at the global scale are: increase in the global atmospheric inventory of carbon dioxide,

■ The disruption of climate as the result of energy-related emissions reaching 14 percent above the 1958 level by 1995. of heat-trapping (greenhouse) gases with long atmospheric Reconstruction of the earlier history of atmospheric carbon dioxide residence times. content (by analysis of air bubbles trapped in layered cores taken

■ The depletion of stratospheric ozone as a result of emissions of from polar ice sheets) has established that the increase from chlorofluorocarbons and related compounds from air-conditioning pre-industrial times to 1958 was about 13 percent. Thus the ratio of and refrigeration equipment (among other sources). the 1995 concentration to the pre-industrial one is $1.14 \times 1.13 = 1.29$,

The character and origins of the first of these are discussed in representing an increase of 29 percent (figure 3.8). The rise in the this section. Stratospheric ozone is not addressed here because it is atmosphere's inventory of carbon dioxide closely tracks the rise in not primarily an energy issue, although it has connections to energy global fossil-fuel burning over the past 150 years. Moreover, studies end-use technologies.⁷ based on relatively abundant carbon isotopes confirm the role of It has been known since the work of Swedish scientist Gustav fossil-fuel-derived carbon in the observed increase. There is reason to Arrhenius at the end of the 19th century that certain gases present believe that the slower increase in the 100 years before that was due in Earth's atmosphere in trace quantities exert a thermal blanketing mainly to widespread deforestation for timber, fuelwood, and charcoal. effect that keeps the planet's surface much warmer than it would Not all of the carbon added to the atmosphere by human activities

otherwise be. These are called 'greenhouse gases' because they stay there. A substantial part is absorbed by dissolution into the work in a way analogous to one of the functions of the glass in a surface layer of the world's oceans (from which oceanic mixing greenhouse, letting sunlight in but trapping outgoing heat by processes gradually transport the dissolved carbon dioxide into the absorbing it and re-radiating some of it back to the ground. much larger volume of water in the deep oceans). And part is

The most important greenhouse gas naturally present in Earth's atmosphere is water vapour. Next in importance is carbon dioxide (CO₂), followed by methane (CH₄) and nitrous oxide (N₂O). The concentrations of these gases in the atmosphere before the start of carbon during the 1980s are summarised in table 3.4. The mean

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FIGURE 3.8. ATMOSPHERIC CONCENTRATIONS OF CARBON DIOXIDE, 1850-1995

370

Ice Core Data Mauna Loa (Hawaii)

360

350

Parts per million

340

330

320

310

300

290

1860 1880 1900 1920 1940 1960 1980 2000

Source: OSTP, 2000.

residence time in the atmosphere of carbon dioxide contributed by human activities, relative to the processes that remove it, is more than 100 years. warming effect of the greenhouse gas increases.¹⁴ This offset, for Measurements and analyses over the past 20 years have revealed which the best estimate is about half of the overall warming effect that the atmospheric concentrations of two other naturally occurring

greenhouse gases—methane (CH₄) and nitrous oxide (N₂O)—
have increased by 145 percent and 14 percent since pre-industrial times. Apparently these increases are at least partly due to direct inputs from human activities as well as to alteration of ecological conditions. The wholly anthropogenic chlorofluorocarbons (CFCs) contained carbon implicated in the depletion of stratospheric ozone are potent greenhouse gases as well. The warming effect of ozone, itself a greenhouse gas, has been increased in the troposphere (as a result of deforestation 1.6 ± 1.0

TABLE 3.4. SOURCES AND FATES OF ANTHROPOGENIC CARBON EMISSIONS, 1980S

Billions of tonnes of Source
Emissions from fossil fuel combustion 5.5 ± 0.5
and cement production 1.6 ± 1.0
Emissions from tropical deforestation 1.6 ± 1.0

Total anthropogenic emissions 7.1 ± 1.1

of anthropogenic emissions of hydrocarbons and nitrogen oxides)

by more than CFCs have decreased it in the stratosphere. Fate

Changes in the atmospheric concentrations of methane, nitrous

oxide, chlorofluorocarbons, and ozone since pre-industrial times Storage in the atmosphere 3.3 ± 0.2

are thought to have increased by about 75 percent the warming Uptake by the oceans 2.0 ± 0.8

Uptake by terrestrial ecosystems 1.8 ± 1.6

potential that would be expected from the observed increases in

carbon dioxide concentrations alone. Increases over this same period Source: IPCC, 1996.

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these models include the observed build-up of greenhouse gases

TABLE 3.5. CHANGES IN EARTH'S ENERGY

BALANCE, PRE-INDUSTRIAL TIMES–1992 (corrected for the effect of atmospheric particulate matter).¹⁷ This

agreement among theory, observation, and computer modelling

Effect

Global average watts extends, moreover, to a variety of subtler trends for which reliable per square metre

measurements have become available only for the past 15–25 years,

Direct effect of increasing carbon dioxide 1.6 ± 0.2 such as cooling in the lower stratosphere, reduction of day-night

temperature differences, and maximum surface warming in northern

Direct effect of increasing methane 0.5 ± 0.1

high latitudes in winter. Taken together, these phenomena are

Direct effect of increasing halocarbons 0.25 ± 0.04 'fingerprints' of greenhouse gas-induced climate change—consistent

with the hypothesis that increases in those gases explain the

Direct effect of increasing

tropospheric ozone 0.4 ± 0.2 observed changes, and inconsistent with alternative hypotheses. Based on the evidence and arguments summarised here, the Direct effect of decreasing stratospheric ozone 0.1 ± 0.02 Intergovernmental Panel on Climate Change (IPCC) concluded in its Second Assessment that “the balance of evidence suggests a Direct effect of tropospheric aerosols -0.5 ± 0.3 discernible human influence on climate” (IPCC, 1996b, p. 4). In Indirect effect of tropospheric aerosols -0.8 ± 0.8 that report the IPCC also extended its earlier analyses of how the human influence on climate would be expected to grow under a Direct effect of natural changes business-as-usual trajectory of future greenhouse gas emissions in solar output (since 1850) 0.3 ± 0.1 and under higher and lower trajectories. The panel found that, Source: IPCC, 1996b. under the range of trajectories considered (and taking into account the uncertainty in the global temperature response to a given that would otherwise have occurred through 1995, is likely to increase in greenhouse gas concentrations), the global average diminish as emissions of particulates and their precursors are more surface air temperature would be 1.0–3.5 degrees Celsius higher tightly regulated.¹⁵ Estimated effects of the various anthropogenic in 2100 than in 1990.¹⁸ greenhouse gases on Earth’s energy balance are shown in table 3.5, In all these cases, according to the IPCC (1996b, p. 5), the average together with estimates of other changing influences on this balance. rate of warming in the 21st century “would probably be greater than Are the changes in climate being observed in the forms and any seen in the last 10,000 years”. In the IPCC’s ‘business-as-usual’ magnitudes that theory predicts for the measured increases emissions scenario, the average global temperature increase in greenhouse gases? Although natural fluctuations in climatic between 1990 and 2100 is 2.0 degrees Celsius—about 2.5 degrees variables would tend to mask human-caused disruption in its Celsius above the temperature a century ago—which would make early stages, a variety of evidence indicates that the ‘signal’ of the world warmer than it has been at any time in the last 125,000 anthropogenic change is becoming visible despite the ‘noise’ of years. In this scenario the sea level would rise 50 centimetres these fluctuations. Specifically: between 1990 and 2100, then “continue to rise at a similar rate in ■ Near-surface air temperatures around the globe have increased future centuries beyond 2100, even if concentrations of greenhouse by 0.3–0.6 degrees Celsius since the late 19th century.¹⁶ The 11 gases were stabilized by that time, and would continue to do so even hottest years since 1860 have all occurred since 1983 (notwith- beyond the time of stabilization of global mean temperature” (IPCC,

standing the multiyear cooling effect of particulate matter injected 1996b, p. 5).

into the stratosphere by a major volcanic eruption in 1991). The IPCC assessment gives due consideration to the range of

■ Directly measured ocean surface-water temperatures have also possible outcomes and to the size of the uncertainties attached to

increased by 0.3–0.6 degrees Celsius on a global average over the group's best estimates. Still, the range of expected ecological and

the past century. In the same period the global sea level, as social impacts of climate changes in the next century leaves little

determined from tidal-range measurements, rose 10–25 room for complacency even at the lower end of the range. And the

centimetres (4–10 inches). Mountain glaciers have generally uncertainties include the possibility of unpleasant surprises that

been in retreat all over the world, and mid- to high-latitude would extend the upper end:

cloudiness and precipitation have generally been increasing. Further unexpected, large and rapid climate system changes (as

These observed changes in climatic variables are consistent, in have occurred in the past) are, by their nature, difficult to predict.

overall magnitudes and in the general pattern of their geographic This implies that future climate changes may also involve "surprises".

distribution, with the predictions of basic theory for the effects of the In particular these arise from the non-linear nature of the climate

changes in greenhouse gas and particulate matter concentrations system. When rapidly forced, non-linear systems are especially

known to have occurred over this period. subject to unexpected behavior. Progress can be made by investigating

The observed climatic changes are also similar to the predictions non-linear processes and subcomponents of the climate system.

of the most sophisticated computer models of global climate, when Examples of such non-linear behavior include rapid circulation

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Mountain glaciers

changes in the North Atlantic and have generally been the productivity of farms, forests, feedbacks associated with terrestrial in retreat all over and fisheries, the geography of ecosystem changes (IPCC, 1996b, human disease, and the abundance

p. 6).¹⁹ the world. and distribution of the plants, animals,

Since the IPCC's Second Assessment, and microorganisms constituting biodiversity,

scientific evidence has continued to accumulate as well as determining the availability of water, concerning human influences on the global climate the frequency and intensity of damage from storms

system.²⁰ In particular, the data and analyses show more and floods, the combination of heat and humidity that determines

compellingly than ever that Earth's average surface temperature is liveability (and requirements for air conditioning) in warm regions

increasing and that this increase can largely be attributed to the in summer, and the potential property loss from rising sea level.

accumulation of greenhouse gases in the atmosphere caused by The average global surface temperature, then, is not a number

human activities. Among the recent findings: that by itself reveals the features of climate that matter most—the

■ 1998 appears to have been the warmest year in a millennium, spatial and temporal patterns of hot and cold, wet and dry, wind and

and the 1990s were the warmest decade in 1,000 years for the calm, frost and thaw that constitute the climate locally and regionally,

Northern hemisphere. Scientists have reconstructed the millennial where people live. Rather, it is a single, highly aggregated index of

temperature record in the Northern hemisphere using proxy data the global climatic state that is correlated in complicated ways with

for temperatures, such as ice cores, lake sediments, corals, and those crucial local and regional climatic features. When the average

tree rings (Mann, Bradley, and Hughes, 1999). global temperature increases, the regional increases will be greater

■ Greenhouse gases from human activities are the driver of on land than on the ocean surface, greater inland than near the

these temperature increases. While solar variability, volcanic coasts, and greater at high latitudes than near the equator. In

eruptions, and El Niño cycles also contribute to these global tem- mid-latitude, mid-continent regions—the midwestern United States,

perature patterns, the 20th century record cannot be explained solely for example—the increase in average temperature is expected to be

by invoking these phenomena (Mann, Bradley, and Hughes, 1998). 1.3–2.0 times the average global increase (hence as much as a

■ Regional patterns of temperature change across Earth's surface 5 degree Celsius increase when the global average has gone up by

and vertical patterns of temperature change in the atmosphere 2.5 degrees; Wigley, 1999 and IPCC, 1996b). At higher latitudes—

provide further evidence of human-induced global warming. central Canada, northern Russia—the increase could be three times

These patterns are consistent with what is expected under the global average, or more.

anthropogenic climate change—and are inconsistent with Evaporation and, hence, precipitation are expected to increase

hypotheses that suggest that solar variability or the urban-heat about 3 percent for each 1 degree Celsius increase in the average

island effect can be used to explain the instrumental temperature global temperature. (Thus a 2.5 degree Celsius increase in the

record (see Wigley, Smith, and Santer, 1998; Wentz and Schabel, average global temperature would produce a 7.5 percent increase

1998; and Peterson and others, 1999). In precipitation.) In addition, a larger fraction of the precipitation is expected to occur during extreme precipitation events, leading to an

Consequences of greenhouse increase in flooding.²¹ Notwithstanding the increase in precipitation, gas-induced climate change the increase in evaporation will likely reduce summer soil moisture. There is a natural tendency to suppose that an average global warming over large regions, increasing the frequency and severity of

of 2.5 degrees Celsius—around the mid-range projection for the droughts in some. At the same time, humid regions will likely

year 2100 relative to 1900—would not be such a bad thing. Raising become more so. Climate simulations conducted at the Geophysical

Earth's average surface temperature from 15.0 to 17.5 degrees Fluid Dynamics Laboratory of the U.S. National Oceanographic and

Celsius (from 59.0 to 63.5 degrees Fahrenheit) does not, at first Atmospheric Administration show that the average heat index (a

glance, seem to be especially problematic. Some regions would discomfort indicator combining temperature and humidity) for the

have longer growing seasons, and some would have shorter seasons southeastern United States in July will increase from about 30

of freezing weather. What would be so bad about that? degrees Celsius (86 degrees Fahrenheit) today to about 35 degrees

Such complacency is unwarranted for several reasons. Most Celsius (95 degrees Fahrenheit) by the time the average global surface

important, small changes in the average global surface temperature temperature has increased 2.5 degrees Celsius (GFDL, 1997).

will cause many changes in other climatic variables—latitudinal As the average temperature and average heat index go up, the

temperature differences, frequency of extreme temperatures, frequency of days with extremely high temperature and humidity

atmospheric circulation patterns, precipitation patterns, humidity, increases disproportionately. An average warming of 1 degree Celsius

soil moisture, ocean currents, and more—that affect human well- might increase the number of days over a particular threshold by 10

being in myriad ways. Climatic conditions are the 'envelope' in percent, while a 2 degree Celsius increase would increase the number

which all other environmental conditions and processes exist and of days over that threshold by substantially more than 20 percent

operate. Thus climate exerts powerful influences over, for example, (Wigley, 1999; IPCC, 1996b). This result portends not only much

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In a range of cases considered by

higher summer discomfort in a the IPCC, the average rate of warming effect on sea level, since floating

greenhouse gas-warmed climate in the 21st century “would probably ice displaces its weight in water, but

but also a possibility of substantial be greater than any seen in with large potential effects on oceanic

increases in death rates from heat stress and atmospheric circulations). Another

in areas that are already hot and humid in the last 10,000 years.” involves the collapse of the largely land-

summer. A decrease in cold-related deaths in borne but ocean-anchored West Antarctic Ice winter would partly offset this effect, but for a variety Sheet, the slow slide of which into the ocean if the

of reasons seems unlikely to offset it entirely (IPCC, 1996a). anchor points melted away could raise sea level by 5 metres

An increase in sea level at the mid-range value of 50 centimetres in 500 years (Oppenheimer, 1998).

between 1990 and 2100 would be devastating to low-lying islands If the ways in which global warming will affect regional climates

and seriously damaging to coastal wetlands and many coastal are uncertain, the ecological consequences of those regional

communities. As with temperature, the damage will come not just changes are even more so. Certainly both the averages and extremes

from the increase in the average but from the increase in extremes. of temperature, humidity, and precipitation are critical in governing

In the case of sea level, this refers to the damage done by storm the distribution and abundance of terrestrial animals, plants,

surges and storm waves starting from a higher baseline (IPCC, 1996a). and microorganisms, just as the averages and extremes of ocean

As for the frequency and intensity of damaging storms them- temperatures, salinities, and current patterns are critical to marine

selves—hurricanes and typhoons in particular—some lines of life. The organisms in question include, of course, the plants that

argument and evidence suggest increases in a greenhouse are the foundation of all terrestrial and oceanic food chains—

gas-warmed world, but the origins and dynamics of such storms including all those that support the human population—and they

are complicated. The higher sea surface temperatures and higher include the pests and pathogens that can destroy crops, fell farm

atmospheric moisture contents associated with a warmer world animals, ravage forests, and bring debilitating diseases to humans.

tend to produce more powerful storms, all else being equal, but Even without the capacity to predict specific effects in detail

other relevant factors that could be affected by climate change might (which is lacking because of inadequacies in ecological as well as

offset this tendency. There is evidence of an increase in the frequency climatological understanding), it is worth noting that:

of Atlantic hurricanes based on a correlation with sea surface ■ The conditions governing what grows where on land are generally temperatures, and there are simulation results indicating higher the result of co-evolution of soils, vegetation, and climate.

wind speeds and lower pressures in tropical storms world-wide Adjusting to climate change is therefore not just a matter of under global temperature increases in the range expected for the 21st allowing cropping patterns and forest characteristics to rapidly century (see Wigley, 1999 and Knutson, Tuleya, and Kurihara, 1998). migrate to a rearranged set of climatic zones; reaching a new

Also subject to considerable uncertainty are the effects of global equilibrium could take centuries. And where drastic changes in warming on the large-scale patterns of atmospheric and oceanic agricultural practices are required to deal with climate change, circulation that are so crucial in determining regional climates. One the capital- and infrastructure-poor developing world will be thinks particularly of the El Niño/Southern Oscillation phenomenon differentially disadvantaged. that affects climates across the central Pacific and the Americas and ■ Winter is the best pesticide (which is why crop pests and many all the way to Africa, the monsoons that are so critical across Africa disease vectors and pathogens are more problematic in the tropics and South Asia, and the North Atlantic thermohaline circulation that than in temperate zones). This means that warmer winters drives the Gulf Stream and greatly moderates the winter climate in outside the tropics will be problematic for food production and Western and Northern Europe. Although there is every reason to for human disease. expect that global warming would influence these phenomena, ■ The warmer, wetter conditions that global warming will bring to neither historical correlations nor the ocean-atmosphere models many of the world's temperate zones will expand the ranges of a used to simulate global climate have proven adequate for predicting variety of diseases (including, quite probably, malaria, cholera, with confidence what the exact effects will be. and dengue fever). Industrialised countries' technological and

There are, however, some suggestive preliminary findings. Some biomedical defences against these diseases may prove less robust modelling results, for example, indicate a substantial weakening of than optimists predict, not least because of the continuing the North Atlantic thermohaline circulation from greenhouse emergence of drug-resistant strains. gas-induced warming, setting in well before the pre-industrial The conclusions of the IPCC Second Assessment about the carbon dioxide concentration has doubled (Broecker, 1997; GFDL, consequences of the greenhouse gas-induced warming expected 1997). Such a weakening would, somewhat paradoxically, make over the 21st century include the following (IPCC, 1996a):

Europe much colder in winter in a world that was warmer overall. ■ Agricultural productivity “is projected to increase in some areas

And even bigger changes, such as some that might ensue from and decrease in others, especially the tropics and subtropics”.²²

ocean-atmosphere-ice interactions, cannot be ruled out. One such “Low-income populations depending on isolated agricultural

scenario involves the complete melting of the Arctic sea ice (with no systems, particularly dryland systems in semi-arid and arid regions,

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are particularly vulnerable to hunger and severe hardship” (p. 6). that mid-continent soil moisture in summer would fall about 50

■ “As a consequence of possible changes in temperature and water percent from 1990s levels; that the sea level rise from thermal

availability under doubled equivalent of CO₂ equilibrium conditions, expansion alone (not allowing for the melting of any of the

a substantial fraction (a global average of one-third, varying by Greenland or Antarctic ice sheets) would approach 2 metres; that

region from one-seventh to two-thirds) of the existing forested the North Atlantic thermohaline circulation would shut down

area of the world will undergo major changes in broad vegetation completely; and that the July heat index for the southeastern United

types—with the greatest changes occurring in high latitudes and States would reach 44 degrees Celsius (110 degrees Fahrenheit).

the least in the tropics” (pp. 5–6). The ecological consequences of such changes—and their influence

■ “Climate change is likely to have wide-ranging and mostly on humans—would be immense. adverse impacts on human health, with significant loss of life...Net Greenhouse gas-induced climate change could also affect energy

climate-change-related increases in the geographic distribution systems, potentially influencing their cost and reliability. The

(altitude and latitude) of the vector organisms of infectious dis- attractiveness of hydropower, windpower, and biomass energy

eases (e.g., malarial mosquitoes, schistosome-spreading snails) systems, for example, depends on favourable and stable, or at least

and changes in the life-cycle dynamics of both vector and infec- predictable, climate conditions at their sites over decades. Energy

tive parasites would, in aggregate, increase the potential trans- demand is also a function of climate, and changes in temperature,

mission of many vector-borne diseases...Increases in non-vec- precipitation, wind, and the like will affect it. Thus it is conceivable

tor-borne infectious diseases such as cholera, salmonellosis, and that climate-change-related reductions in the attractiveness of

other food- and water-related infections also could occur, partic- renewables combined with increases in energy demand could act

ularly in tropical and subtropical regions, because of climatic as positive feedback mechanisms—increasing greenhouse gas

impacts on water distribution, temperature, and microorganism emissions faster than they would otherwise because of greater use

proliferation... [H]otter temperatures in urban environments of fossil fuels.

would enhance both the formation of secondary pollutants (e.g.,

ozone) and the health impact of certain air pollutants. There Alternative energy futures

would be increases in the frequency of allergic disorders and of and greenhouse gas emissions

cardiorespiratory disorders and deaths caused by various air According to the IPCC, in 1990 global

emissions of carbon dioxide

pollutants”. from fossil fuel burning totalled about 5.9 billion tonnes of

■ “Climate change and the resulting sea-level rise can have a num- contained carbon. (It is

customary to keep track of carbon dioxide

ber of negative impacts on energy, industry, and transportation emissions in terms of their carbon

content rather than their total

infrastructure; human settlements; the property insurance industry; mass, to facilitate comparisons

with other stocks and flows in

tourism; and cultural systems and values”. the global carbon cycle, in which carbon may be in a

variety of

Nearly all attempts to predict the consequences of greenhouse chemical compounds.) Carbon

dioxide emissions from tropical

gas-induced climate change, including those of the IPCC, have deforestation totalled about 1.5

billion tonnes, with an uncertainty

confined themselves to addressing the changes associated with of plus or minus 1.0 billion tonnes.

The IPCC assumes that rates of

roughly a doubling of pre-industrial carbon dioxide concentrations. tropical deforestation will

decline in the 21st century, becoming

This has been done so that the results of studies by different even smaller relative to fossil fuel

carbon dioxide emissions. In

investigators would be readily comparable, inasmuch as they were 1997 fossil fuel combustion

produced about 6.3 billion tonnes of

all looking at a similar degree of climate change—not because carbon emissions (table 3.6).

there is any particular reason to believe that no more than a The geographic distribution of

industrial emissions of carbon—

doubling of pre-industrial carbon dioxide will occur. Indeed, as the

next section indicates, the world could end up with carbon dioxide

levels three or even four times the pre-industrial value. But the

TABLE 3.6 SOURCES OF INDUSTRIAL CARBON

prevalence of studies that look only at the effects of a doubling **EMISSIONS, 1997 (BILLIONS OF TONNES)**

seems to have led many people to suppose that these are the most

Combustion of petroleum products 2.70

that could occur.

In reality, as the few studies of higher levels of warming make Combustion of coal 2.40 plain, the uncertainties and controversies surrounding whether a Combustion of natural gas for energy use 1.20

doubling of atmospheric carbon dioxide would have overwhelmingly negative consequences are of much less importance when one Cement manufacturing 0.20 contemplates a quadrupling. A study of quadrupling by one of the Flaring of natural gas 0.05 main U.S. climate study centres concluded, for example, that the average temperature increases in Northern hemisphere mid-continent Total 6.60 regions would be 8–12 degrees Celsius (15–22 degrees Fahrenheit); Source: Authors' calculations based on energy data from BP, 1998; USEIA, 2000.

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FIGURE 3.9. SOURCES OF INDUSTRIAL CARBON DIOXIDE EMISSIONS, 1995 AND 2035

Developing Developing

world – 27% world – 50%

Other Asia – 6%

China – 11% Other Asia – 14% United States – 15%

Latin

America –

11% United States – 22%

Western

Africa – 3% China – 17% Europe – 12%

Middle East – 3%

Western Europe – 17%

Latin

Asia – 7% America – 6% Eastern Europe/

Former Soviet

Union – 19%

Eastern Europe/ Africa – 8%

Former Soviet Union – 27%

Middle East – 5% Asia – 4%

Industrialised Industrialised

world – 73% world – 50%

1995 2035

Source: OSTP, 2000.

that is, emissions from fossil fuel combustion (including flaring of developed by the IPCC in 1992, the 'central' scenario, designated

natural gas) and cement manufacturing—is shown in figure 3.9 for IS92a, increases in the effects of atmospheric particles over the next

1995 and projects for 2035 under a business-as-usual energy 100 years continue to roughly counterbalance the effects of increases in future. In 1995 nearly three-quarters of these emissions came from in the non-carbon dioxide greenhouse gases. Thus the net increase in industrialised countries. Under the business-as-usual scenario, the in the heat-trapping effect over this period is about what would be expected from the carbon dioxide build-up alone.

by 2035. (The cumulative contribution of developing countries to The IS92a scenario is very similar to the 'unsustainable' IIASA- the atmospheric burden of anthropogenic carbon dioxide will WEC (International Institute for Applied Systems Analysis – World remain smaller than that of industrialised countries for some time Energy Council) A2 scenario presented in chapter 9. Both are based thereafter, however, and per capita emissions from developing on World Bank 'medium' population projections; the IS92a scenario countries will remain smaller than those from industrialised ones nario uses an older median projection in which the world for longer still.) population reaches 11.3 billion by 2100 and the IIASA-WEC uses a The IPCC analysis and its scenarios for future emissions also take newer projection of 10.7 billion by 2100. The IS92a scenario into account the other anthropogenic greenhouse gases—methane, assumes that real economic growth world-wide averages 2.9 percent tropospheric ozone, nitrous oxide, and halocarbons—and a year from 1990 to 2025 and 2.0 percent a year from 2025 to anthropogenic particulate matter in the atmosphere that partly 2100, resulting in overall growth of 2.3 percent a year from offsets the heat-trapping effect of the greenhouse gases by screening 1990–2100 compared with 2.5 percent a year in the A2 scenario. out incoming sunlight. As noted, the IPCC found that, as of the Both scenarios assume that the energy intensity of economic activity mid-1990s, the non-carbon dioxide greenhouse gases had added (energy per unit of real GDP) declines by 1.0 percent a year from about 75 percent to the heat-trapping effect that would have resulted 1990–2100 and that the carbon intensity of energy supply (kilograms from the build-up of carbon dioxide alone. But the IPCC's best of carbon emitted in carbon dioxide per unit of energy supplied) estimate of the effect of increasing particle concentrations was that decreases by 0.2 percent a year over this period. The result is that global these had approximately cancelled out the effect of the increases in carbon emissions increase in both scenarios from 7.4 billion tonnes non-carbon dioxide greenhouse gases. In one of the six scenarios a year in 1990 to 20.0 billion tonnes a year in 2100, and cumulative

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carbon emissions between 1990 and 2100 total 1,500 billion tonnes. what the numbers above imply for the stabilisation target for

The carbon content of the atmosphere in 2100 under the IS92a atmospheric carbon dioxide of 550 ppmv, about twice the pre-

and A2 scenarios would be some 1,500 billion tonnes, or about 715 industrial level. (While there can be no confidence that this level

parts per million of carbon dioxide by volume, 2.5 times the pre- would avoid climate change seriously disruptive of human well-

industrial level, and still rising steeply. (Only about half of the 1,500 being throughout much of the world, a doubled carbon dioxide

billion tonnes of carbon added between 1990 and 2100 would have target is widely discussed because it is, at least arguably, near the

remained in the atmosphere, the rest having been taken up by upper limit of what is tolerable and near the lower limit of what

oceans and by vegetation according to the IPCC's carbon cycle seems achievable.) This would require that cumulative emissions

model.) This is the scenario for which the IPCC obtained the surface between 1990 and 2100 be less than two-thirds those in the IS92a

temperature and sea level estimates mentioned above. Because of scenario. It would also require that emissions begin to decline after

the thermal lag time of the oceans and the continuing melting of peaking at about 11 billion tonnes of carbon a year around 2030.

polar ice under warmer conditions, both temperature and sea level This more sustainable development path and the challenge of

would continue to rise after 2100 even if the growth of atmospheric achieving the transition towards such a path are illustrated by the

carbon dioxide were halted at that point. IIASA-WEC C scenario presented in chapter 9. That scenario leads to

The challenge of stabilising the carbon dioxide content of the the stabilisation of atmospheric carbon concentrations at about 430

atmosphere is illustrated in the IPCC's Second Assessment with ppmv and cumulative emissions of some 540 billion tonnes of

emission trajectories that would be able to achieve stabilisation at carbon from 1990-2100.

Perhaps more important, the development

concentrations ranging from 450-1,000 parts per million by volume path that leads to atmospheric stabilisation of carbon at a relatively

(ppmv). (The pre-industrial concentration was about 280 ppmv; benign level also leads to the fulfilment of most of the other criteria

today's is 365 ppmv.) These trajectories can be characterised by the for sustainable development discussed in this report.

cumulative and average emissions they entail between 1990 and The difficulty of achieving this goal becomes particularly apparent

2100 (although what happens after that also matters). The results when one views it in terms of the roles of industrialised and

are summarised in table 3.7.

developing countries. In 1990 industrialised countries emitted about 4.5 billion tonnes of carbon from fossil fuel burning (three-cumulative emissions of 1,500 billion tonnes of carbon between 1990 and 2100 and annual emissions of 20 billion tonnes of carbon quarters of the world total, or 3.6 tonnes per inhabitant of these in 2100, are both above even the highest of these stabilisation countries). Developing countries emitted 1.5 billion tonnes (about trajectories. Such emissions would nearly triple pre-industrial 0.37 tonnes per capita). In 1992, as part of the UNFCCC, industrialised atmospheric carbon content by 2100 and create a situation in which countries agreed to try to limit their carbon emissions in 2000 to an eventual quadrupling or more could not be avoided no matter 1990 levels (see below). But few are on a track towards achieving what measures were taken thereafter. (These levels are so high as to this. For example, in 1997 U.S. carbon emissions were about 9 percent render irrelevant the current controversies over exactly how severe higher than in 1990. the climatic consequences of a doubling of atmospheric carbon Considerably more effort is required. For example, assume that might be; a quadrupling would transform Earth's climate beyond industrialised countries were willing and able to return to their recognition; see GFDL, 1997.) 1990 carbon emissions by 2010—a decade after the initial UNFCCC To illustrate the challenge associated with reducing emissions to target (and a performance considerably weaker than called for in the levels being debated in the context of the United Nations the 1997 Kyoto Protocol; see below)—and were also willing and Framework Convention on Climate Change (UNFCCC), consider able to reduce these levels by 10 percent a decade thereafter. Even

TABLE 3.7. IPCC SCENARIOS FOR STABILISING CARBON DIOXIDE LEVELS, 2075–2375

To stabilise concentrations at (parts per million by volume)	450	550	650	750	1,000
By about the year	2075	2125	2175	2200	2375
Cumulative emissions in 1990–2100 would need to be in the range of (billions of tonnes of carbon)	550–750	750–1,100	970–1,270	1,090–1,430	1,220–1,610
Average emissions in 1990–2100 would be in the range of (billions of tonnes of carbon per year)	5.7–5.9	7.9–9.0	10.2–10.8	10.0–11.8	12.7
And peak emissions (billions of tonnes of carbon per year)	9.5	11	12.5	13.5	15

In the year 2012 2030 2050 2060 2075

Source: IPCC, 1996b.

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then, stabilising atmospheric carbon dioxide concentrations at 550 C scenario, which achieves the required emissions reduction

ppmv would still require that per capita emissions in developing countries not exceed 1 tonne of carbon in the global peak-emissions year of 2030. (This assumes that emissions from International agreements deforestation have been eliminated by 2030 and that the population to address global climate change of developing countries is about 7.5 billion at that time, consistent with the 'medium' World Bank projection.) As shown in later chapters, with vigorous promotion of renewables and energy-efficient technologies, such a per capita level could produce much higher living standards. came into force in March 1994.

Even more challenging, given the justifiable economic aspirations of developing countries, the unwillingness of many industrialised countries to take the steps needed to reduce emissions, and the gas concentrations in the atmosphere at a level that would prevent common expectations of all countries to rely heavily on expanded fossil fuel use, is that the per capita emissions of both industrialised and developing countries would need to fall sharply after 2030 to stay on this 550 ppmv stabilisation trajectory. (See chapter 9 for climate change; to promote the sustainable management and conservation more discussion of carbon emission scenarios, particularly the of greenhouse gas 'sinks' (such as forests); to develop adaptation

BOX 3.7. GREENHOUSE GASES AND NATURAL DEBT

Countries take on national debt when they remain because of long residence times. responsibility for reducing emissions based spend money faster than their economies produce it. Building up national debt is of a number of countries that together on current emissions, essentially a way of borrowing from the account for 55 percent of the world's since natural debt (cumulative greenhouse country's future earnings. A little national debt population and about 75 percent of global gases in the atmosphere) is more closely can be beneficial, by providing resources carbon dioxide emissions. related to actual climate impact. Such

so that the economy grows faster than it has been argued that it would be proposals are not welcomed by negotiators otherwise might. But a lot of national debt can more appropriate to determine a country's for most industrialised countries.

be quite disruptive. Countries can also take One argument against using natural debt on 'natural debt' by putting pollutants into Natural debt: Carbon as as a measure of responsibility is that it would

the environment faster than they are naturally carbon dioxide remaining in the global be unfair. After all, it is argued, the ancestors

removed. In this way they borrow from the atmosphere from fossil fuel combustion of today's populations in industrialised

environment's future assimilative capacity. countries did not know they were causing Humanity has been adding greenhouse Current Cumulative a problem by emitting greenhouse gases.

gases to the atmosphere faster than they can emissions emissions Thus today's populations, who did not do

be naturally removed. As a result the global 1996 1950–96 the polluting, should not have to pay for past

atmospheric burden of carbon dioxide and (tonnes (tonnes mistakes. This view is partly accounted for

other greenhouse gases has been rising. per capita) per capita) in the table, which only shows emissions from

The extra burden of greenhouse gases in the United States 5.3 119 1950. But there are two important counter-

atmosphere above pre-industrial levels is a arguments:

measure of the global natural debt. Indeed, Canada 4.2 91 • Today's rich populations have enjoyed this natural greenhouse gas debt is the Germany 3.3 87 the (considerable) benefits derived from principal driver of climate change, since past use of fossil fuels and other green-

it determines the extra radiative forcing Russia 3.9 78 house gas-generating activities and (warming). United thus should accept the debts that go

Although most discussions of greenhouse Kingdom 2.7 78 along with those benefits. It is not a matter

gas control address the current emissions of Australia 4.1 70 of punishment, but one of recognising countries, cumulative emissions (natural debt) the debits as well as the credits (the are the chief determinant of the impact on the Sweden 1.7 54 polluter pays principle).

climate. The natural debts of countries differ France 1.8 49 • Saying that if someone did not know they

substantially more than do current emissions, were doing a risky thing that they need

because some countries have been emitting Japan 2.4 41 not be held responsible is a sure way to large amounts for much longer than others Korea, encourage people not to make the effort

have. The largest natural debts have been Dem. Rep. 3.0 32 to discover whether their activities might

accrued by industrialised countries, which cause problems for future generations. It

Korea, Rep. 1.7 16

started burning large amounts of fossil fuels essentially rewards ignorance. A sustainable early in the 20th century. Some of those China 0.6 8 world energy system, on the other hand, greenhouse gases have been removed India 0.2 3 is one in which cross-generational naturally from the atmosphere, but some responsibility is accepted by all.

Source: Smith, 1991a; Hayes and Smith, 1994; Smith, 1997.

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Climate change is likely

strategies; to take climate change to have wide-ranging and mostly probably the chief source of into account when setting relevant adverse impacts on human health from energy social, economic, and environmental health, with significant systems. At the workplace scale, in the policies; to cooperate in technical, scientific, form of coal dust for example, they are and educational matters; and to promote loss of life. a principal contributor. At the regional scientific research and information exchange. scale, secondary particulates from sulphur The UNFCCC also establishes more demanding and nitrogen gases contribute to ill health and obligations for industrialised countries, which agreed to try form the basis for acid deposition. At the global scale, they

to reduce emissions of carbon dioxide and other greenhouse gases to contribute to climate change through local warming or cooling,

1990 levels by 2000. OECD countries are also committed to facilitating depending on particle composition and ground characteristics.

the transfer of financial and technological resources to developing Overall, it is believed that human-caused aerosols had a net cooling

countries, beyond those already available through existing development effect during the 20th century, masking some of the warming that

assistance. The convention requires industrialised countries to take would have occurred through greenhouse gas build-up.

the lead in adopting measures to combat climate change, recognising The transfer of aerosols from one scale to another is governed

that they are mainly responsible for historic and current emissions of by complicated processes involving geography, elevation, wind,

greenhouse gases, and that developing countries will need assistance moisture, size, composition, and so on. Nevertheless, in general it

to meet the treaty's obligations (box 3.7). can be said that reducing emissions at one scale will have impacts

The structure provided by the UNFCCC is being built on with additions at other scales. In most cases these benefits are beneficial. As

agreed in a series of conferences of the parties. Most notably: particulates and their precursors such as sulphur oxides are

■ The first conference of the parties, held in Berlin (Germany) in brought under control because of concerns at other scales, however,

March 1995, focused on the need to reinforce the commitments greenhouse gas warming may actually become greater than in

in article 4 of the UNFCCC with “quantified limitation and reduction” the immediate past because of the apparent net cooling effect in

objectives for annex 1 countries after 2000.²⁴ The mandate did the atmosphere.

not call for new commitments for developing country parties but

only for enhanced efforts at implementing the existing commitments Environmental risk transition relating to these countries in article 4. During development, societies tend to push off environmental problems

■ The third conference of the parties, held in Kyoto (Japan) in from smaller to larger scales (Smith, 1990, 1997). For energy,

December 1997, produced a protocol to the framework convention household hazards dominate fuel cycles in the poorest parts of the

codifying commitments for reductions in greenhouse gas emissions

after 2000. The protocol commits annex I parties (except

Australia, Iceland, New Zealand, Norway, Russia, and Ukraine) to FIGURE 3.10. ENVIRONMENTAL RISK TRANSITION

reduce greenhouse gas emissions by 5–8 percent below 1990

levels between 2008 and 2012 and to make “demonstrable

progress” towards achieving these commitments by 2005. Household sanitation

Severity

Overall emissions are to be computed on a net basis, accounting

for afforestation, reforestation, and deforestation as well as emis- Urban air pollution

Greenhouse

sions from energy supply and other industrial activities. (See box gas emissions

3.7 for a discussion of another approach to measuring a country’s

greenhouse gas contributions.)

The Kyoto Protocol, having not yet been ratified by the requisite

number of nations, is not in force. It has been criticised by some

(especially in the United States) as demanding too much too soon

of industrialised nations while not requiring anything of developing

countries. It has been criticised by others as not requiring enough Increasing wealth

of anyone, representing only a small ‘down payment’ on the kinds

of emission reductions that will be required over the 21st century to

avoid intolerable climate change from greenhouse gases. Shifting environmental burdens

Local Global

Cross-scale impacts Immediate Delayed

Some types of pollutants are created and create problems at every Threaten health Threaten ecosystems

scale and easily transfer between scales. Aerosols (particulates) are

a good example. At the household and community scales, they are Source: McGranahan and others, 2000; Smith and Akbar, 1999.

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world, while community impacts dominate fuel cycles in middle- impacts of climate change will be greater than the costs. Among the income cities through industrial and vehicular pollution. In the important benefits are avoiding or reducing the direct impacts on richest countries, household and community problems have mostly human health that might accompany climate change, and avoiding been pushed off to the global level in the form of greenhouse gases ecosystem effects that could have significant indirect impacts (figure 3.10). In all countries, however, occupational risks per on humanity. Reducing current greenhouse gas emissions could worker-hour tend to be much higher than risk per hour of public generate long-term health benefits such as fewer malarial mosquitoes, exposure. As with other exposures, however, occupational risks also fewer extreme climatic events (including tropical cyclones and heat tend to be higher in poorer countries (box 3.8). episodes), shifts in atmospheric composition towards less pollution, The environmental risk transition curves in figure 3.10 should reduced impacts on food production, and decreasing refugee not be considered fixed in the sense that today's developing nations populations from reduced sea level rise and other factors will be forced to go through them in the same way that today's (McMichael and others, 1996). Similarly, reduced greenhouse gas industrialised countries have done. Rather, the curves should be emissions could lead to less damage to important ecosystems. viewed as a management framework by which to judge the progress Each step of the causal chain from greenhouse gas emissions to of development policy. The task in developing countries is to avoid global warming to direct effects on health and ecosystems is not the excesses of the past, to continue to push down the household understood with certainty, however. As a result of that uncertainty, curve, and to not let the community curve rise out of hand. This many observers are still unconvinced that the potential but might be considered a kind of 'tunnelling' through the curves to distant health and ecosystem benefits justify large spending on avoid climbing over the peaks by applying cleaner, more efficient greenhouse gas reductions today. Although the consensus scientific energy supply and use technologies earlier in the development opinion (as represented by the IPCC) is that such ill effects are process than has occurred to date. likely if current trends in greenhouse gas emissions continue, scepticism holds back international agreements to significantly Win-win strategies to link environmental alter current greenhouse gas emissions. This is particularly so in improvements at different scales developing countries, which must contend with many urgent problems

The most convincing argument for spending resources to reduce related to human health and welfare. But it also applies to many current greenhouse gas emissions is that the benefits from reduced groups in industrialised countries.

BOX 3.8. THE KUZNETS ENVIRONMENTAL CURVE: FACT OR FALLACY?

An illustration of the environmental risk transition between scales is seen in the figure Urban particulate concentrations, human exposure, and national development

below, which plots the relationship between urban PM10 (particulates smaller than 10 microns in diameter) concentrations and

Human exposure

countries' development status as indicated

by their UNDP human development index (a function of income, literacy, and life expectancy).

community concentrations

Relative pollution

Superficially, urban PM10 concentrations seem to follow the so-called Kuznets environmental

curve—that is, they first rise during development, reach a peak, then decline (Grossman and Kruger, 1994). (The curve is named after the

4 Nobel Prize-winning economist Simon Kuznets, who noted in the 1960s that many countries go through a period of increasing

2 income inequality during development before becoming more equitable.) From the standpoint of the risk transition, however, this curve only addresses the community 0

Lowest Low Medium High Highest

scale in the form of ambient urban air pollution. It ignores what happens at other Development level

scales, which may be more important. The main concern about particulates time indoors and in other places close to concentrations. Instead of rising and then falling,

is their impact on human health. From a local sources of pollution, exposure patterns exposures decline continuously—illustrating

health standpoint, it is not so much urban can be quite different from patterns of ambient that the Kuznets curve misses the actual

concentrations that are critical but human pollution. Thus, as shown in the figure, because trend, which is that overall risk tends to fall

exposure, which is a function of not only household sources dominate exposure in the even though community risk rises, because where the pollution is but also where the poorest countries, the pattern of exposures of the shift of household to community people are. Because people spend a lot of is quite different than that of urban ambient impacts (Smith, 1993).

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BOX 3.9. WIN-WIN CROSS-SCALE ENVIRONMENTAL STRATEGIES IN CHINA

A recent study found that greenhouse gas In many developing country households by 2020. Other countries with high household reductions resulting from changes in energy the particulate emissions from burning and industrial dependence on solid fuels, use would generally be accompanied by unprocessed solid fuels (biomass and coal) such as India, could be expected to see substantial near-term human health benefits in routinely exceed by an order of magnitude similar benefits. China (Wang and Smith, 1998). But the level or more the safe levels specified by the Acid deposition is increasingly serious in of health benefits would vary greatly with the World Health Organization. Thus a 15 percent many regions, damaging forests, crops, and choice of energy technologies and sectors. reduction by 2020 in Chinese households' aquatic animals. The RAINS-ASIA model Shifting from conventional coal-fired power greenhouse gas emissions below the 'business- (Foell and others, 1995) indicates that large plants to natural gas, for example, has larger as-usual' level would avoid more than areas in Asia have acidity levels in excess health benefits than reducing global warming 100,000 premature deaths a year. Reduced of critical loads due to sulphur deposition, potential, while shifting from coal power to emissions of health-damaging pollutants posing significant environmental threats to hydropower results in the same percentage through household improvements in energy a variety of ecosystems. The model also reduction in emissions of health-damaging efficiency and changes in the fuel mix would projects that sulphur deposition will eventually pollutants and greenhouse gases. This variation in health benefits is even also reduce greenhouse gases. Thus reducing exceed critical loads by a factor of more than larger between sectors. The conservative greenhouse gas emissions at the global 10 in many parts of Asia as a result of the estimates in the study show that the health scale could significantly ease human health growing dependence on fossil fuels. This benefits of a 1 tonne reduction in particulate impacts at the household scale. increase will threaten the sustainability of

emissions from household stoves are at least On a national scale, a 15 percent reduction many natural and agricultural ecosystems in 40 times those from reduction of the same in greenhouse gases below business as usual the region. The model develops a series of amount from coal-fired power plants. In terms by 2020 would result in 125,000–185,000 emission control scenarios and shows that of human health benefits, the choice of energy fewer premature deaths in both the power energy efficiency and fuel substitution technologies and sectors in which to conduct and household sectors, depending on pathways, which are also the main mitigation greenhouse gas reduction efforts is more different control pathways (energy efficiency options for greenhouse gases, can be important than choice of a particular target for or fuel substitution). This range represents important instruments for controlling acid-greenhouse gas reduction. about 1 percent of the total mortality in China forming emissions. One approach to resolving this impasse is to promote ‘no regrets’ regional scales. This is a win-win strategy to link environmental scenarios for reducing greenhouse gases. Such scenarios achieve improvements between scales. significant near-term benefits for human health and ecosystems in The potential health benefits from reduced greenhouse gases can addition to reducing greenhouse gases. Thus such immediate be estimated from the global burden of ill health from air pollution. actions can be justified even if climate sensitivity to additional Using airborne particulates as the indicator pollutant, the World greenhouse gases turns out to be less than is now thought (Repetto Health Organization estimates that air pollution causes 2.7–3.0 million and Austin, 1997). Examples of such near-term benefits include the premature deaths a year, or 5–6 percent of global mortality (WHO, environmental and energy-security advantages that would accrue 1997). Since most of this pollution comes from the combustion of through less dependence on fossil fuels, and the human welfare fossil and biomass fuels, which would be among the main targets of benefits that could emerge if an international greenhouse gas control any effort to control greenhouse gases, the potential reduction in regime were oriented towards assisting economic development and health-damaging emissions would seem to be at least as great as the reducing vulnerability among poor populations (Hayes and Smith, 1994). target reduction in greenhouse gas emissions. Arguably it is even Among the significant near-term benefits of greenhouse gas greater, however, since switching from dirty, less efficient fuels reductions are the human health benefits resulting from changes in (such as coal) to cleaner, more efficient fuels (such as natural gas) the efficiency and structure of energy use that would be a large part reduces emissions of health-damaging pollutants even more than

of most greenhouse gas reduction scenarios. Although fuel cycles greenhouse gas emissions. With greenhouse gas reduction targets have several effects on health and ecosystems—for example, on the order of 10–20 percent, the scale of emissions of health-through water pollution, the potential for large accidents, and damaging pollutants and associated reduction of ill health could be occupational health and safety—the largest and most sensitive to in the same range or somewhat higher—perhaps a 250,000–change are probably those related to pollutant emissions from the 750,000 annual reduction in premature deaths world-wide. processing and burning of fuels. The same combustion processes To more accurately estimate these near-term health benefits, it is that produce greenhouse gas emissions such as carbon dioxide and necessary to link each technological option in a particular green-methane also generate local and regional airborne health-damaging house gas reduction scenario with the accompanying reduction in and acid-precursor pollutants such as particulates and sulphur emissions of health-damaging pollutants. The health impact of these oxides. Thus a reduction in greenhouse gases at the global scale, emissions, however, depends on the sector of the economy in which through improvements in energy efficiency and changes in the mix they are taken. This is because the degree of human exposure of fuels, can be expected to reduce health-damaging and acid- created by a unit of emissions of health-damaging pollutants depends precursor pollutants as well, potentially bringing immediate on where they are released relative to where people spend time, environmental improvements at the household, community, and their ‘exposure effectiveness’. Thus a tonne of emissions averted

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The World Health Organization

in the household or transportation estimates that air pollution causes Assessment methods sectors close to where people spend 2.7–3.0 million premature deaths A number of methods have been much of the time will generally cause a year, or 5–6 percent of developed to compare the disparate a much greater reduction in human effects of energy systems on a common exposure (and improvement in health) global mortality. basis. Here we discuss perhaps the two than a tonne averted in the industrial or power most common, economic valuation and fuel-sectors. An example of such a win-win possibility in cycle analysis. In both cases there is still much China is presented in box 3.9. uncertainty and some controversy on the fundamental On a community scale, more than 1.1 billion people living in nature of the analyses and on the data inputs required. Thus we urban areas world-wide are exposed to particulate or sulphur present these examples not as definitive, but as illustrative of the

dioxide levels in excess of World Health Organization guidelines type of information they can provide.

(Schwela, 1995). These pollutants are released by industrial, How much is clean air worth? Because it generally becomes

household, and transportation energy use. Air pollution is particu- increasingly expensive to reduce pollution as emission controls

larly severe in megacities. Again, reducing greenhouse gas tighten, a fundamental question is how much money each bit of

emissions by changing energy use and structure can also reduce the emissions control is worth. If the value of averting damage were

particulate and sulphur dioxide emissions that cause severe urban known, then the degree of control could be set such that the total

air pollution. cost (control cost plus residual damage cost) is minimised.

The same principle applies in the cities of the industrialised Normally, the cost of control is fairly well known. Similarly, damage

world, although the scale of absolute benefits is less because air is fairly easy to value for simple property destruction, such as corro-

pollution levels are lower. From an economic standpoint, however, sion of buildings and reduction of crop yields. But valuing damage to

there can still be substantial secondary benefits from greenhouse critical ecosystems or human health is not straightforward. As a

gas controls through associated reductions in health-damaging result there are no universally accepted methods.

pollutants, acid precipitation, and the like. The value of these Several approaches are used to value human health, including:

benefits could in many cases rival the costs of the greenhouse gas ■ Human capital—the value of lost wages and associated medical

controls, making a win-win result (Elkins, 1996; Burtraw and costs from illness or injury and premature death.

Mansur, 1999). ■ Value of statistical life—the imputed value from extrapolating

As noted, the near-term health and ecosystem benefits of reducing human risk-averting behaviour. For example, if the labour market

greenhouse gases provide the opportunity for a true no-regrets shows that inducing a worker to accept a job with an additional

reduction policy in which substantial advantages accrue at various death risk of 1/1,000 a year requires extra wages of X dollars a

scales even if the risk of human-induced climate change turns out year, then X multiplied by 1,000 would equal the value of a lost life.

to be less than many people now fear. Increases in energy production ■ Willingness to pay—in which people are asked how much they

and end-use efficiency and changes in the mix of fuels can reduce would be willing to pay to avoid certain risks. These amounts are

environmental impacts at the household, community, regional, and then extrapolated to find the equivalent value of a life.

global scales, while meeting greenhouse gas targets. To achieve In recent years the willingness to pay approach has become more these benefits effectively, however, considerations of health and widely used, because it holds more theoretical appeal for economists other secondary benefits should be included from the start in than other approaches. But beyond the obvious difficulty of finding designing greenhouse gas control strategies. accurate data, there are three important problems with this

This no-regrets strategy also has important implications for approach. First, measured willingness to pay can be quite different emissions trading in the form, for example, of joint implementation for the same issue depending on how the question is phrased, raising and clean development mechanisms. Because the near-term health doubt about the measure's intrinsic utility. Second, willingness to improvements are local, they accrue almost entirely to the nation in pay to avoid a certain risk depends on the respondent's knowledge which greenhouse gas control projects are undertaken—unlike the about the risk, which varies dramatically by geography, demography, benefits of greenhouse gas reductions, which accrue globally. These education, and time, and which is quite difficult to account for in large local benefits may provide a significant extra incentive for surveys. Finally, although it is clear that willingness to pay varies with other developing countries to enter into arrangements in which income, it is not clear by how much. As people grow richer, their local greenhouse gas controls are financed externally and the willingness to pay to avoid a given risk generally increases even faster. emission credits are shared. Indeed, a greenhouse gas reduction But this relationship is not clear across different periods and cultures. strategy can be consistent with such critical national development Consider some examples of how willingness to pay has been objectives as reducing local air pollution, increasing energy used to value air pollution in developing and industrialised efficiency, and improving social equity by providing energy services countries. In China, the dirtiest fossil fuel, coal is widely used, its to remote areas through renewable energy. use is expected to grow rapidly in the next few decades, and effective

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pollution controls are not widely used. A recent study estimated that pay considerable amounts to avoid this life shortening. Table 3.8 air pollution cost China about \$48 billion in 1995 (7 percent of presents recent estimates of the health impacts of coal and natural GDP), including impacts of acid deposition as well as health effects gas power plants equipped with the best available control technologies.

from outdoor and indoor air pollution (World Bank, 1997). The For natural gas combined-cycle plants the only significant health study found that the dominant cost came from health costs for urban costs are associated with nitrogen oxide emissions, and these costs residents, some \$32 billion (5 percent of GDP). Moreover, it are relatively low (typically about 5 percent of the electricity generation projected that under business-as-usual conditions, pollution-related cost). But for coal the estimated health costs (mostly due to health costs for urban residents would increase to \$98 billion by health damage from fine particle air pollutants) are large and 2020 at current income levels, or \$390 billion (13 percent of GDP) comparable to the electricity generation cost. These estimates are with adjustments to willingness to pay related to growth in income. quite uncertain, however. This is substantially more than the estimated economic damage of Table 3.9 presents estimates of health costs in France for air greenhouse gas emissions from the same facilities. pollution from gasoline-fuelled cars equipped with pollution China's high pollution-related health costs are partly due to controls and for diesel-fuelled cars. The estimated health impacts, relatively limited pollution controls. Even in countries with relatively measured per litre of fuel sold, are high (though, as with power extensive pollution controls, however, pollution-related health costs generation, quite uncertain), especially for urban driving—about can be high because aggregate energy-related emissions can be high twice the retail price (excluding retail taxes) for gasoline cars and even if emissions per unit of energy provided are low. Moreover, the 25 times the retail price for diesel cars. As in China, the economic willingness to pay to avoid pollution damages will be high in high- costs of greenhouse gas emissions from European cars and power income countries. And in densely populated regions such as Europe plants would seem to be much lower. and Japan, large populations are generally exposed to air pollution. These costs will vary significantly by region depending on the mix Recent studies in Europe have shown that health impacts of rural and urban driving, whether emissions are at ground level or dominate the external social costs of pollution, and estimate that the from tall stacks, local and regional population densities, and costs of health impacts due to fine particle air pollution are income, which affects willingness to pay. Applying the results in especially high (Rabl and Spadaro, 2000; Spadaro and Rabl, 1998; table 3.9 to developing countries, where per capita income Spadaro and others, 1998; Krewitt and others, 1999). These averaged about \$2,800 in 1995, the imputed health costs would be economic calculations reflect recent epidemiological studies indicating 0.1–0.5 times those estimated for France when all other factors are

that fine particles are associated with serious health effects, including equal, depending on how willingness to pay varies with income.

premature death (see the section on community scale). If willingness to pay continues to increase more rapidly than

Although considerable uncertainty remains about health impacts income, health impacts will become increasingly important

from small particles, the economic value of these impacts is expected for developing countries even if emission controls are put in place.

to be high—at least in densely populated regions of high-income That is because their income and energy consumption levels will

countries, where large populations are exposed to air pollution that rise more rapidly than energy consumption levels, even with emis-

can shorten lives by a few months. These populations are willing to sion controls in place. To illustrate, consider the WEC projection

TABLE 3.8. AIR POLLUTANT EMISSIONS AND ESTIMATED HEALTH COSTS FOR EUROPEAN POWER PLANTS EQUIPPED WITH THE BEST AVAILABLE CONTROL TECHNOLOGIES

Emission rate (grams per kilowatt-hour)		Unit health cost (cents per kilowatt-hour)	
Unit health cost		Natural	Natural
(cents per gram)		Pulverised coal	gas com-
Siting		steam-electric	Pulverised coal
bined		steam-electric	bined
cycle		cycle	cycle
Sulphur	Nitrogen	Sulphur	Nitrogen
PM10	PM10	PM10	Total
dioxide	oxides	dioxide	oxides
Typical	1.0	1.6	1.7
Urban	1.6	2.3	5.1
Rural	0.7	1.1	0.5

Note: These calculations were carried out as part of the European Commission’s ExternE Program. Studies under the program have estimated the economic values of health impacts by assessing people’s willingness to pay to avoid adverse health effects. The health cost estimates shown are median values; the 68 percent confidence interval is 0.25–4.0 times the median cost. Source: Rabi and Spadaro, 2000.

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TABLE 3.9. AUTOMOTIVE NITROGEN OXIDE AND PARTICULATE EMISSIONS AND ASSOCIATED PUBLIC HEALTH COSTS IN FRANCE

Emission rate	Health costs (dollars)
Fuel	
Fuel and (grams	

economy Per gram Per kilometre Per litre of fuel
driving per kilometre)
(kilo-
environ-
metres

ment Nitrogen Particu- Nitrogen Particu- Nitrogen Particu- Nitrogen Particu-
per litre) Total Total
oxides lates oxides lates oxides lates oxides lates

Gasoline

Urban 8.7 0.68 0.017 0.022 2.750 0.015 0.047 0.062 0.13 0.41 0.54

Rural 10.3 0.79 0.015 0.027 0.188 0.021 0.003 0.024 0.22 0.03 0.25

Diesel

Urban 10.4 0.75 0.174 0.022 2.750 0.017 0.479 0.496 0.17 4.98 5.15

Rural 12.7 0.62 0.150 0.027 0.188 0.017 0.028 0.045 0.21 0.36 0.57

Note: These calculations were carried out as part of the European Commission's ExternE Program. Studies under the program have estimated the

economic values of health impacts by assessing people's willingness to pay to avoid adverse health effects. The health cost estimates shown are median

values; the 68 percent confidence interval is 0.25–4.0 times the median cost. The gasoline cases are for cars equipped with catalytic converters.

Source: Spadaro and Rabl, 1998; Spadaro and others, 1998.

that in developing countries the number of cars will increase 6-fold ash disposal. Each of these facilities has environmental impacts and

between 1990 and 2020 (WEC, 1995). Even if all were gasoline- is, in a sense, 'switched on' whenever the appliance is used even

fuelled cars equipped with three-way catalytic converters, health though it may be physically unconnected and thousands of kilometres

costs in developing countries would increase to \$40–120 billion in away. Furthermore, environmental impacts occur not only during

2020, depending on the rate of increase in willingness to pay. If all normal operations of these facilities, but also during their construction,

cars were diesel, health costs would be six times as high. These repair, and dismantling—what is called their life cycle. Even non-

health cost estimates do not include health impacts associated with fuel energy systems, such as photovoltaic power plants, have fuel

buses, trucks, and locomotives. cycles and life cycles, including the harvesting, processing, and Health costs might end up being much higher than these transport of the materials used to construct the facility.

estimates because real world emission levels tend to be considerably Comparative risk assessment is one term used for studies of the

higher than regulated emission levels (Ross and others, 1995). That life-cycle impacts of alternative fuel cycles, such as different ways to

happens for a variety of reasons, including that regulated emissions produce electric power. Such studies are usually normalised are for well-maintained cars and that regulations tend to be for driving according to an appropriate unit of energy output—for example, cycles that often do not reflect the way people actually drive. impacts may be scaled per kilowatt-hour or per barrel of oil

These high estimated future health costs argue for much tighter equivalent. The idea is that in this way all the impacts can be fairly emission controls than can be achieved with widely used current compared across alternative energy systems, giving full and consistent technologies. How much additional control will be needed? This is information to decision-makers. Such studies first account for one of the critical questions for providing sustainable transport insults over the life cycle of each part of the fuel cycle, such as land systems for the world's cities. Nevertheless, despite large uncertain- used, tonnes of pollution released, long-term waste generated, ties due to the willingness to pay method as well as to basic under- water consumed, and labour required per unit of output. Then most standing of air pollution health effects, it seems safe to conclude that comparative risk assessments try to convert as many insults as the economic value of air pollution abatement is substantial in possible into impacts with common measures, such as deaths, developing and industrialised countries alike. injuries, illnesses, and financial damage costs. Since the occupational Fuel cycle analysis. Supplying modern energy often involves impacts of energy systems can be significant (see the section on processes at a chain of facilities that may be quite physically distinct workplace scale), often both public and worker risks are from one another. These processes are usually referred to as 'fuel determined (box 3.10). cycles', although they rarely rely on any cycling. Comparisons based In addition to the methodological and other problems of fuel- on fuel cycles are useful for organising impact analyses of energy cycle analysis and comparative risk assessment mentioned in box supply and demand. 3.10, there are some fundamental concerns with these kinds of Consider the fuel cycle supporting the operation of an electric comparisons that revolve around this unit of analysis—that is, the appliance. It may involve a coal mine, coal washery, coal train, coal production of a certain amount of energy. For occupational impacts, power plant, and transmission lines, as well as ancillary facilities for example, the number of accidents or illnesses per unit of energy such as coal tailings piles, washery settling lagoons, and power-plant output is just as much a measure of labour intensity as of safety.

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It is difficult to envisage a sustainable energy future in which unprocessed

Indeed, from a societal standpoint, solid fuels remain an important Community scale putting many people to work in low- source of energy for a significant Fuel use is the chief cause of urban air

risk activities is much better than employing pollution, though there is substantial a few people in high-risk activities. But both

fraction of the world's variation among cities in the relative contri-

may look the same in the comparisons. households. butions of vehicles and stationary sources.

In addition, by using energy as the output measure, Diesel-fuelled vehicles, which are more prominent

such analyses reveal little about the impact of such facilities in developing countries, pose a growing challenge to meet-

on overall public health, which uses time as the risk denominator. ing urban health-related pollution guidelines (responsible for about 1

It may be that one way of producing electricity has slightly different percent of the global burden of disease).

public health implications than another, but neither may have much To be sustainable, mean urban air pollution around the world

significance overall or, conversely, neither may be acceptable. will need to be no greater than what is common in rich countries

Entirely different kinds of analyses are needed to evaluate these today—for example, less than 30 micrograms per cubic metre of

kinds of questions. PM10. An additional requirement for sustainability is that urban ambient ozone levels not rise as vehicle fleets grow. Sufficiently

Implications for the future clean power generation by fossil and nuclear sources is technically

As noted in the introduction, there has not been space in this chapter feasible, although costs are uncertain (see chapter 8). Similarly,

to present what is known and suspected about the many environmental hybrid vehicles are substantially cleaner than current types (see box

insults of different energy systems and the resulting impacts on 3.5). Achieving sustainability, however, will probably require mov-

ecosystems and human health. But we have addressed a large fraction ing most of the world's fleet to fuel cells or comparably clean sys-

of the most important ones. The central task of this report is to tems by the middle of the 21st century.

outline the main characteristics of a sustainable energy future. Thus

it is appropriate to examine the main categories of insults and Regional scale

impacts discussed in this chapter to see what requirements they The problem of regional atmospheric emissions will not go away

impose on such a future. quickly. The increasing demand for energy, especially in developing countries, will put heavy pressure on cheap and easily obtainable

Household scale fossil fuels such as coal and oil. Prospects for constraining increases
 About half of the world's households use solid fuels (biomass and in regional emissions are better
 for some pollutants and source
 coal) for cooking and heating in simple devices that produce large types. An ambitious goal for
 sulphur dioxide emissions, for example,
 amounts of air pollution. Because the pollution is released at the would be a 50 percent reduction
 world-wide by 2050. This goal
 times and places where people spend time, the health impact is could be achieved by reducing the
 sulphur content of fossil fuels
 high, accounting for 4–5 percent of the global burden of disease.
 and using emission controls on new, large power plants and industrial
 The chief ecosystem impact relates to charcoal production and
 facilities. Switching to natural gas in developing countries would
 urban fuelwood harvesting, which puts pressure on forests near cities
 also considerably aid the achievement of this goal. Increases in
 and may account for a few percentage points of global deforestation.
 sulphur dioxide emissions in the developing world will be offset by
 It is difficult to envisage a sustainable energy future in which
 legislatively driven reductions in industrialised countries.
 unprocessed solid fuels remain an important source of energy for a
 Nitrogen oxide emissions are a bigger problem. The expansion
 significant fraction of the world's households. Gases, liquids, and
 electricity are the main clean alternatives. Although today these of transportation systems in
 developing countries will add to the
 alternatives mostly derive from fossil fuels, this need not be the case nitrogen oxide burden from
 industrial production and power
 in the future. In the future these alternatives may be made from generation. Moreover, nitrogen
 oxides are much harder to control
 renewable biomass fuels such as wood and crop residues (see than sulphur dioxide. An ambitious
 goal would be to stabilise
 chapters 7, 8, and 10). Indeed, a further criterion for sustainable nitrogen oxide emissions at
 current levels by 2050. Only a major
 energy is that any biomass harvested to make household fuels shift away from fossil fuels in all
 parts of the world or a shift to
 should be done on a renewable basis to ease pressure on forests alternative energy carriers such
 as hydrogen derived from fossil
 and other natural ecosystems. fuels will enable this goal to be achieved.
 Carbon monoxide emissions will likely fall significantly as
 Workplace scale developing countries move away from biofuels and as automobiles
 The harvesting of solid fuels (biomass, coal, uranium) creates the become more efficient
 world-wide. Emissions of volatile organic
 highest risks per energy worker and the largest impacts on the energy compounds from energy
 sources will likely be reduced, but large

workforce world-wide. Risks to coal miners, for example, are many increases can be expected from non-energy sources, particularly as times those for the average industrial worker. To be sustainable, the commercial and residential use of solvents increases in the average miner risks will probably have to be lowered to those in the developing world. Holding global emissions of volatile organic compounds in the safest mines today. pounds to a 20–50 percent increase by 2050 will be a challenge.

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BOX 3.10. COMPARATIVE RISK ASSESSMENT USING FUEL-CYCLE ANALYSIS

A number of concerns drive the need to assess economic parameters such as the discount rate, the nature of its production and use, the environmental and human health damage rate, and the treatment of accident scenarios is extremely site-specific, and its associated costs are associated with electricity production. These for which no actuarial data exist (such as damages and benefits depend on other local factors) include informing utility planning decisions in expected and worst case). activities such as agriculture (Lazarus and others, 1995). terms of total social costs, enlightening cost- A fuel-cycle analysis typically results in a 1995). Similarly, hydroelectric utilities elude concise benefit analyses of pollution mitigation technologies, list of impacts per unit of output in the form of generalisation due to the wide range of sophistication in facilitating formulation of regulatory procedures, premature deaths, ecosystem damage, global warming among technologies and the site-specificity of and delineating the secondary benefits of reducing warming, and the like. To provide a common ecological and human health risks. Fuel-cycle analyses of greenhouse gas emissions. Attempts to quantify metric for comparison, many studies then analyses yield widely disparate conclusions for damages incurred by electricity generation attempt to monetise these impacts. This solar thermal and dispersed photovoltaic technologies date to the 1970s and are known process introduces substantially more gies. Some studies, particularly those focusing as fuel-cycle analyses or comparative risk uncertainty and controversy into the process. on operation of energy systems, suggest that assessments. Most fuel-cycle analyses have Fuel-cycle analyses have typically generated these solar technologies confer negligible human costs. Most have been undertaken in industrialised countries, 'total cost' figures in terms of m\$ (\$0.001) or health and ecological risks (Rabl and Spadaro, 1995). but a few have considered electricity production mECU per kilowatt-hour, with recent (post-1990) (2000). Other studies assert that the occupational health and safety risks in developing countries (such as Lazarus and European and U.S. estimates ranging from risks and short-term environmental damage to others, 1995). 0.016–80 m\$ per kilowatt-hour for coal and from associated with solar technologies can exceed

Fuel-cycle analyses attempt to account for all 0.002–23 m\$ per kilowatt-hour for nuclear (Rabl those of conventional electricity generation damages caused by physical and chemical and Spadaro, 2000). Given the four orders of methods (Hallenbeck, 1995; Bezdek, 1993).

processes and activities undertaken to generate magnitude spanned by these fuel chain damage Pitfalls associated with fuel-cycle analysis

electricity from a specific fuel or resource, from costs, it is clear that they are sensitive to the include the use of poorly defined or inconsistent

fuel acquisition to waste disposal in a steady- particular designs and metrics of the studies. study boundaries, confusion of average and

state operations approach, or from construction Accordingly, interpretation of studies' results marginal effects, underestimation of the

to decommissioning in a facility lifetime approach. requires extensive supplementary information to uncertainty associated with quantification of

Because different insults exert their impacts illuminate, for example, the specific impacts damages, neglect or inadequate treatment

over different temporal and spatial scales, the assessed and the study-specific approaches to of environmental stochasticity, and focus on

geographic extent and time horizon of an analysis valuation of life, valuation of non-fatal outcomes, what is easily quantified rather than on what

must be carefully defined to cover all significant weighting of public and occupational health is actually significant (Koomey, 1990).

impacts. Contemporary fuel-cycle analyses outcomes, and discount rates. Thus fuel-cycle A number of outstanding issues remain for

generally follow a damage function or impact analysis damage costs cannot stand alone and streamlining approaches to fuel-cycle analysis.

pathway approach whereby dominant impacts are difficult to compare, despite the fact that These issues include identifying the functional

are identified; stresses (incremental population they may superficially appear to be based on relations and key parameters defining uncertainty,

exposures to air pollution, occupational hazards, comparable metrics (dollars per kilowatt-hour). the variation in damages with key parameters,

transportation risks) are quantified; stresses are Although comparative risk assessment using the degree of accuracy and resolution with

translated to impacts, typically through exposure- fuel-cycle analysis cannot yet indicate unambiguous regard to atmospheric modelling and receptor

response functions or actuarial data; and impacts preferences or even readily allow for comparisons distribution (needed to capture the site-specificity

are quantified and aggregated in terms of the between studies, recent studies suggest that of impacts), and the magnitude of error incurred

study's chosen metric (ORNL and RFF, 1992; Curtiss public health and occupational health effects by using 'typical' average values rather than

and Rabl, 1996). The study design stage entails dominate the externalities associated with the detailed, site-specific data (Curtiss and Rabl,

a number of choices on impacts to be considered nuclear fuel chain. Health effects and global (1996). In addition, metrics used in fuel-cycle

(public and occupational health, ecological damage, warming dominate conventional coal technologies analysis to deal with incommensurate impacts—

agricultural losses, material corrosion, visual (Rabl and Spadaro, 2000). Biomass as an energy such as the soiling of buildings, crop damage,

amenity), temporal and spatial assessment resource is less easily characterised even in and human morbidity and mortality—are not

boundaries, models and hypotheses for analysis, terms of dominant impacts because it is highly uniform between studies.

The faster that clean and efficient vehicles and fuels can replace current agricultural production. vehicles and fuels, the greater will be the reduction in emissions of Large dams will continue to provide potential for significant benefits

nitrogen oxides, carbon monoxide, and volatile organic compounds. and severe environmental impacts, depending on their location and

It seems that acid deposition world-wide will increasingly become design. For sustainability, much better evaluation will be needed to

a nitrogen oxide (and possibly an ammonia) problem rather than a maximise the benefits and minimise the environmental impacts.

sulphur dioxide problem. On balance, reductions in acidification in

Europe and North America are likely to continue, but hotspots of Global scale

damage in the developing world (such as southwestern China) may Energy systems generate two-thirds of human-caused greenhouse

persist for years and worsen. Regional ozone will increasingly be the gases. Thus energy use is the human activity most closely linked to

biggest problem because of the expected difficulties in mitigating the potential for climate change. Climate change is feared to entail

emissions of nitrogen oxides and volatile organic compounds, significant direct impacts on human health as well as on Earth's

the two main precursors of ozone. Only small improvements in ecosystems. As noted, there has been a tendency for environmental

regional ozone levels may occur in North America and Europe in problems at the local level to be solved partly by pushing off the

the next 10–20 years. And considerable deterioration is likely in impacts to larger scales.

Greenhouse gases and their potential for

Asia, Africa, and Latin America, endangering human health and global climate change represent the final and, in many ways, most

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A sampling of results from fuel-cycle tude than those associated with fission tech- the technologies assessed—for example, biomass

analyses, which have been used in a variety nologies and comparable to those for fusion.

configurations include a wood steam boiler,

of contexts, follows: In the public health domain, with central esti- an atmospheric fluidised bed combustor, and

- An investigation of externalities of electricity makes of about 2 fatalities per GWy(e), coal an integrated gasifier combined-cycle turbine.

production from biomass and coal in the and oil appear to confer greater delayed mor- But methodological issues and assumptions

Netherlands suggests that while average fatality burdens by a factor of two (relative to associated with activities outside the generation

private costs for the biomass strategy photovoltaic systems) to three or four orders stage account for a large portion of the

assessed are projected to be about twice of magnitude (relative to wind, hydroelectric, variability. For example, one study credits

those for coal in 2005, internalisation of

external damages and benefits would yield and nuclear technologies). While acute risks product heat from cogeneration cycles for

about equal social costs. The most important associated with renewable energy technolo-displacing greenhouse gases from gas

distinguishing factors between coal and gases are highly uncertain, this study places heating systems (Fritzsche, 1992). In this

biomass are differences in carbon dioxide them as comparable to or somewhat higher framework the greenhouse gas intensity of

emissions and indirect economic effects such than those associated with fuel-intensive biomass can become negative, and that of

as employment (Faaij and others, 1998). combustion technologies, at 0.1–1.0 fatalities natural gas fuel cycles can be reduced 50

- A comparison of fuel-intensive combustion- per GWy(e) (Fritzsche, 1989). percent below the next lowest estimate.

based utilities (coal, oil, gas, and biomass), • A study by the Stockholm Environmental Among fossil fuels, the greenhouse gas

selected renewable energy technologies (solar Institute suggests that in terms of greenhouse intensity of natural gas is most variable,

thermal, photovoltaic, wind, and hydroelectric), gas emissions, natural gas is preferred to primarily due to different assumptions about

and nuclear technologies (light water reactor, residual fuel for electricity generation in methane emissions during drilling, processing,

fast breeder reactor, and hypothetical fusion Venezuela, even under the assumption of and transport. For non-fossil fuels, estimates

reactor) suggests that coal inflicts the greatest relatively high methane emissions through generally span at least an order of magnitude,

delayed occupational health burden (such as natural gas system losses. In this context the primarily because of the sensitivity of these

disease), with a central estimate of 0.1 fatalities global warming potential per kilowatt-hour of cycles to assumptions on the operation life of

per gigawatt-year of electricity and an upper natural gas electricity generation is projected the facility and the greenhouse gas intensity

estimate of about 3 fatalities per GWy(e). to be 12–27 percent lower than that associated of the electric and manufacturing sectors on

Acute occupational risks (such as accidents) with residual fuel (Lazarus and others, 1995). which equipment production depends. In

posed by combustion technologies are purported • Since the early 1990s several studies have addition, the hydroelectric cycle's greenhouse

to be marginally less than those associated tried to quantify greenhouse gas emissions gas intensity is sensitive to the area of

with renewable energy technologies—with associated with different fuel cycles (see land flooded and, for projects with multiple

central estimates on the order of 1 fatality per table below). Some of the variability arises generating units per reservoir, the boundary

GWy(e)—but greater by an order of magni- from the different conversion efficiencies of of the system considered.

Greenhouse gas emission intensities for selected fuels (grams of carbon dioxide equivalent per kilowatt-hour)

Conventional Advanced

coal coal Oil Gas Nuclear Biomass Photovoltaic Hydroelectric Wind

960-1,300 800-860 690-870 460-1,230 a 9-100 37-166 a 30-150 2-410 11-75

Note: These estimates encompass a range of technologies and countries as described in Pearce and Bann, 1992; Fritsche, 1992; Yasukawa and others,

1992; ORNL and RFF, 1992–98; Gagnon and van de Vate, 1997; and Rogner and Khan, 1998.

a. Natural gas and biomass fuel cycles were also analysed in cogeneration configurations, with product heat credited for displacing greenhouse gas emissions

from gas heating systems. That approach reduced greenhouse gas emissions to 220 grams of carbon dioxide equivalent per kilowatt-hour for natural gas

and –400 for biomass (Fritsche, 1992). Other cycles could incorporate cogeneration and be analysed in this manner.

challenging of the stages. Although there are promising technologies sea level rise is more sensitive to magnitude, and ecosystem damage

for fossil systems that capture and sequester the greenhouse gases is more sensitive to rate. Perhaps the most worrisome aspect of

resulting from combustion, as well as fossil, nuclear, and solar human-engendered warming, however, is that it threatens to cause

systems releasing no greenhouse gases, their prospects are not warming at rates completely unprecedented in Earth's recent

entirely understood (see chapters 7 and 8). geologic history. The magnitude of potential change is somewhat

It is difficult to define a sustainable level of greenhouse warming less unprecedented. Thus it may be reasonable to establish a

above the natural background. Achieving something akin to the somewhat less stringent definition of sustainability for greenhouse

natural background, on the other hand, will not be possible for many gas emissions—one that calls for stabilising atmospheric levels as

centuries, barring major, unprecedented, and unforeseen technical quickly as possible, recognising that the resulting levels (and their breakthroughs, global catastrophes, or changes in human behaviour. warming) will be considerably higher than the natural background.

What then, might be considered a workable definition of 'sustainable' Achieving stable atmospheric levels during the 21st century will

for the climate change impacts of the world energy system? require bringing human greenhouse gas emissions to annual rates

The coming climate change can be considered in two parts: substantially below those today. Doing so will not be easy. Indeed, it

magnitude (total warming) and rate (annual increase). Some types will require major commitments of resources and political will (see

of impacts are more sensitive to one than the other. For example, the section on the global scale and chapter 9). The longer such

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efforts are delayed, the higher and longer will be the eventual stable 5. PM10 are particles less than 10 microns (millionths of a meter) in size,

which penetrate deeper into the respiratory system than larger particles.

warming level and accompanying impacts.

6. The World Commission on Dams, which began deliberating in 1998,

Reaching emission levels in 2050 below those in 2000 will is publishing its reports in mid-2000 (WCD, 1999). These will include

probably require annual declines in energy intensity of at least 1.4 8-10 case studies examining social, economic, environmental, energy,

percent and in carbon intensity of energy of at least 0.4 percent. financial, managerial, and other aspects plus a database of 150 dams in

different countries.

With the assumptions in table 9.1, even these major accomplishments 7. This section draws heavily on IPCC (1996a, b). The IPCC was

would still allow emissions growth of 0.4 percent a year to 22 percent established by the World Meteorological Organization and United

above 2000 emissions by 2050. With such modest growth, however, Nations Environment Programme in 1988 to assess the scientific,

technological, economic, and social aspects of anthropogenic climate

and 50 years of experience promoting efficient and low-carbon change. Some 2,000 scientists and other specialists from more than

energy sources, it might be possible to achieve emissions below 40 countries served as authors and reviewers of the 17 volumes of

2000 levels within a few years after 2050. exposition and analysis issued by the IPCC through 1996. The IPCC's

first assessment report, completed in late 1990, served as the basis for

the negotiations of the United Nations Framework Convention on

Cross-scale Climate Change, concluded in 1992 and discussed below.

There are important opportunities for 'no regrets' strategies that 8. The mean global surface temperature of the Earth is about 15 degrees Celsius (59 degrees Fahrenheit). Without greenhouse gases, it would achieve benefits at more than one scale. For example, if greenhouse be -18 degrees Celsius (0 degrees Fahrenheit).

gas controls are targeted towards reducing solid fuel use in house- 9. Combustion emits water vapour and carbon dioxide in comparable holds and in other energy systems with large health impacts (such quantities. But the rate of water addition to the global atmosphere by as vehicle fleets), significant improvements can occur at the local, combustion is tens of thousands of times smaller than the rates of addition and removal by evaporation and precipitation. And because the community, and global scales. Fine particles are generated and have added water remains in the atmosphere only a few days, these human impacts at all scales, so control measures will benefit from integrated additions cause at most local effects and no long-term build-up. In approaches. Similarly, the regional impacts from sulphur and nitrogen contrast, the quantity of carbon dioxide added by combustion is only about 10 times smaller than what is added and removed by natural emissions can be reduced in conjunction with control efforts at the photosynthesis and decomposition, and a large part of the anthro-community and global scales. Much additional effort is needed to identify pogenic increment remains in the atmosphere for decades. Thus it has time to become uniformly distributed around the globe, irrespective of environmental control pathways that optimise these multiple benefits. where it was emitted, and to accumulate over long periods (as long as the sum of the natural and anthropogenic addition rates is greater than Conclusion the removal rate).

Impacts other than those discussed in this chapter need to be 10. There is no net accumulation of carbon dioxide in the atmosphere from combustion of wood and other biomass fuels, as long as new considered, particularly in local situations. But if the environmental plant growth replaces what is burned. This is because a growing plant insults and their ecosystem and health impacts focused on here removes from the atmosphere exactly as much carbon dioxide as is released when the plant decomposes or burns. When new growth were controlled as indicated, the world would have moved most of does not replace what is burned or decomposed, as in deforestation, the way towards a sustainable energy system. a net addition of carbon dioxide to the atmosphere results. (See the

Among the other impacts requiring careful consideration are the section on greenhouse gas emissions at the household scale, for a discussion of non-carbon dioxide greenhouse gas emissions from incom-relationships between energy systems and military, political, and plete biomass combustion.)

economic security—the subjects of the next chapter. ■ 11. Feedbacks are phenomena wherein the consequences of a disturbance act back on its cause, making the disturbance either bigger (positive feedback) or smaller (negative feedback) than it started out.

Notes 12. The first and longest-running series of measurements was initiated by Charles Keeling at a monitoring station atop the Mauna Loa volcano

1. Insult is defined here as the physical stressor (such as air pollution) on the island of Hawaii. produced by an energy system. Impact, in contrast, is defined as the potential negative (or positive) outcome (such as respiratory disease or 13. The main such terrestrial ‘sinks’ for atmospheric carbon are in the forest destruction) affecting humanity. As with other useful terms Northern hemisphere (see Houghton, 1996; Fan and others, 1998). (diagnosis, prognosis, pathology) the term insult is borrowed from 14. Particles in the atmosphere exert both cooling and warming effects medicine, where it is defined as “a generic term for any stressful on the Earth’s surface temperature, depending on the characteristics of stimulus, which under normal circumstances does not affect the host the specific particles in terms of absorption and scattering of incoming organism, but may result in morbidity when it occurs in a background of solar and outgoing terrestrial radiation, and on the roles of different pre-existing compromising conditions” (Segen, 1992). It has been used particles in cloud formation. Averaged over the globe and the different in environmental discussions, however, since at least the mid-1970s types of anthropogenic particles, the net effect is cooling. (see Ehrlich and others, 1977).

15. Because particulate matter and its gaseous precursors have much

2. Modern fuels involve extensive fuel cycles with relevant environmental shorter residence times in the atmosphere than any of the major green-impacts and energy efficiencies at several points. The air pollution house gases, its offset of part of the greenhouse effect will shrink in line exposures per meal are still lower than that from solid fuels, however. with declining particulate and precursor emissions.

3. The burden of disease refers to the total healthy life years lost due to 16. This range corresponds to 0.54–1.08 degrees Fahrenheit, but the this risk factor. It is composed of two parts that are added together: life two-significant-figure precision resulting from applying the exact years lost to deaths and life years lost to diseases and injuries weight- conversion (1 degree Celsius = 1.8 degrees Fahrenheit) is illusory. ed by a severity factor. The warming is not uniform, however, being generally greater near the

4. These include the main gaseous pollutants as well as particulates. For poles than near the equator. And because of the complexity of the heat lead emissions, the overall reduction in hazard per vehicle mile was transfer processes of the climatic system, some regions may get colder about 75 percent (US Census Bureau, 1996; USEPA, 1996). even as the globe gets warmer on average.

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17. The most sophisticated models demonstrate the fundamental Bezdek, R. H. 1993. "The Environmental, Health, and Safety

soundness of their representations of global climatic processes by Implications of Solar Energy in Central Station Power Production."

simulating quite accurately the undisturbed climate of the planet in Energy 18 (6): 681–85.

respects such as the variation of geographic patterns of temperature Bolin, B. 1998. "The Kyoto Negotiations on Climate Change: A Science

and precipitation with the changes of the seasons. Perspective." Science 279: 330–31.

18. Temperatures would continue to rise thereafter, even in the cases in Bourke, G. 1988.

"Subduing the Sea's Onslaught." South (July).

which the atmospheric concentrations of greenhouse gases had been

stabilised by then, because of the climate-response lag time caused by Bower, B.T., and others.

1987. "Guide for Analysis for Integrated

Residuals Management." Draft final report. Resources for the

the thermal inertia of the oceans.

Future, Washington, D.C.

19. Non-linear means that small disturbances can have large

BP (British Petroleum). 1998. BP Statistical Review of World Energy

consequences. Forcing is the technical term for an externally imposed 1998. London.

disturbance, such as a change in greenhouse gas concentrations.

An example of a potential positive feedback on global warming from Brauer, M., and J.

Hisham-Hashim. 1998. "Fires in Indonesia: Crisis and

terrestrial ecosystems is that the warming could increase the rate of Reaction." Environmental Science and Technology 9: 404A–407A.

release of greenhouse gases into the atmosphere from decomposition Braunstein, H.M., and others.

1981. Biomass Energy Systems and the

of dead organic matter in forests and swamps. Environment. New York: Pergamon Press.

20. The discussion here is drawn from a treatment prepared by one of Brimblecombe, P. 1987. The Big Smoke: A History of Air Pollution in

the authors (Holdren) for the report of the Panel on International London since Medieval Times. London: Methuen.

Cooperation in Energy Research, Development, Demonstration, and Broecker, W.S. 1997.

"Thermohaline Circulation, the Achilles Heel of Our

Deployment, President's Committee of Advisors on Science and Climate System: Will Man-made CO₂ Upset the Current Balance?"

Technology (PCAST, 1999). Science 278: 1582–88.

21. See IPCC (1996b). Both trends—an increase in rainfall and an Brouwer, I.D. 1994. "Food and Fuel: A Hidden Dimension in Human

increase in the fraction of it occurring in extreme events—have been Nutrition." Diss. Wageningen University, Netherlands.

convincingly documented for the period since 1900 in the United States

- (Karl and Knight, 1998). Burtraw, D., and E. Mansur 1999. "Environmental Effects of SO₂ Trading and Banking." *Environmental Science & Technology* 33 (20): 3489–94.
22. The improvements in agricultural productivity foreseen for some California Energy Commission. 1998. "Electric Vehicles." http://www.energy.ca.gov/afvs/ev/q_a.html
- regions by the IPCC are due partly to carbon dioxide fertilisation of plant growth, partly to increased water availability from increased precipitation, and partly to technological change. Although the IPCC discusses California Environmental Protection Agency. 1999. "Fuel Cell Electric Vehicles Fact Sheet." Air Resources Board Zero-Emission Vehicles
- lematic in a greenhouse gas-warmed world, these possibilities do not Program. http://www.arb.ca.gov/msprog/zevprog/fuelcell/fcell_fs.pdf
- seem to be fully reflected in the productivity projections. Carpenter, L., and others. 1990. "Health Related Selection and Death
23. This section draws heavily on material written by one of the authors Rates in the United Kingdom Atomic Energy Authority Workforce."
- (Holdren) for the report of the Panel on Federal Energy Research and British Journal of Industrial Medicine 47 (4): 248–58.
- Development, President's Committee of Advisors on Science and Carpenter, R.A., and others. 1990. Environmental Risk Assessment:
- Technology (PCAST, 1997). Dealing with Uncertainty in Environmental Impact Assessment. Manila: Asian Development Bank.
24. Annex 1 countries, as defined in the UNFCCC, are OECD countries plus the countries of Eastern Europe and some of those of the former Chameides, W.L., P.S. Kasibhatla, J. Yienger, and H. Levy. 1994.
- Soviet Union (the Baltics, Belarus, Russia, Ukraine). "Growth of Continental-scale Metro-agro-plexes, Regional Ozone Pollution, and World Food Production." *Science* 264: 74–77.
- Charlson, R.J., S.E. Schwartz, J.M. Hales, R.D. Cess, J.A. Coakley, J.E. Hansen, and J. Hoffman. 1992. "Climate Forcing by Anthropogenic References Aerosols." *Science* 255: 423–30.
- Agarwal, B. 1985. Cold Hearths, Barren Slopes. New Delhi: Allied Pubs. Charpak, G., and R.L. Garwin. 1997. Feux Follets et Champignons Nucleaires. Paris: Editions Odile Jacob.
- Akimoto, H., H. Mukai, M. Nishikawa, K. Murano, S. Hatakeyama, C. Liu, M. Buhr, K.J. Hsu, D.A. Jaffe, L. Zhang, R. Honrath, J.T. Merrill, and Collier, M., R.H. Webb, and J.C. Schmidt. 1996. "Dams and Rivers:
- R.E. Newell. 1996. "Long-range Transport of Ozone in the East Asian Primer on the Downstream Effects of Dams." Circular 1126. U.S.
- Pacific Rim Region." *Journal of Geophysical Research* 101: 1999–2010. Geological Survey, Branch of Information Services, Tucson, Ariz., and Denver, Colo.
- Albright, D., and H.A. Feiveson. 1988. "Plutonium Recycling and the Problem
- Crane, M. 1998. "Power Generation and Distribution." In J.M. Stellman,

of Nuclear Proliferation." Annual Review of Energy 13: 239-65. ed., Encyclopaedia of Occupational Health and Safety. 4th ed.

American Honda Motor Company. 1999. "Honda Hybrid To Be Called Geneva: International Labour Organization.

'Insight'—Will Feature (IMA) Integrated Motor Assist System, Curren, T., and A. Nixon. 1992. "Current Issue Review: Acid Rain."

Lightweight Aluminum Body Structure." <http://www.honda2000.com/> Library of Parliament, Research Branch, Ottawa, Canada.

insight/press.html

Curtiss, P.S., and A. Rabl. 1996. "Impacts of Air Pollution: General

Armstrong, J., and R. Menon. 1998. "Mining and Quarrying." In J.M. Relationships and Site Dependence." Atmospheric Environment

Stellman, ed., Encyclopaedia of Occupational Health and Safety. 30: 3331-47.

4th ed. Geneva: International Labour Organization.

Dixon, J.A., and others. 1989. Dams and the Environment: Considerations

Arndt, R.L., G.R. Carmichael, D.G. Streets, and N. Bhatti. 1997. "Sulfur in World Bank Projects. Washington, D.C.: World Bank.

Dioxide Emissions and Sectoral Contributions to Sulfur Deposition Dohan, M.R., and others. 1974. "The Effect of Specific Energy Uses on

in Asia." Atmospheric Environment 31: 1553-72. Air Pollutant Emissions in New York City." Brookhaven National

Arthur, E.A., ed. 1997. "Technology Development Examples Applicable Laboratory, Upton, NY. to Nuclear Energy and Fuel-Cycle Improvement." Report LA-UR- Dunn, S. 1998. "Hydroelectric Power Up Slightly." In Vital Signs. New

97-2989. Los Alamos National Laboratory and Sandia National York: Norton and Co.

Laboratory, Los Alamos, N.M.

Dvorak, A.J., and B.G. Lewis. 1978. "Impacts of Coal-fired Power Plants

Bascom, R., and others. 1996. "Health Effects of Outdoor Air Pollution. on Fish, Wildlife, and Their Habitats." Argonne National Laboratory,

American Journal of Respiratory Critical Care Medicine 153: 477-98. Division of Environmental Impact Studies, Ann Arbor, Mich.

WORLD ENERGY ASSESSMENT: ENERGY AND THE CHALLENGE OF SUSTAINABILITY 105

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Dynesius, M., and C. Nilsson. 1994. "Fragmentation and Flow Gupta, N. 1989. "Health Implications of Nuclear Energy." Economic and

Regulation of River Systems in the Northern Third of the World." Political Weekly 24 (39): 2166.

Science 266 (4): 759-62. Habegger, L.J., and others. 1982. Direct and Indirect Health and Safety

Ehrlich, P.R., and others. 1977. Ecoscience: Population, Resources, Impacts of Electrical Generation Options. Vienna: International

Environment. San Francisco, Calif.: W.H. Freeman. Atomic Energy Agency.

EIA (Energy Information Administration). 1995. Electricity Generation Haefele, W. 1990. "Energy from Nuclear Power." Scientific American 9:

and Environmental Externalities: Case Studies. Washington, D.C.: 136-44.

U.S. Department of Energy. Hallenbeck, W.H. 1995. "Health Impact of a Proposed Waste-to-Energy
Elkins, P. 1996. "How Large a Carbon Tax Is Justified by the Secondary Facility in Illinois." *Bulletin of Environmental Contamination and Benefits of CO2 Abatement?* *Resource and Energy Economics Toxicology* 54 (3): 342-48.
18: 161-87. Hanisch, C. 1999. "Powering Tomorrow's Cars." *Environmental Science*
Environment Canada. 1999. "A Primer on Environmental Citizenship." and *Technology* 33 (11):
458A-62A.
<http://www.ns.ec.gc.ca/aeb/ssd/acid/acidfaq.html> Hayes, P., and K.R. Smith, eds. 1994. *Global Greenhouse Regime: Who Pays?*
Evans, J.S. 1991. "Radiological Health Effects Models for Nuclear London and Tokyo: Earthscan and United Nations University Press.
Power Plant Accident Consequence Analysis: An Update (1990)." Hiraoka, T. 1994. "Nuclear Electricity Generation by Seawater Uranium.
Nuclear Technology 94 (2): 204-12. *Journal of the Atomic Energy Society of Japan* 36(7): 644-45.
Faaij, A., and others. 1998. "Externalities of Biomass Based Electricity Holdren, J.P. 1983. "Nuclear Power and Nuclear Weapons: The
Production Compared with Power Generation from Coal in the Connection Is Dangerous." *Bulletin of the Atomic Scientists* 1.
Netherlands." *Biomass and Bioenergy* 14 (2): 125-47.
———. 1989. "Civilian Nuclear Technologies and Nuclear Weapons
Fearnside, P. 1995. "Hydroelectric Dams in the Brazilian Amazon as Proliferation." In C. Schaerf, B. Holden-Reid, and D. Carlton, eds.,
Sources of 'Greenhouse' Gases. *Environmental Conservation* 22(1). New Technologies and the Arms Race. London: Macmillan.
Florig, H.K. 1997. "China's Air Pollution Risks." *Environmental Science* ———. 1992. "The Transition to Costlier Energy." In L. Schipper and S.
and *Technology* 31(6): 276A-79A. Meyers, eds., *Energy Efficiency and Human Activity: Past Trends*, Foell, W., M. Amann, G. Carmichael, M. Chadwick, J. Hettelingh, L. Future Prospects. Cambridge: Cambridge University Press.
Hordijk, and D. Zhao, eds. 1995. *RAINS-ASIA: An Assessment* Holgate, S.T., and others, eds. 1999. *Air Pollution and Health*. San
Model for Air Pollution in Asia. Washington, D.C.: World Bank. Diego, Calif.: Academic Press.
Ford Motor Company. 1998. "1998 Ford Ranger Electric Vehicle Holland, M. 1996. "Quantifying the Externalities of Fuel Cycle Activities
Specifications." <http://www.ford.com/electricvehicle/rangspecs.html> outside the European Union." Maintenance Note 1 for the ExternE
———. 2000. "Crown Victoria Natural Gas Vehicle Specifications." CORE Project, Energy Technology Support Unit. Didcot, Oxfordshire.
<http://www.ford.com/default.asp?pageid=429> Hong Kong Municipal Government. 1999. "Air Pollution Components,
Fritsche, U. 1992. "TEMIS: A Computerized Tool for Energy and Effects and Causes." Planning, Environment, and Lands Bureau.
Environmental Fuel and Life-Cycle Analysis." Paper presented at
<http://www.plb.gov.hk/cleanair/cne.htm>

an International Energy Agency expert workshop on Life-Cycle Horii, N. 1999. "Evolution of Energy Demand after China's Economic Reform."

Analysis of Energy Systems: Methods and Experience, Paris.

Horne, A. J., and C.R. Goldman. 1994. Limnology. New York: McGraw-Hill.

Fritzsche, A.F. 1989. "The Health Risks of Energy Production." Risk Analysis 9 (4): 565–77. Houghton, J.T., and others, eds. 1994. Climate Change 1994: Radiative Forcing of Climate Change and an Evaluation of the IPCC IS92

Fthenakis, V.M., and others. 1995. "An Assessment of Mercury Emission Scenarios. Cambridge: Cambridge University Press.

Emissions and Health Risks from a Coal-Fired Power Plant." Journal of Hazardous Materials 44 (2–3): 267–83. Huang, M., Z. Wang, D. He, H. Xu, and L. Zhou. 1995. "Modeling Studies on Sulfur Deposition and Transport in East Asia. Water, Air

Funk, K., and A. Rabl. 1999. "Electric Versus Conventional Vehicles: and Soil Pollution 85: 1921–26. Social Costs and Benefits in France." Transport and Environment 4 (6): 397–411. IARC (International Agency for Research on Cancer). 1984. Carcinogenicity Monographs 1984.

Gagnon, L. 1998. "Greenhouse Gas Emissions from Hydro Reservoirs: The Level of Scientific Uncertainty in 1998." Hydro-Quebec, Ichikawa, Y., and S. Fujita. 1995. "An analysis of Wet Deposition of Sulfate Montreal. Using a Trajectory Model for East Asia." Water, Air and Soil Pollution 85: 1927–32.

Gagnon, L., and J.P. van de Vate. 1997. "Greenhouse Gas Emissions from Hydropower." Energy Policy 25 (1): 7–13. ILO (International Labour Organization). 1998. Yearbook of Labour Statistics. Geneva.

General Motors Corporation. 1999. "EV1 Electric: Specs/Pricing." <http://www.gmev.com/specs/specs.htm> IPCC (Intergovernmental Panel on Climate Change). 1990. Climate Change: The IPCC Scientific Assessment. Cambridge: Cambridge

GFDL (Geophysical Fluid Dynamics Laboratory). 1997. Climate Impact University Press. of Quadrupling Atmospheric CO₂. National Oceanic and

Atmospheric Administration, Princeton, N.J. ———. 1995. Climate Change 1994: Radiative Forcing of Climate Change. Cambridge: Cambridge University Press.

Gilinsky, V. 1977. "Plutonium, Proliferation, and Policy." Technology Review 79(4): 58–65. ———. 1996a. Climate Change 1995: Impacts, Adaptations, and Mitigation of Climate Change: Scientific-Technical Analyses.

Gleick, P.H. 1992. "Environmental Consequences of Hydroelectric Cambridge: Cambridge University Press.

Development: The Role of Size and Type". Energy 17:735–47. ———. 1996b. The Science of Climate Change: Summary for

- Global Toyota. 1999. "Future Vehicles: RAV4 EV." <http://www.toyota.com> Policymakers and Technical Summary of the Working Group I
- Gould, J., and T.F. Golob. 1997. "Clean Air Forever? A Longitudinal Report. Cambridge: Cambridge University Press.
- Analysis of Options about Air Pollution and Electric Vehicles." Jaffe, D.A., R.E. Honrath, L. Zhang, H. Akimoto, A. Shimizu, H. Mukai. K. University of California at Irvine, Institute of Transportation Studies. Murano, S. Hatakeyama, and J. Merrill. 1996. Journal of
- Grossman, G., and A. Krueger. 1994. "Economic Growth and the Geophysical Research 101: 2037-48.
- Environment." NBER Working Paper 4634. National Bureau of Jenkins, S.A., and others. 1988. "The Impacts of Dam Building on the
- Economic Research, Cambridge, Mass. California Coastal Zone." California Waterfront Age (September).
- 106 WORLD ENERGY ASSESSMENT: ENERGY AND THE CHALLENGE OF SUSTAINABILITY
- Chapter 3: Energy, the Environment, and Health
- Jennings N.S. 1998. "Mining: An Overview." In J.M. Stellman, ed., Lew, C.S., and others. 1996. "Use of the Pisces Database: Power Plant
- Encyclopaedia of Occupational Health and Safety. 4th ed. Geneva: Aqueous Stream Compositions." Water, Air and Soil Pollution
- International Labour Organization. 90: 113-22.
- Johnson, C.E., and R.G. Derwent. 1996. "Relative Radiative Forcing Lovei, M. 1998. Phasing Out Lead from Gasoline: Worldwide Experience
- Consequences of Global Emissions of Hydrocarbons, Carbon and Policy Implications. World Bank Technical Paper 397.
- Monoxide and Nitrogen Oxide from Human Activities Estimated Washington, D.C. with a Zonally-Averaged Two-Dimensional Model." Climatic Lydersen, E., V. Angell, O. Eilertsen, T. Larssen, J. Mulder, I.P. Muniz,
- Change 34 (3-4): 439-62.
- H.M. Seip, A. Semb, R.D. Vogt, and P. Aagaard. 1997. "Planning of
- Johnson, F.R., and W. H. Desvousges. 1997. "Estimating Stated an Integrated Acidification Study and Survey on Acid Rain Impacts
- Preferences with Rated-pair Data: Environmental, Health, and in China." Norwegian Institute for Water Research Report SNO
- Employment Effects of Energy Programs." Journal of Environmental 3719-97. Oslo, Norway.
- Economics and Management 34 (1): 79-99.
- MacKenzie, J. 1997. Climate Protection and the National Interest: The
- Karl, T.R., and R.W. Knight. 1998. "Secular Trends of Precipitation Links among Climate Change, Air Pollution, and Energy Security.
- Amount, Frequency, and Intensity in the USA." Bulletin of the Washington, D.C.: World Resources Institute.
- American Meteorological Society 79: 231-41.
- Mage, D.R., and O. Zali, eds. 1992. "Motor Vehicle Air Pollution: Public

Kiehl, J.T., and others. 2000. "Radiative Forcing due to Sulfate Aerosols Health Impact and Control Measures." WHO/PEP/92.4. World
from Simulations with the National Center for Atmospheric Health Organization, Geneva.
Research Community Climate Model, Version 3." Journal of
Mann, M.E., R.S., Bradley, and M.K. Hughes. 1998. "Global-scale
Geophysical Research-Atmospheres 105 (D1): 1441-57.
Temperature Patterns and Climate Forcing over the Past Six
Kjellstrom, T. 1994. "Issues in the Developing World." In L. Rosenstock Centuries." Nature 392:
779-87.
and M. Cullen, eds., Textbook of Clinical Occupational and
———. 1999. "Northern Hemisphere Temperatures during the Past
Environmental Medicine. Philadelphia, Pa.: W.B. Saunders and Co.
Millennium: Inferences, Uncertainties, and Limitations."
Kjellstrom, T., J.P. Koplan, and R.B. Rothenberg. 1992. "Current and Geophysical Research Letters
(15 March).
Future Determinants of Adult Ill Health. In R. Feachem and others,
eds., The Health of Adults in the Developing World. Oxford: Oxford Mark, J., J..M. Ohi, and D.V.
Hudson, Jr. 1994. "Fuel Savings and
University Press. Emissions Reductions from Light Duty Fuel Cell Vehicles."
NREL/TP-463-6157. National Renewable Energy Laboratory,
Knutson, T.R., R. E. Tuleya, and Y. Kurihara. 1998. "Simulated Increase Golden, Colo.
of Hurricane Intensities in a CO2-warmed Climate." Science
279: 1018-20. Matsumoto, S., A. Inaba, and Y. Yanagisawa. 1997. "Technology
Assessment of Alternative Fuels by CO2 Fixation Use in Passenger
Koomey, J. 1990. "Comparative Analysis of Monetary Estimates of Cars." Energy Conversion and
Management 38: S455-60.
External Environmental Costs Associated with Combustion of
Fossil Fuels." University of California, Lawrence Berkeley McCully, P. 1996. Silenced Rivers: The
Ecology and Politics of Large Dams.
Laboratory, Applied Science Division. London: Zed Books.
Kraus, R.S. 1998. "Oil Exploration and Drilling." In J.M. Stellman, ed., McGranahan, G., and others.
2000. Citizens at Risk: From Urban Sanitation
Encyclopaedia of Occupational Health and Safety. 4th ed. Geneva: to Sustainable Cities. London:
Earthscan.
International Labour Organization. McManus, N. 1998. "Hydroelectric Power Generation." In J.M.
Stellman,
Krebs, M. 1999. "Promise, and Pitfalls, of Fuel Cells." The New York ed., Encyclopaedia of
Occupational Health and Safety. 4th ed.
Times, 2 April, F1. Geneva: International Labour Organization.
Kremer, J. 2000. "MOBILE6 and Compressed Natural Gas Vehicles." McMichael, A.J., E. Haines, R.
Slooff, and S. Kovats, eds. 1996.
U.S. Environmental Protection Agency, Office of Mobile Sources, "Climate Change and Human
Health." Assessment Prepared by a

Washington, D.C. Task Group on behalf of the World Health Organization, World Meteorological Organization, and United Nations Environment

Krewitt, W., T. Heck, A. Truckenmueller, and R. Friedrich. 1999. Programme. Geneva, Switzerland.

"Environmental Damage Costs from Fossil Electricity Generation in Germany and Europe." *Energy Policy* 27: 173–83.

Mendler, C. 1997. "Taking a New Look at Hybrid Electric Vehicle Efficiency." *Automotive Engineering* 105 (2): 67–70.

Krewitt, W., and others. 1998. "Health Risks of Energy Systems." *Risk Analysis* 18 (4): 377–83.

Mills, W.B., and others. 1998. "Predictions of Potential Human Health and Ecological Risks from Power Plant Discharges of Total

Lal, M, K. Gauri, and G.C. Holdren. 1981. "Pollutant Effects on Stone Residual Chlorine and Chloroform into Rivers." *Environmental*

Monuments." *Environmental Science and Technology* 15: 386–90.

Science & Technology 32 (14): 2162–71.

Lal, M., U. Cubasch, R. Voss, and J. Waszkewitz. 1995. "Effect of Moffit, A. 1998. "Iron and Steel." In J.M. Stellman, ed., *Encyclopaedia of*

Transient Increase in Greenhouse Gases and Sulphate Aerosols on Occupational Health and Safety. 4th ed. Geneva: International

Monsoon Climate." *Current Science* 69: 752–63.

Labour Organization.

Lazarus, M., and others. 1995. The SEI/UNEP Fuel Chain Project: Morison, W.G. 1998. "Nuclear Power Generation." In J.M. Stellman, ed.,

Methods, Issues and Case Studies in Developing Countries. *Encyclopaedia of Occupational Health and Safety*. 4th ed. Geneva:

Boston, Mass.: Stockholm Environment Institute–Boston. International Labour Organization.

Leach, Gerald. 2000. Personal communication. Morris, S.C., and others. 1980. "Health and Environmental Effects of the

Leigh, J., P. Macaskill, C. Corvalan, E. Kuosma, and J. Mandryk. 1996. *National Energy Plan: A Critical Review of Some Selected Issues.*"

"Global Burden of Disease and Injury Due to Occupational Brookhaven National Laboratory, Upton, NY.

Factors." WHO/EHG/96.20. World Health Organization, Geneva.

Moskowits, P.D., M.D. Rowe, and others. 1977. "Workshop Report on

Lelieveld, J., and others. 1997. "Terrestrial Sources and Distribution of Quantifying Environmental Damage from Energy Activities."

Atmospheric Sulphur." *Philosophical Transactions of the Royal Brookhaven National Laboratory, National Center for Analysis of*

Society of London Series B-Biological Sciences 352 (1350): 149–58.

Energy Systems, Office of Environmental Policy, Upton, NY.

Lerman, Y., A. Finkelstein, and others. 1990. "Asbestos Related Health MRC (Medical Research Council). 1996. *Electrification and Health: The*

Hazards among Power Plant Workers." *British Journal of Industrial Interface between Energy, Development, and Public Health.*

Medicine 47 (4): 281–82.

Community Health Research Group. Tygerberg, South Africa.

WORLD ENERGY ASSESSMENT: ENERGY AND THE CHALLENGE OF SUSTAINABILITY 107

Chapter 3: Energy, the Environment, and Health

- Murray, C., and A. Lopez, eds. 1996. Global Burden of Disease. World OTA (Office of Technology Assessment). 1992. Fueling Development: Health Organization, World Bank, and Harvard School of Public Energy Technologies for Developing Countries. OTA-E516.
- Health. Cambridge, Mass.: Harvard University Press. Washington, D.C.: U.S. Government Printing Office.
- Nakićenović, N., A. Grübler, and A. McDonald, eds. 1998. Global Energy Page, W.P., and others. 1982. "Direct Monetary Losses to the Perspectives. Cambridge: Cambridge University Press. Agricultural Sector from Coal-fired Electricity Generation and NAPAP (National Acid Precipitation Assessment Program). 1998. Background: A Regional Analysis." Energy 7 (9): 759–68.
- Biennial Report to Congress: An Integrated Assessment. National PCAST (President's Committee of Advisors on Science and Science and Technology Council, Committee on Environment and Technology). 1997. "Federal Energy Research and Development for Natural Resources. Washington, D.C. <http://www.nnic.noaa.gov/> the Challenges of the 21st Century." Office of Science and CENR/NAPAP/NAPAP_96.htm Technology Policy, Panel on Energy Research and Development, Nash, L. 1993. "Water Quality and Health." In P.H. Gleick, ed., Water in Washington, D.C. Crisis: A Guide to the World's Fresh Water Resources. New York: ———. 1999. "Powerful Partnerships: The Federal Role in International Oxford University Press. Cooperation on Energy Innovation." Office of Science and National Academy of Sciences. 1994. Management and Disposition of Excess Technology Policy, Panel on International Cooperation in Energy Weapons Plutonium. Washington, D.C.: National Academy Press. Research, Development, Demonstration, and Deployment, Washington, D.C.
- . 1995. Management and Disposition of Excess Weapons Plutonium: Reactor-Related Options. Washington, D.C.: National Pearce, D., and C. Bann. 1992. "The Social Costs of Electricity Academy Press. Generation in the U.K." Paper presented at an International Energy National Fuel Cell Research Center. 1999. "What Is a Fuel Cell?" Agency expert workshop on Life-Cycle Analysis of Energy <http://www.nfrcr.uci.edu/fcinfo/what.htm> Systems: Methods and Experience, Paris.
- National Research Council. 1992. Nuclear Power: Technical and Penner, J.E., S.J. Ghan, and J.J. Walton. 1991. "The Role of Biomass Institutional Options for the Future. Committee on Future Nuclear Burning in the Budget and Cycle of Carbonaceous Soot Aerosols Power Development. Washington, D.C.: National Academy Press. and Their Climate Impact." In J.S. Levine, ed., Global Biomass Burning: Atmospheric, Climatic, and Biospheric Implications.
- . 1998. Research Priorities for Airborne Particulate Matter. Cambridge, Mass.: MIT Press.

National Academy of Sciences, Washington, D.C.

Peterson, T.C., K.P. Gallo, J. Lawrimore, T.W. Owen, A. Huang, and D.A.

Natural Gas Vehicle Coalition. 2000a. "Questions and Answers about McKittrick. 1999. "Global Rural Temperature Trends." *Geophysical*

Natural Gas Vehicles." <http://www.ngvc.org/qa.html> *Research Letters* 26: 329-32.

———. 2000b. "The NGV Message: Environmental and Health Benefits." Petrini, C., and P. Vecchia. 1993. "Health Problems Related to High-

<http://www.ngvc.org/ngvmess.html>

Power Transmission Lines [in Italian]." *Aei Automazione Energia*

Neil, D. 1999. "Pump This." *Car and Driver* 44 (10): 110-16. *Informazione* 80 (4): 430-33.

Nriagu, J.O. 1989. "A Global Assessment of Natural Sources of Piccot, S.D., J.J. Watson, and J.W. Jones. 1992. "A Global Inventory of

Atmospheric Trace Metals." *Nature* 338 (6210): 47-49. Volatile Organic Compound Emissions from Anthropogenic

———. 1990. "Global Metal Pollution: Poisoning the Biosphere." *Sources." Journal of Geophysical Research* 97: 9897-9912.

Environment 32 (7): 6-16. Poschen, P. 1998. "Forestry." In J.M. Stellman, ed., *Encyclopaedia of Office of Science and Technology Policy*. 2000. White House Initiative Occupational Health and Safety. 4th ed. Geneva: International

on Global Climate Change, Washington D.C . Labour Organization.

www.whitehouse.gov/Initiatives/Climate/greenhouse.html Pottinger, I. 1997. "The Environmental Impacts of Large Dams."

Olivier, J.G.J., A.F. Bouwman, K.W. Van der Hoek, and J.J.M. Berdowski. International Rivers Network. <http://www.irn.org/basics/impacts.shtml>

1998. "Global Air Emission Inventories for Anthropogenic Sources of Rabl, A. 1998. "Mortality Risks of Air Pollution: The Role of Exposure-

NOx, NH3, and N2O in 1990." *Environmental Pollution* 102: 135-48. response Functions." *Journal of Hazardous Materials* 61 (1-3): 91-98.

Olivier, J.G.J., A.F. Bouwman, C.W.M. van der Maas, J.J.M. Berdowski, Rabl, A. and J.V. Spadaro. 1999. "Damages and Costs of Air Pollution: An

C. Veldt, J.P.J. Bloos, A.J.H. Visschedijk, P.Y.J. Zandveld, and J.L. Analysis of Uncertainties." *Environment International* 25 (1): 29-46.

Haverlag. 1996. "Description of EDGAR Version 2.0." RIVM Report

771060 002. Bilthoven, the Netherlands. ———. 2000. "Public Health Impact of Air Pollution and Implications for the

Energy System." *Annual Review of Energy and the Environment* 25.

Oppenheimer, M. 1998. "Global Warming and the Stability of the West

Antarctic Ice Sheet." *Nature* 393: 325-32. Rabl, A., and others. 1998. "Health Risks of Air Pollution from Incinerators:

A Perspective." *Waste Management & Research* 16 (4): 365-88.

ORNL and RFF (Oak Ridge National Laboratory and Resources for the

Future). 1992. "U.S.-EC Fuel Cycle Study: Background Document Reddy, A.K.N., R.H. Williams, and T.B. Johansson. 1997. *Energy after*

to the Approach and Issues.” Oak Ridge, Tenn. Rio: Prospects and Challenges. New York: United Nations

Development Programme.

———. 1994a. Estimating Externalities of Coal Fuel Cycles. Oak Ridge, Tenn.

———. 1994b. Estimating Externalities of Hydroelectric Fuel Cycles. Repetto, R., and D. Austin. 1997. The Costs of Climate Protection: A Guide

Oak Ridge, Tenn. for the Perplexed. Washington, D.C.: World Resources Institute.

———. 1994c. Fuel Cycle Externalities: Analytical Methods and Issues. Rogner, H.-H. and A. Khan. 1998. “Comparing Energy Options: The

Oak Ridge, Tenn. Inter-agency DECADES Project.” IAEA Bulletin 40 (1).

———. 1995. Estimating Externalities of Nuclear Fuel Cycles. Oak Ross, M., R. Goodwin, R. Watkins, M.Q. Wang, and T. Wenzel. 1995.

Ridge, Tenn. “Real-World Emissions from Model Year 1993, 2000, and 2010

Passenger Cars.” Report prepared for Energy Foundation and U.S.

———. 1996. Estimating Externalities of Oil Fuel Cycles. Oak Ridge, Tenn. Department of Energy through Lawrence Berkeley Laboratory and

———. 1998a. Estimating Externalities of Biomass Fuel Cycles. Oak Oak Ridge National Laboratory. American Council for an Energy-

Ridge, Tenn. Efficient Economy, Berkeley, Calif.

———. 1998b. Estimating Externalities of Natural Gas Fuel Cycles. Oak Rowe, M.D. 1981. “Human Exposure to Particulate Emissions from

Ridge, Tenn. Power Plants.” Brookhaven National Laboratory, Upton, N.Y.

Osepchuk, J.M. 1996. “Health and Safety Issues for Microwave Power Rubin, E.S. 1999. “Toxic Releases from Power Plants.” Environmental

Transmission.” Solar Energy 56 (1): 53-60. Science and Technology 33 (18): 3062-67.

108 WORLD ENERGY ASSESSMENT: ENERGY AND THE CHALLENGE OF SUSTAINABILITY

Chapter 3: Energy, the Environment, and Health

Rudd, J., and others. 1993. “Are Hydroelectric Reservoirs Significant ———. 2000b. “Greenhouse Implications of Household Stoves: An

Sources of Greenhouse Gases?” Ambio 22 (4). Analysis for India.” Annual Review of Energy and Environment

Saez, R.M., and others. 1998. “Assessment of the Externalities of 25: 741-63.

Biomass Energy, and a Comparison of Its Full Costs with Coal.” Smithsonian Institution. 1996.

“Ocean Planet: Oil Pollution.” <http://seawifs>.

Biomass and Bioenergy 14 (5/6): 469-78.

gsfc.nasa.gov/OCEAN_PLANET/HTML/peril_oil_pollution.html

Saksena, S., and V. Dayal. 1997. “Total Exposures As a Basis for the Spadaro, J.V., and A. Rabl. 1998. “External Costs of Energy: Application

Economic Valuation of Air Pollution in India.” Energy Environment of the ExterneE Methodology in France.” ARMINES (Ecole des

Monitor 13 (2): 93-102. Mines), Paris.

Schwela, D. 1995. “Public Health Implications of Urban Air Pollution in Spadaro, J.V., and others.

1998. “External Costs of Air Pollution: Case

Developing Countries." Paper presented at the Tenth World Clean Study and Results for Transport between Paris and Lyon."

Air Congress, 28 May–2 June, Erjos, Finland. *International Journal of Vehicle Design* 20 (1–4): 274–82.

Schwela, D., and O. Zali, eds. 1999. *Urban Traffic Pollution*. New York: Spinrad, B. 1983. "Nuclear Power and Nuclear Weapons: The E & FN Spon. Connection Is Tenuous." *Bulletin of the Atomic Scientists* 2.

Segen, J. C. 1992. *The Dictionary of Modern Medicine*. Park Ridge, Steinbugler, M.M., and R.H. Williams. 1998. "Beyond Combustion: Fuel N.J.: Parthenon. Cell Cars for the 21st Century." *Forum for Applied Research and Public Policy* 13 (4): 102–07.

Seigneur, C., and others. 1996. "Multipathway Health Risk Assessment of Power Plant Water Discharges." *Water Air and Soil Pollution* Stellman, J.M., ed. 1998. *Encyclopaedia of Occupational Health and* 90 (1–2): 55–64. Safety. 4th ed. Geneva: International Labour Organization.

Seip, H., D. Zhao, J. Xiong, D. Zhao, T. Larssen, B. Liao, and R. Vogt. Streets, D.G., G.R. Carmichael, M. Amann, and R.L. Arndt. 1999. 1995. "Acidic Deposition and Its Effects in Southwestern China." "Energy Consumption and Acid Deposition in Northeast Asia." *Water, Air and Soil Pollution* 85: 2301–06. *Ambio* 28: 135–43.

Shiklomanov, I. 1998. *Assessment of Water Resources and Water Sweet*, W. 1984. "The Nuclear Age: Power, Proliferation, and the Arms Availability in the World. Report for the Comprehensive Assessment Race." *Congressional Quarterly*, Washington, D.C.

of the Fresh Water Resources of the World, United Nations, New York. Thornton, E.W. 1994. "Health Risks of Energy Saving." *Nature* 372

Sims, J. 1994. *Women, Health and Environment: An Anthology*. (6504): 327.

WHO/EHG/94.11. Geneva: World Health Organization. UN (United Nations). 1995. *Framework Convention on Climate Change*. United Nations Conference of the Parties, First Session, Berlin, Germany.

Smith, D.R., and A.R. Flegal. 1995. "Lead in the Biosphere: Recent Trends." *Ambio* 24 (1): 21–23. UNECE (United Nations Economic Commission for Europe). 1996. "The State of Transboundary Air Pollution." *Air Pollution Studies* 12. Geneva.

Smith, K.R. 1990. "Risk Transition and Global Warming." *Journal of Energy Engineering-Asce* 116 (3): 178–88. UNEP (United Nations Environment Programme). 1998. *Protecting Our Planet, Securing Our Future*. United Nations Environment

———. 1991a. "Allocating Responsibility for Global Warming: The Programme, U.S. National Aeronautics and Space Administration, Natural Debt Index." *Ambio* 20 (2): 95–96. and World Bank.

———. 1991b. "Managing the Risk Transition." *Toxicology and Industrial* UNEP and WHO (United Nations Environment Programme and World Health 7 (5–6): 319–27. Health Organization). 1992. *Urban Air Pollution in Megacities of the*

———. 1993. "Fuel Combustion, Air Pollution Exposure, and Health: World. Blackwell Publishers. The Situation in Developing Countries." Annual Review of Energy USAEC (U.S. Atomic Energy Commission). 1974. "Comparative Risk- and Environment 18: 529-66. Cost-Benefit Study of Alternative Sources of Electrical Energy." ———. 1997. "Development, Health, and the Environmental Risk Office of Energy Systems Analysis, Washington, D.C. Transition." In G. Shahi, B.S. Levy, A. Binger, and others, eds., U.S. Census Bureau. 1996. Statistical Abstract of the US: National Data International Perspectives in Environment, Development, and Book. Washington D.C.: U.S. Department of Commerce. Health. New York: Springer Publications. USDOE (U.S. Department of Energy). 1980. Nuclear Proliferation and ———. 1998. The National Burden of Disease from Indoor Air Pollution Civilian Nuclear Power: Report of the Non-Proliferation Alternative in India. Mumbai: Indira Gandhi Institute of Development Research. Systems Assessment Program. Washington, D.C.: U.S. Government ———. 2000. "National Burden of Disease in India from Indoor Air Printing Office. Pollution." Proceedings of the National Academy of Sciences 97. ———. 1995. "Executive Summary: Encouraging the Purchase and Use Smith, K.R., and S. Akbar. 1999. "Health-damaging Air Pollution: A of Electric Motor Vehicles." Office of Transportation Technologies, Matter of Scale." In G. McGranahan and G. Murray, eds., Health Washington, D.C. and Air Pollution in Rapidly Developing Countries. York, U.K.: ———. 1997. Final Nonproliferation and Arms Control Assessment of Stockholm Environment Institute of York. Weapons-Usable Fissile Material Storage and Excess Plutonium Smith, K.R. J.M. Samet, I. Romieu, and N. Bruce. 2000. "Indoor Air Disposition Alternatives. Washington, D.C. Pollution in Developing Countries and ALRI in Children." Thorax 55: US Energy Information Administration. 2000. International Energy 518-32. Annual 1998. US Department of Energy: Washington D.C. Smith, K.R., J. Weyant, and J.P. Holdren. 1975. "Evaluation of USEPA (U.S. Environmental Protection Agency). 1994. "Motor Vehicles Conventional Power Systems." University of California at Berkeley, and the 1990 Clean Air Act." Office of Mobile Sources, Washington, D.C. Energy and Resources Program. ———. 1996. National Air Pollution Emission Trends: 1900-1995. Smith, K.R., C. Corvalan and T. Kjellstrom. 1999. "How Much Global Ill- Washington, D.C. health Is Attributable to Environmental Factors?" Epidemiology 10. ———. 1997. "Regulatory Impact Analyses for the Particulate Matter Smith, K.R. and others. 2000a., "Greenhouse Gases from Small-scale and Ozone National Ambient Air Quality Standards and Proposed Combustion Devices in Developing Countries, Phase IIa: Regional Haze Rule." Research Triangle Park, N.C. Household Stoves in India." EPA-600/R-00-052 U.S. Environmental ———. 1998a. "Study of Hazardous Air Pollutant Emissions from

Protection Agency, Office of Research and Development, Electric Utility Steam Generating Units." Final report to Congress, Washington, D.C. vol. 1. Washington, D.C.

WORLD ENERGY ASSESSMENT: ENERGY AND THE CHALLENGE OF SUSTAINABILITY 109

Chapter 3: Energy, the Environment, and Health

———. 1998b. "Toyota Prius Hybrid System (THS)." Office of Air and Radiation. <http://www.epa.gov/orcdizux/reports/adv-tech/priustst.pdf>

———. 1999. "Environmental Effects of Acid Rain." <http://www.epa.gov/docs/acidrain/effects/envben.html>

van Aardenne, J.A., G.R. Carmichael, H. Levy, D. Streets, and L. Hordijk.

1999. "Anthropogenic NO_x Emissions in Asia in the Period 1990–2020." *Atmospheric Environment* 33: 633–46.

Wang, W., and T. Wang. 1995. "On the Origin and the Trend of Acid Precipitation in China." *Water, Air and Soil Pollution* 85: 2295–2300.

Wang, X., and K.R. Smith. 1998. "Near-term Health Benefits of Greenhouse Gas Reductions: A Proposed Method for Assessment and Application to China." WHO/EHG/98.12. World Health Organization, Geneva.

WCD (World Commission on Dams). 1999. "Work Programme in Brief." Cape Town, South Africa. <http://www.dams.org>

WEC (World Energy Council). 1993. *Energy for Tomorrow's World*. London: Kogan Page.

———. 1995. *Global Transport Sector Energy Demand towards 2020*. London.

Weeks, J.L. 1998. "Health Hazards of Mining and Quarrying." In J.M. Stellman, ed., *Encyclopaedia of Occupational Health and Safety*. 4th ed. Geneva: International Labour Organization.

Wentz, F.J., and M. Schabel. 1998. "Effects of Orbital Decay on Satellite-derived Lower-tropospheric Temperature Trends." *Nature* 394: 661–64.

WHO (World Health Organization). 1994. "Ultraviolet Radiation." *Environmental Health Criteria* 160. Geneva.

———. 1996. "Health Effects of the Chernobyl Accident." WHO/EHG/96.X. Geneva.

———. 1997. "Health and Environment in Sustainable Development: Five Years after the Earth Summit." WHO/EHG/97.8. Geneva.

———. 1998a. "Chrysotile Asbestos." *Environmental Health Criteria* 190. Geneva.

———. 1998b. *Healthy Cities Air Information Management System 2.0*. Geneva.

———. 1999. *Provisional Global Air Quality Guidelines*. Geneva.

Wigley, T.M. 1999. *The Science of Climate Change: Global and U.S. Perspectives*. Arlington, Va.: Pew Center on Global Climate Change.

- Wigley, T.M., R.L. Smith, and B.D. Santer. 1998. "Anthropogenic Influence on the Autocorrelation Structure of Hemispheric-mean Temperatures." *Science* 282: 1676–79.
- Willrich, M., and T. Taylor. 1974. *Nuclear Theft: Risks and Safeguards*. Cambridge, Mass.: Ballinger.
- WMO (World Meteorological Organization). 1995. *Scientific Assessment of Ozone Depletion: 1994*. Geneva.
- . 1999. *Scientific Assessment of Ozone Depletion: 1998*. Geneva.
- World Bank. 1992. *World Development Report 1992: Development and the Environment*. New York: Oxford University Press.
- . 1993. *China: Involuntary Resettlement*. Washington, D.C.
- . 1997. *Clear Water, Blue Skies: China's Environment in the New Century*. China 2020 series. Washington, D.C.
- WRI (World Resources Institute). 1998. *World Resources Report 1998–99: Environmental Change and Human Health*. New York: Oxford University Press.
- Yasukawa, S., and others. 1992. "Life Cycle CO₂ Emission from Nuclear Power Reactor and Fuel Cycle System." Paper presented at an International Energy Agency expert workshop on Life-Cycle Analysis of Energy Systems: Methods and Experience, Paris.
- Zhao, D., and J. Xiong. 1988. "Acidification in Southwestern China." In H. Rodhe and R. Herrera, eds., *Acidification in Tropical Countries*. Chichester, U.K.: John Wiley and Sons Ltd.

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Chapter 3: Energy, the Environment, and Health

PWE 206: Energy, power and Environment

CHAPTER ONE: Principles, Demands, and Outlook of Energy, Power, and Environment

Understanding the principles governing energy, power, and the environment is critical in addressing the challenges of sustainability and resource management. In this concept, we will focus on the fundamental principles, the evolving demands, and the future outlook of energy, power, and their implications on the environment.

Principles of Energy

Energy is the capacity to do work and is essential for all forms of life and technological advancements and innovations. There are various types and/or forms of energy which include Kinetic, potential, thermal, chemical, nuclear, electromagnetic, etc.

Conservation of energy principle; The total energy in a closed system remains constant over time, though it may change forms, as highlighted in the laws of thermodynamics.

Laws of thermodynamics:

1. First law: Energy cannot be created or destroyed, only transformed from one form to another.
2. Second law: The entropy of an isolated system always increases over time.

Principles of Power concept

Power is the rate at which energy is transferred, converted, or expended. Power is measured in watts (W) or joules per second (J/s).

Power = Energy Expended / Time

Power generation and consumption are central to modern civilization and industrial processes.

Sustainable power sources namely renewable power sources (solar, wind, hydro, biomass, etc.) and non-renewable (fossil fuels energy, nuclear energy, etc.).

Efficiency: The ratio of useful power output to the total power input, is essential for optimizing energy use.

Principles of Environment Concept

The environment encompasses the surroundings in which organisms, including humans, exist and interact.

- Environmental sustainability: Meeting the needs of the present without compromising the ability of future generations to meet their own needs.
- Ecosystem dynamics: Interactions between biotic and abiotic components of ecosystems, regulating factors like climate, soil composition, and biodiversity.
- Environmental degradation: Human activities, such as pollution, deforestation, and habitat destruction, negatively impact ecosystems and biodiversity.
- Climate change: The result of increased greenhouse gas emissions, primarily from human activities, leading to global warming, rising sea levels, and extreme weather events.

Greenhouse Gas Emissions

Greenhouse Gases (GHGs) concept: Greenhouse gases are substances in the Earth's atmosphere that trap heat, leading to the greenhouse effect, which includes carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), fluorinated gases, and water vapor. GHGs play a crucial role in regulating the Earth's temperature and maintaining a habitable climate.

Sources of Greenhouse Gas Emissions: There are various sources of producing greenhouse gases such as;

- I. Fossil Fuel Combustion: The burning of coal, oil, and natural gas for electricity generation, transportation, and industrial processes releases CO₂ into the atmosphere.
- II. Deforestation: Clearing of forests for agriculture, urbanization, and logging releases stored carbon and reduces the Earth's capacity to absorb CO₂.
- III. Agriculture: Livestock farming, rice cultivation, and the use of synthetic

fertilizers emit methane and nitrous oxide, potent greenhouse gases.

IV. Industrial Processes: Manufacturing, cement production, and chemical manufacturing release CO₂ and other greenhouse gases as byproducts.

V. Waste Management: Landfills and waste treatment processes emit methane from decomposing organic waste.

Impacts of Greenhouse Gas Emissions; The Presence of GHGs in the Atmosphere has led to some phenomena which has contributed negatively to nature's existence. Greenhouse gas emissions pose significant challenges to global climate stability, human health, and environmental sustainability.

- a. Climate Change: Accumulation of GHGs in the atmosphere leads to global warming, resulting in changes in temperature patterns, shifts in precipitation, sea level rise, and more frequent extreme weather events.
- b. Ocean Acidification: Increased CO₂ levels in the atmosphere lead to ocean acidification, which harms marine ecosystems and threatens coral reefs, shellfish, and other marine life.
- c. Air Quality: Some GHGs, such as methane and volatile organic compounds (VOCs), contribute to air pollution and smog formation, posing risks to human health and the environment.

Mitigation Strategies for Greenhouse Gas Emissions: Diverse ways of minimizing the emission of greenhouse gases by adopting mitigation strategies, transitioning to low-carbon technologies, and promoting sustainable practices. Curbing GHG emissions requires collective action, innovative solutions, and policy interventions at local, national, and international levels as highlighted below. These will ensure a reduction in the impacts of climate change, ecosystem protection, and building of a more resilient and sustainable future for generations to come.

- Transition to Renewable Energy: Investing in renewable energy sources such as solar, wind, hydro, and geothermal power reduces reliance on fossil fuels and lowers CO₂ emissions.
- Energy Efficiency: Improving energy efficiency in buildings, transportation, and industries reduces energy consumption and associated GHG emissions.
- Afforestation and Reforestation: Planting trees and restoring forests helps to capture carbon dioxide from the atmosphere, mitigating climate change and enhancing biodiversity.
- Shift in Agricultural Practices: Implementing sustainable agricultural practices such as organic farming, agroforestry, and conservation tillage reduces emissions of methane and nitrous oxide from agricultural activities.
- Policy Instruments: Enacting policies such as carbon pricing, emissions trading schemes, renewable energy incentives, and regulations on industrial emissions can drive emission reductions and promote sustainable development.

Technology and Innovation Initiatives: The roles of technology advancement in

mitigating carbon emissions are highlighted as follows;

- I. Carbon Capture and Storage (CCS): CCS technologies capture CO₂ emissions from industrial processes and power plants and store them underground, reducing emissions to the atmosphere.
- II. Clean Energy Technologies: Advancements in clean energy technologies, such as battery storage, hydrogen fuel cells, and advanced nuclear reactors, offer low-carbon alternatives to fossil fuels.
- III. Climate-Smart Agriculture: Innovative agricultural practices, precision farming technologies, and crop breeding techniques can reduce emissions from agriculture and enhance resilience to climate change.
- IV. Data and Analytics: Utilizing data analytics, remote sensing, and satellite technology enables better monitoring of GHG emissions, identification of emission hotspots, and tracking of progress toward emission reduction goals.

Demands

Demands as related to energy, power, and environment. The rise in energy demand is associated with the following factors;

- a. Increasing global population and economic development drive escalating demands for energy and power.
- b. Urbanization and industrialization result in higher energy consumption and environmental pressures.
- c. Energy demand varies across sectors: transportation, residential, commercial, and industrial.
- d. Developing countries experience rapid energy demand growth, requiring innovative solutions for sustainable development.
- e. The demand for clean and renewable energy sources is rising due to environmental concerns and resource depletion of fossil fuel-based energy alternatives.

Outlook

The future expectation for sustainable and reliable energy supply initiatives as well as improving environmental sustainability includes

- a) Transition to renewable energy: Governments, industries, and individuals are increasingly investing in renewable energy technologies to mitigate climate change and reduce dependence on fossil fuels.
- b) Technological advancements: Innovations in energy storage, smart grids, and energy-efficient technologies are shaping the future of energy production and consumption.
- c) Policy and regulation: Governments worldwide are implementing policies and regulations to promote renewable energy adoption, reduce emissions, and preserve natural resources.
- d) Public awareness and education: Raising awareness about energy

conservation, environmental stewardship, and sustainable living practices is crucial for fostering a culture of sustainability.

CHAPTER TWO: Electrical Energy from Fossil Fuels

Fossil fuels are organic materials formed from the remains of plants and animals over millions of years. These fuels are extracted from the Earth's crust and are rich in carbon and hydrogen. Fossil fuels consist mainly of three, which are coal, oil (petroleum), and natural gas. Each type has its unique properties and applications.

Significance of Fossil Fuels in Energy Production

Fossil fuels play a crucial role in meeting the world's energy needs. They have been the primary source of energy for various applications, including electricity generation, transportation, heating, and industrial processes. The abundance and relatively low cost of fossil fuels have made them attractive for energy production, leading to their widespread use globally.

Electrical Energy Generation Concept from Fossil Fuels

Fossil fuels are a major source of electrical energy worldwide, accounting for a significant portion of electricity generation in many countries. The process of generating electricity from fossil fuels typically involves burning the fuel to produce heat, which is then used to generate steam. The steam drives a turbine connected to a generator, producing electricity. Electrical energy generation from fossil fuel sources has been a cornerstone of modern civilization, powering economic growth and development.

Different types of fossil fuel power plants exist, including coal-fired, oil-fired, and gas-fired power plants, each utilizing their respective fuel source to generate electricity.

Environmental and Social Impact of Fossil Fuel Energy

- Though fossil fuels have provided reliable and affordable energy for decades, their use has raised concerns about environmental pollution and climate change.
- Burning fossil fuels releases carbon dioxide (CO₂) and other greenhouse gases into the atmosphere, contributing to global warming and climate change.
- Fossil fuel extraction, transportation, and combustion also pose risks to local ecosystems and human health, including air and water pollution and habitat destruction.

CHAPTER THREE: Hydroelectricity Generation

Hydroelectricity, also known as hydroelectric power, is one of the oldest and most widely used renewable energy sources. It involves generating electricity from flowing or falling water. This process harnesses the kinetic energy of water to produce electrical energy, making it a clean and sustainable power generation method. Hydroelectricity generation plays a vital role in the global energy mix, providing clean, reliable, and sustainable power. Despite its challenges, advancements in

technology and environmental management continue to improve the efficiency and sustainability of hydroelectric power plants, contributing to a greener and more resilient energy future.

Principle of Operation

Hydroelectric power plants convert the potential energy of water stored in dams or the kinetic energy of flowing water into electrical energy. The basic principle behind hydroelectricity generation involves the following steps:

1. **Water Intake:** Water is collected from rivers, lakes, or reservoirs and directed towards the dam or power plant intake.
2. **Dam or Reservoir:** The water is stored in a dam or reservoir, creating a head or height difference, which stores potential energy in the water.
3. **Penstock:** The stored water is released from the reservoir and flows through a large pipe called a penstock, which directs the water towards the turbine.
4. **Turbine:** The flowing water drives the turbine blades, causing the turbine to rotate. The turbine converts the kinetic energy of the water into mechanical energy.
5. **Generator:** The rotating turbine shaft is connected to a generator, which converts the mechanical energy into electrical energy through electromagnetic induction.
6. **Transmission:** The generated electricity is transmitted through power lines to homes, businesses, and industries for various applications.

Types of Hydroelectric Power Plants

1. **Impoundment or Conventional Hydroelectric Plants:** These plants use dams to store water in reservoirs. When electricity demand is high, water is released from the reservoir, flowing through turbines to generate electricity.
2. **Run-of-River Hydroelectric Plants:** Unlike impoundment plants, run-of-river plants do not require large reservoirs. They utilize the natural flow of rivers or streams to generate electricity, minimizing environmental impacts.
3. **Pumped Storage Hydroelectric Plants:** These plants have two reservoirs situated at different elevations. During periods of low electricity demand, excess electricity is used to pump water from the lower reservoir to the upper reservoir. When demand increases, water is released from the upper reservoir, flowing through turbines to generate electricity.

Advantages of Hydroelectricity

1. **Renewable and Sustainable:** Hydroelectric power relies on the water cycle, making it a renewable energy source.
2. **Clean Energy:** It produces minimal greenhouse gas emissions and pollutants compared to fossil fuels.
3. **Reliable and Flexible:** Hydroelectric plants can quickly respond to changes in electricity demand and provide continuous power generation.
4. **Water Management:** Hydroelectric dams offer flood control, water storage for irrigation, and recreational opportunities like boating and fishing.

Challenges and Considerations

1. Environmental Impact: Hydroelectric dams can disrupt ecosystems, alter river habitats, and affect fish migration patterns.
2. Capital Intensive: Building hydroelectric dams and power plants requires significant upfront investment and long lead times.
3. Land Use and Resettlement: Large-scale hydro projects may involve land acquisition, displacement of communities, and cultural impacts on indigenous populations.
4. Climate Change: Changing precipitation patterns and water availability due to climate change can affect hydroelectric power generation.

Problems of Hydroelectricity Generation

While hydroelectricity is a clean and renewable energy source, it also presents various challenges and problems that need to be addressed. These issues range from environmental impacts to social and economic considerations. Understanding these problems is essential for ensuring the sustainable development and operation of hydroelectric power projects.

Addressing the problems associated with hydroelectricity generation requires a multidisciplinary approach that considers environmental, social, economic, and governance aspects. Sustainable hydroelectric development involves careful planning, stakeholder engagement, and adaptive management practices to minimize negative impacts and maximize benefits for both present and future generations.

Environmental Impacts

1. Habitat Destruction: The construction of dams and reservoirs can lead to the inundation of large areas of land, resulting in the loss of terrestrial and aquatic habitats. This can disrupt ecosystems and threaten biodiversity.
2. Fish Migration: Dams can obstruct the natural migration of fish species, affecting their breeding patterns and population dynamics. Fish passage facilities are often required to mitigate these impacts, but their effectiveness varies.
3. Water Quality: Reservoirs can alter water temperature, flow patterns, and sediment transport, affecting downstream ecosystems and water quality. Nutrient loading, sedimentation, and algal blooms are common issues associated with hydroelectric reservoirs.
4. Methane Emissions: Decomposing organic matter in flooded areas can release methane, a potent greenhouse gas, contributing to climate change. Reservoir management strategies can influence methane emissions but may not eliminate them entirely.

Social and Economic Considerations

1. Displacement of Communities: Large-scale hydro projects often require the resettlement of communities living in affected areas. This can lead to social unrest, loss of livelihoods, and cultural disruptions for indigenous populations.
2. Land Use Conflicts: Hydroelectric projects may compete with other land uses such

as agriculture, forestry, and recreation. Conflicts over land tenure, resource rights, and compensation can arise between different stakeholders.

3. Economic Viability: The economic feasibility of hydroelectric projects depends on factors such as construction costs, electricity prices, and financing arrangements.

Delays, cost overruns, and market uncertainties can undermine project profitability.

4. Reservoir Sedimentation: Over time, sediment accumulation in reservoirs can reduce storage capacity and affect water availability for power generation and other purposes. Sediment management strategies, such as dredging and flushing, may be necessary but can be expensive and environmentally disruptive.

Regulatory and Governance Challenges

1. Environmental Regulation: Compliance with environmental regulations and permitting requirements can be complex and time-consuming for hydroelectric developers. Balancing environmental protection with energy development objectives requires effective regulatory frameworks and stakeholder engagement.

2. Transboundary Issues: Hydroelectric projects located on international rivers may involve transboundary impacts and require cooperation among neighboring countries. Negotiating agreements on water sharing, environmental protection, and project operation can be challenging.

3. Public Participation**: Meaningful public participation and consultation are essential for addressing community concerns, ensuring project transparency, and building trust. However, achieving consensus among diverse stakeholders with competing interests can be difficult.

CHAPTER FOUR: Cost of Hydroelectricity Generation

The cost of hydroelectricity generation encompasses various factors, including initial capital investment, operation and maintenance expenses, environmental mitigation measures, and financing costs. Understanding these cost components is essential for evaluating the economic feasibility and competitiveness of hydroelectric projects compared to other energy sources.

The cost of hydroelectricity generation depends on various factors, including project size, location, technology, regulatory requirements, and financing arrangements.

While hydroelectricity offers numerous benefits, including renewable energy generation, grid stability, and water management, addressing the associated costs is crucial for ensuring the economic viability and sustainability of hydroelectric projects in the long term.

Capital Costs

1. Infrastructure Investment: The construction of hydroelectric power plants involves significant upfront capital expenditure, including the costs of dams, turbines, generators, penstocks, and transmission infrastructure.

2. Site Preparation: Clearing land, excavating foundations, and constructing access roads and other infrastructure add to the overall project costs.

3. Environmental Mitigation: Compliance with environmental regulations and mitigation measures, such as fish passage facilities, sediment management, and habitat restoration, can increase project costs.

Operation and Maintenance Costs

1. Turbine Maintenance: Regular inspection, repair, and maintenance of turbines and associated equipment are necessary to ensure reliable and efficient operation.
2. Reservoir Management: Reservoir operations, including water level management, sedimentation control, and environmental monitoring, require ongoing maintenance and management efforts.
3. Security and Safety: Ensuring the safety and security of dams, facilities, and personnel involves additional operational expenses, including surveillance, emergency preparedness, and risk management measures.

Environmental and Social Costs

1. Environmental Impact Mitigation: Addressing the environmental impacts of hydroelectric projects, such as habitat restoration, fish passage facilities, and water quality management, can incur significant costs.
2. Social Impacts: Resettlement of communities, compensation for land acquisition, and addressing social conflicts and grievances can contribute to project costs.
3. Regulatory Compliance: Meeting environmental, social, and safety regulations may require additional investments in monitoring, reporting, and compliance activities.

Financing Costs

1. Interest Payments: Financing hydroelectric projects through loans or bonds involves interest payments, which add to the overall project costs over the repayment period.
2. Capital Recovery: Recovering the initial investment through electricity sales and revenue generation is essential for covering financing costs and achieving project profitability.
3. Risk Premiums: Uncertainties related to project delays, cost overruns, market fluctuations, and regulatory changes may increase borrowing costs and risk premiums for investors.

Levelized Cost of Electricity (LCOE)

The levelized cost of electricity (LCOE) is a commonly used metric for comparing the cost of electricity generation from different sources. It represents the average cost of producing electricity over the lifetime of a power plant, expressed in dollars per megawatt-hour (MWh). Calculating LCOE involves considering all relevant costs, including capital, operating, maintenance, and financing costs, as well as the expected electricity output over the plant's lifetime.

CHAPTER FIVE: Capacity in Relation to Hydroelectricity Generation

Hydroelectricity generation capacity refers to the maximum amount of electrical power that a hydroelectric power plant can produce under specific operating

conditions. Understanding capacity is essential for assessing the potential output and reliability of hydroelectric projects, as well as for planning energy infrastructure and grid integration.

Capacity is a key parameter in the planning, operation, and evaluation of hydroelectricity generation. By understanding installed capacity, capacity factors, load factors, and the roles of peak and base load capacity, policymakers, energy planners, and operators can optimize the contribution of hydroelectric resources to the overall energy system, enhance grid reliability, and promote sustainable development.

Installed Capacity

Installed capacity, often expressed in megawatts (MW) or gigawatts (GW), represents the total power generation capacity of a hydroelectric plant or a portfolio of plants within a region or country.

Types of Installed Capacity

- Nameplate Capacity: The maximum rated output of a hydroelectric plant under ideal operating conditions, typically based on the manufacturer's specifications.
- Effective Capacity: The actual output of a hydroelectric plant under real-world operating conditions, accounting for factors such as maintenance, availability, and seasonal variations in water availability.
- Net Capacity: The portion of nameplate capacity that is available for electricity generation after deducting losses due to internal consumption, auxiliary loads, and transmission losses.

Capacity Factor

The capacity factor is a measure of the actual electricity generation relative to the maximum potential generation over a specific period, typically expressed as a percentage.

Capacity Factor = (Actual Electricity Generation / (Installed Capacity × Hours in the Period)) × 100%

Factors Affecting Capacity Factor:

- Hydrological Conditions: Variations in water availability, precipitation patterns, and seasonal runoff can impact the capacity factor of hydroelectric plants.
- Operational Efficiency: Factors such as turbine efficiency, reservoir management, and maintenance practices influence the capacity factor.
- Grid Integration: Market demand, dispatch instructions, and grid stability considerations affect the utilization of hydroelectric capacity and its contribution to overall electricity supply.

Load Factor

The load factor measures the ratio of the average electricity demand to the installed capacity over a specific period, indicating the utilization rate of the capacity.

Load Factor = (Average Electricity Demand / Installed Capacity) × 100%

A high load factor indicates efficient utilization of capacity, while a low load factor

may suggest underutilization or excess capacity relative to demand.

Peak Capacity and Base Load Capacity

1. **Peak Capacity:** The maximum capacity that can be deployed to meet periods of high electricity demand, typically during peak hours or seasons. Hydroelectric plants can provide valuable peaking capacity due to their ability to ramp up quickly and respond to fluctuations in demand.
2. **Base Load Capacity:** The minimum capacity required to meet constant or relatively stable electricity demand, typically provided by baseload power plants such as hydroelectric facilities with steady water availability and predictable output.

CHAPTER SIX: Storage of Hydroelectricity Generation

Storage reserve, also known as reservoir storage or hydroelectric storage, refers to the capacity of a hydroelectric power plant to store water in a reservoir for future use in electricity generation. This stored water serves as a flexible energy resource, allowing hydroelectric facilities to respond to fluctuations in electricity demand, optimize generation schedules, and provide grid stability.

Storage reserve is a valuable asset in hydroelectricity generation, offering flexibility, reliability, and grid stability benefits. By effectively managing reservoir storage and incorporating storage technologies like pumped storage hydroelectricity, policymakers and operators can maximize the value of hydroelectric resources, optimize energy system operations, and accelerate the transition to a sustainable and resilient energy future.

Functions of Storage Reserve

1. **Load Balancing:** Storage reservoirs enable hydroelectric plants to balance electricity supply and demand by storing excess energy during periods of low demand and releasing stored water to generate electricity during peak demand periods.
2. **Peak Shaving:** Hydroelectric facilities with storage capacity can provide peaking power to meet short-term spikes in electricity demand, reducing the need for fossil fuel-based peaking plants and enhancing grid reliability.
2. **Grid Stability:** Storage reserves contribute to grid stability by providing fast-responding, dispatchable generation capacity that can adjust output rapidly to maintain grid frequency and voltage within acceptable limits.
3. **Energy Arbitrage:** Hydroelectric operators can take advantage of price differentials in electricity markets by storing water when electricity prices are low and releasing it for generation when prices are high, thereby maximizing revenue.
5. **Renewable Integration:** Storage reservoirs facilitate the integration of variable renewable energy sources like wind and solar by providing backup capacity and enabling the smoothing of intermittent generation.

Factors Affecting Storage Reserve

1. **Reservoir Capacity:** The size of the reservoir determines the amount of water that

can be stored and the duration of storage available for electricity generation.

2. Hydrological Conditions: Variations in precipitation, snowmelt, and runoff patterns affect water availability and reservoir levels, influencing the effectiveness of storage reserves.

3. Operational Constraints: Regulatory requirements, environmental considerations, and operational limitations may restrict the use of storage reserves for electricity generation, particularly during periods of drought or low water availability.

4. Reservoir Management: Effective reservoir management practices, including flood control, sedimentation management, and ecological considerations, play a crucial role in optimizing storage reserves for multiple purposes.

Types of Hydroelectric Storage

1. Conventional Storage: Hydroelectric plants with large reservoirs store water for long-term use, providing significant storage capacity and operational flexibility.

2. Pumped Storage: Pumped storage hydroelectric plants use surplus electricity during off-peak hours to pump water from a lower reservoir to an upper reservoir. During periods of high demand, water is released from the upper reservoir to generate electricity through turbines, effectively serving as a large-scale energy storage system.

Hydroelectricity Generation Efficiency

Efficiency in hydroelectricity generation refers to the ratio of useful electrical energy output to the total energy input from the water source. Maximizing efficiency is essential for optimizing the performance and economic viability of hydroelectric power plants while minimizing environmental impacts.

Efficiency is a critical performance indicator in hydroelectricity generation, reflecting the effectiveness of converting water's energy into electricity. By optimizing turbine design, operational practices, and technology innovation, hydroelectric operators can maximize efficiency, enhance economic competitiveness, and minimize environmental impacts, contributing to a sustainable energy future.

Components of Efficiency

1. Conversion Efficiency: Conversion efficiency represents the effectiveness of converting the potential or kinetic energy of water into electrical energy. It includes losses associated with turbine, generator, and transmission system inefficiencies.

2. Overall Efficiency: Overall efficiency accounts for losses at each stage of the hydroelectric generation process, including turbine, generator, transformer, and transmission losses. It provides a comprehensive measure of the entire power generation system's performance.

Factors Affecting Efficiency

1. Turbine Design: The design and efficiency of the turbine play a crucial role in determining overall efficiency. Factors such as turbine type (e.g., Pelton, Francis, Kaplan), size, operating conditions, and maintenance affect turbine performance.

2. Hydrological Conditions: Variations in water flow, head, and seasonal runoff

patterns impact the efficiency of hydroelectric plants. Droughts, floods, sedimentation, and water quality issues can affect turbine performance and overall efficiency.

3. **Generator Efficiency:** The efficiency of the generator in converting mechanical energy from the turbine into electrical energy influences overall efficiency. Generator design, electrical losses, and maintenance practices affect generator performance.

4. **Transmission Losses:** Electrical losses during transmission and distribution of electricity from the power plant to end-users reduce overall efficiency. Minimizing transmission distances and using efficient transmission technologies can help mitigate these losses.

5. **Operational Practices:** Operating hydroelectric plants at optimal conditions, scheduling maintenance activities, and implementing efficiency improvement measures contribute to maximizing efficiency. Regular monitoring, troubleshooting, and performance optimization are essential.

Measurement and Evaluation

1. **Efficiency Metrics:** Efficiency is typically expressed as a percentage and can be measured at various stages of the hydroelectric generation process, including turbine efficiency, generator efficiency, and overall plant efficiency.

2. **Performance Monitoring:** Continuous monitoring of key performance indicators, such as flow rate, head, power output, and energy losses, enables operators to identify inefficiencies, diagnose problems, and implement corrective actions.

Improving Efficiency

1. **Technology Innovation:** Advances in turbine and generator design, materials, and control systems can improve efficiency and performance.

2. **Optimized Operations:** Implementing best practices in plant operation, maintenance, and management can enhance efficiency and reliability.

3. **Modernization and Upgrades:** Retrofitting older hydroelectric plants with modern equipment, automation systems, and control strategies can improve efficiency and extend their operational lifespan.

4. **Integrated Resource Planning:** Integrating hydroelectricity with other renewable energy sources, energy storage technologies, and demand-side management strategies can enhance system efficiency and flexibility.

New Environmental Effects on Hydroelectricity Generation

While hydroelectricity generation is often touted as a clean and renewable energy source, it is not without environmental impacts. As technology advances and our understanding of ecological systems deepens, new environmental effects associated with hydroelectricity generation have emerged. It is essential to recognize and mitigate these effects to ensure the sustainability and long-term viability of hydroelectric projects.

As hydroelectricity continues to play a significant role in the global energy mix, it is

essential to address the new environmental effects associated with hydroelectric generation. By integrating ecosystem-based approaches, adaptive management strategies, and stakeholder engagement into project planning and management, policymakers, developers, and communities can minimize negative impacts, enhance environmental sustainability, and promote the responsible development of hydroelectric resources.

1. Climate Change Impacts

Altered Flow Regimes: Changes in precipitation patterns, snowmelt timing, and hydrological cycles due to climate change can affect water availability and flow regimes, influencing the operation and performance of hydroelectric plants.

Reservoir Methane Emissions: Warmer water temperatures and increased organic matter decomposition in reservoirs can lead to higher methane emissions, exacerbating greenhouse gas emissions and climate change.

2. Ecosystem Disturbances

Invasive Species: Hydroelectric projects can facilitate the spread of invasive species, disrupting native ecosystems and biodiversity. Invasive species may outcompete native species, alter food webs, and degrade habitat quality.

Habitat Fragmentation: Dams and reservoirs fragment river systems, obstructing fish migration routes and altering aquatic habitats. Fragmentation can lead to population declines, genetic isolation, and reduced biodiversity.

3. Water Quality Concerns

Nutrient Loading: Reservoirs can accumulate nutrients from upstream sources, leading to eutrophication, algal blooms, and degraded water quality. Excessive nutrient loading can harm aquatic ecosystems and impair water treatment processes.

Sedimentation: Sedimentation in reservoirs can reduce storage capacity, alter downstream flow regimes, and degrade aquatic habitats. Sediment management practices are necessary to mitigate the impacts of sedimentation.

4. Social and Cultural Impacts

Community Displacement: Large-scale hydroelectric projects often involve the resettlement of communities and displacement of indigenous populations, leading to social unrest, loss of livelihoods, and cultural disruption.

Cultural Heritage Loss: Submergence of archaeological sites, cultural landscapes, and sacred sites due to reservoir inundation can result in the loss of cultural heritage and intangible cultural values.

5. Regulatory and Policy Challenges

Environmental Regulation: Compliance with environmental regulations and permitting requirements can be challenging for hydroelectric developers, particularly in ecologically sensitive areas. Balancing environmental protection with energy development objectives requires robust regulatory frameworks and stakeholder engagement.

Transboundary Impacts: Hydroelectric projects located on international rivers may involve transboundary impacts and require cooperation among neighboring countries. Negotiating agreements on water sharing, environmental protection, and project operation can be complex and contentious.

CHAPTER SEVEN: Electrical Energy Generation from Nuclear Reactors

Nuclear power plants utilize the energy released during nuclear fission to generate electricity. The process involves splitting of atomic nuclei, typically uranium or plutonium, releasing a tremendous amount of energy in the form of heat. This heat is then used to produce steam, which drives turbines connected to generators, ultimately generating electricity.

Nuclear power plays a crucial role in the global energy mix, providing a reliable and low-carbon source of electricity. However, it is essential to address safety, waste management, and cost concerns to ensure the sustainable development of nuclear energy in the future.

Nuclear Reactor Fundamentals:

Nuclear reactors are designed to sustain and control nuclear fission reactions. The core of a nuclear reactor contains fuel rods, usually enriched uranium or plutonium, surrounded by a moderator such as water or graphite. Neutrons, released during fission, are slowed down by the moderator, increasing the likelihood of further fission reactions. Control rods made of materials like boron or cadmium are used to regulate the rate of fission by absorbing excess neutrons.

Electricity Generation Process:

1. Heat Generation: Nuclear fission reactions release a large amount of heat energy.
2. Heat Transfer: The heat produced in the reactor core is transferred to a coolant, typically water, which flows through the reactor core and absorbs the heat.
3. Steam Production: The coolant water is then pumped to a heat exchanger where it heats a separate water loop, turning it into steam.
4. Turbine Operation: The steam is directed towards turbines, causing them to spin.
5. Electricity Generation: The spinning turbines are connected to generators, which convert mechanical energy into electrical energy.

Safety Measures:

- I. Nuclear power plants incorporate numerous safety systems to prevent accidents and mitigate their consequences.
- II. These include control rod systems, emergency coolant systems, containment buildings, and redundant safety features.
- III. Strict regulatory standards and regular inspections ensure the safety and reliability of nuclear power plants.

Advantages of Nuclear Energy:

1. Low Greenhouse Gas Emissions: Nuclear power generation produces minimal greenhouse gas emissions, making it a cleaner alternative to fossil fuels.

2. High Energy Density: Nuclear fuel contains a high energy density, meaning a small amount of fuel can generate a large amount of electricity.
3. Reliability: Nuclear power plants can operate continuously for long periods, providing a stable source of electricity.

Challenges and Concerns:

1. Nuclear Waste: The disposal of radioactive waste generated by nuclear reactors remains a significant challenge.
2. Safety Concerns: Accidents such as Chernobyl and Fukushima have raised concerns about the safety of nuclear power.
3. Cost: The construction and decommissioning of nuclear power plants can be expensive, and the long-term costs associated with waste disposal and plant maintenance add to the overall expense.

CHAPTER EIGHT: Energy in the Fast breeder reactors (FBRs)

Fast breeder reactors (FBRs) are a type of nuclear reactor designed to produce more fissile material than they consume by "breeding" fissile material from fertile material through fast neutron reactions. Here's a breakdown of energy aspects in fast breeder reactors:

Fuel Utilization:

- i. FBRs use fast neutrons to cause fission in fertile material like uranium-238 (U-238) or thorium-232 (Th-232).
- ii. Fast neutrons have higher energy compared to thermal neutrons, enabling them to induce fission in fertile materials.
- iii. This breeding process converts fertile material into fissile material, such as plutonium-239 (Pu-239) or uranium-233 (U-233), which can sustain a nuclear chain reaction.

Energy Production:

- I. FBRs produce energy through nuclear fission reactions, where the nucleus of an atom splits into smaller fragments, releasing a significant amount of energy.
- II. The fission of fissile material like Pu-239 or U-233 in the reactor core generates heat, which is used to produce steam.
- III. The steam drives turbines connected to generators, ultimately producing electricity.

Breeding Ratio:

- i. The breeding ratio in a fast breeder reactor refers to the ratio of fissile material produced to the fissile material consumed.
- ii. Ideally, FBRs aim for a breeding ratio greater than one, indicating that they produce more fissile material than they consume.
- iii. Achieving a breeding ratio greater than one is crucial for the sustainability and efficiency of FBRs, as it enables them to generate more fuel than they use.

Advantages:

FBRs offer several advantages, including:

- I. High fuel utilization efficiency: FBRs can utilize fertile materials, significantly extending the available nuclear fuel resource.
- II. Reduced long-term nuclear waste: By converting fertile materials into fissile materials, FBRs can reduce the volume and radiotoxicity of nuclear waste.
- III. - Potential for closing the nuclear fuel cycle: FBRs have the potential to close the nuclear fuel cycle by recycling spent nuclear fuel and utilizing plutonium as fuel.

Challenges:

Despite their potential benefits, FBRs face various challenges, including:

- a. Technical complexity: FBRs are technologically challenging to design and operate due to the high-energy neutrons involved and the need for advanced materials.
- b. Safety concerns: FBRs pose unique safety challenges, including the risk of sodium fires and the potential for core disruptions.
- c. Cost: FBRs have historically been more expensive to construct and operate compared to conventional light-water reactors, which has limited their commercial deployment.

Overall, fast breeder reactors represent a promising technology for enhancing the sustainability and efficiency of nuclear energy production. However, addressing technical, safety, and economic challenges is essential to realizing their full potential.

CHAPTER NINE: Energy in Fusion Power

Fusion power is the process of harnessing energy from the fusion of atomic nuclei. It holds the promise of clean, abundant energy with minimal environmental impact.

Fusion power has the potential to revolutionize the world's energy landscape, providing a clean, safe, and virtually limitless source of power for future generations.

Basics of Fusion

Fusion is the process by which two atomic nuclei combine to form a heavier nucleus, releasing energy in the process. It is the same process that powers the sun and other stars.

Energy Release in Fusion:

In fusion reactions, the mass of the resulting nucleus is slightly less than the sum of the masses of the original nuclei. This "missing" mass is converted into energy according to Einstein's famous equation, $E=mc^2$, where E is energy, m is mass, and c is the speed of light.

Fusion Reactions:

The most promising fusion reaction for power generation on Earth is the fusion of isotopes of hydrogen, particularly deuterium and tritium.

The primary reaction is:



The resulting helium nucleus is stable, and the released neutron carries significant kinetic energy.

Conditions for Fusion:

- ◆ Fusion requires extremely high temperatures and pressures to overcome the electrostatic repulsion between positively charged nuclei.
- ◆ Temperatures of millions of degrees Celsius are required to achieve the necessary kinetic energy for fusion.

Confinement Methods

Two primary methods are used to confine the fusion fuel:

1. Magnetic Confinement: Uses strong magnetic fields to confine the hot plasma.
2. Inertial Confinement: Uses high-energy lasers or particle beams to compress and heat the fusion fuel rapidly.

Challenges and Progress:- Fusion research faces significant technical challenges, including maintaining plasma stability, achieving high enough temperatures and pressures, and sustaining fusion reactions for long periods. Despite challenges, significant progress has been made, with experimental fusion reactors demonstrating controlled fusion reactions and generating energy.

Advantages of Fusion Power:

- I. Fusion power offers numerous advantages, including:
- II. Abundant fuel supply (deuterium can be extracted from water, tritium can be bred from lithium).
- III. Minimal radioactive waste compared to fission.
- IV. No greenhouse gas emissions or air pollution.
- V. Inherent safety features (no risk of meltdown or runaway reactions).

Current Status and Future Prospects:

Experimental fusion reactors such as ITER (International Thermonuclear Experimental Reactor) are under construction to demonstrate the feasibility of sustained fusion power. Commercial fusion power plants are still several decades away, but continued research and development hold promise for realizing this clean energy source.

CHAPTER TEN: Energy in Solar Power

Solar power is a renewable energy source derived from the sun's radiation. It is one of the cleanest and most abundant sources of energy available to us. Understanding the principles of solar energy is essential for harnessing its potential for electricity generation, heating, and other applications.

Solar power offers a sustainable and scalable solution to meet the world's energy needs while reducing dependence on fossil fuels and mitigating climate change. Continued research and innovation in solar technology are essential for further improving efficiency, reducing costs, and expanding access to clean energy worldwide.

Solar Radiation:

The sun emits electromagnetic radiation across a wide spectrum of wavelengths, including ultraviolet (UV), visible, and infrared (IR) light. Solar radiation intensity varies with factors such as time of day, season, latitude, and atmospheric conditions.

Solar Cells:

Solar cells, also known as photovoltaic (PV) cells, convert sunlight directly into electricity through the photovoltaic effect. When photons of sunlight strike the semiconductor material in a solar cell, they excite electrons, creating an electric current.

Solar Panel Systems:

Solar panels consist of multiple interconnected solar cells housed within a protective enclosure. Solar panel systems may include additional components such as inverters, batteries, charge controllers, and mounting structures.

Types of Solar Power Systems:

1. Grid-Tied Systems: Solar panels are connected to the utility grid, allowing excess electricity to be exported and providing backup power during grid outages.
2. Off-Grid Systems: Solar panels are coupled with battery storage to provide electricity independent of the grid, commonly used in remote locations or for standalone applications.
3. Hybrid Systems: Combining solar power with other renewable energy sources or conventional power generation methods to optimize energy production and reliability.

Energy Conversion Efficiency:

The efficiency of solar panels refers to the percentage of sunlight that is converted into usable electricity.

Factors affecting efficiency include the quality of solar cells, sunlight intensity, temperature, shading, and system design.

Environmental Impacts:

Solar power is a clean and environmentally friendly energy source, producing no greenhouse gas emissions or air pollution during operation. However, the manufacturing process of solar panels and associated components may have environmental impacts, including resource consumption, waste generation, and emissions.

Economic Considerations:

The cost of solar power has declined significantly in recent years due to technological advancements, economies of scale, and policy incentives.

Factors influencing the economics of solar power include installation costs, financing options, government subsidies, and electricity prices.

CHAPTER ELEVEN: Energy in Geothermal Power

Geothermal power harnesses heat energy from within the Earth for electricity generation and heating. It is a renewable and environmentally friendly source of

energy with potential for widespread utilization.

Geothermal power offers a reliable, sustainable, and locally available source of energy for electricity generation and heating applications. Continued research, development, and investment are essential for unlocking the full potential of geothermal resources worldwide.

Basics of Geothermal Energy :

Geothermal energy originates from the Earth's core, where radioactive decay and residual heat from planetary formation produce high temperatures. This heat is transferred through the Earth's layers, creating geothermal gradients and thermal anomalies near the surface.

Geothermal Resources:

Geothermal resources can be classified into:

i. Hydrothermal Resources: Underground reservoirs of hot water or steam.

Enhanced Geothermal Systems (EGS): Engineered reservoirs created by injecting water into hot, dry rock formations.

ii. Geopressured Resources: Hot brines located in deep sedimentary formations.

iii. Magma Resources: Heat extracted directly from molten rock (not yet commercially viable).

Geothermal Power Plants:

There are several types of geothermal power plants:

I. Dry Steam Plants: Tap into underground steam reservoirs and use it to drive turbines.

II. Flash Steam Plants: Extract high-pressure hot water and convert it to steam for turbine operation.

III. Binary Cycle Plants: Transfer heat from lower-temperature fluids to a secondary fluid with a lower boiling point, which then drives a turbine.

Geothermal Heating Systems:

a. Geothermal heat pumps use the Earth's stable underground temperature to heat and cool buildings efficiently.

b. Ground-source heat pumps circulate fluid through underground pipes to transfer heat to or from the ground, depending on the season.

Environmental Benefits:

a) Geothermal power has minimal environmental impact compared to fossil fuels, with low greenhouse gas emissions and no air pollution during operation.

b) It can help reduce dependence on imported fuels and mitigate climate change by displacing coal and natural gas-fired power plants.

. Economic Considerations:

■ The cost of geothermal energy depends on factors such as resource quality, drilling depth, and plant size.

■ While upfront costs can be significant, geothermal power has low operating and

maintenance costs and can provide long-term price stability compared to fossil fuels.

Challenges and Opportunities:

Challenges in geothermal development include resource exploration and assessment, drilling technology, reservoir management, and regulatory hurdles.

Opportunities for innovation include enhanced reservoir engineering techniques, advanced drilling technologies, and integrated geothermal district heating systems.

Global Geothermal Potential:

Geothermal energy has significant global potential, particularly in regions with active tectonic activity or high geothermal gradients. Countries such as Iceland, the United States, Philippines, Indonesia, and Kenya are leaders in geothermal power generation.

CHAPTER TWELVE: Energy in Tidal Power

Tidal power harnesses energy from the gravitational forces between the Earth, moon, and sun to generate electricity. It is a renewable and predictable source of energy with the potential to contribute significantly to global electricity generation. Tidal power represents a promising renewable energy source with the potential to contribute to a sustainable energy future. Continued investment in research, development, and deployment is essential for realizing the full potential of tidal energy and addressing global energy challenges.

Basics of Tidal Energy

Tidal energy is primarily driven by the gravitational pull of the moon and, to a lesser extent, the sun, causing the periodic rise and fall of ocean tides. This movement of water creates kinetic energy that can be captured and converted into electricity using various technologies.

Tidal Energy Resources: Tidal energy resources can be classified into two:

1. Tidal Stream Energy: Extracts energy from the kinetic motion of water currents using underwater turbines similar to wind turbines.
2. Tidal Range Energy: Captures energy from the vertical movement of water between high and low tides using barrage or dam structures.

Tidal Stream Energy:

Tidal stream turbines are submerged in areas with strong tidal currents, such as channels between islands or near coastal areas. Turbines are designed to rotate with the flow of water, generating electricity as they spin.

Tidal Range Energy:

Tidal barrage or dam structures are built across estuaries or coastal bays to capture and store water during high tides. During low tides, water is released through turbines to generate electricity as it flows back to the sea.

Environmental Impacts:

Tidal power projects can have both positive and negative environmental impacts.

- Positive impacts include minimal greenhouse gas emissions and habitat creation in artificial reef structures.
- Negative impacts may include changes in tidal flow patterns, sediment transport, and potential disruption to marine ecosystems.

Economic Considerations:

The cost of tidal energy depends on factors such as site selection, technology type, and project scale. While tidal energy projects may have higher upfront costs compared to other renewable energy sources, they offer long-term predictability and low operating costs.

Global Tidal Energy Potential:

Tidal energy has significant potential in regions with strong tidal currents, such as the United Kingdom, Canada, France, China, and South Korea. The global technical potential for tidal energy is estimated to be substantial, although only a fraction of this potential has been developed to date.

Research and Development:

Ongoing research and development efforts focus on improving tidal energy technologies, optimizing project design and operation, and mitigating environmental impacts. Innovations in turbine design, materials, and deployment strategies aim to increase efficiency and reduce costs.

Research findings and advancements in the field of tidal power energy:

1. Tidal Turbine Optimization:

Research is ongoing to improve the design and efficiency of tidal turbines. Recent studies have focused on optimizing blade shape, orientation, and materials to maximize energy extraction from tidal currents while minimizing maintenance requirements and environmental impacts.

2. Array Layout Optimization:

Researchers are investigating optimal layouts for tidal turbine arrays to maximize energy capture while minimizing interactions between turbines and reducing environmental effects. Advanced modeling techniques and machine learning algorithms are being employed to optimize array configurations.

3. Environmental Impact Assessment:

Recent research has focused on assessing and mitigating the environmental impacts of tidal energy projects. Studies have examined the effects of tidal turbines on marine ecosystems, fish migration patterns, sediment transport, and water quality to inform sustainable project development and regulatory decisions.

4. Materials and Manufacturing Innovations:

- Advances in materials science and manufacturing technologies are driving improvements in tidal turbine performance and reliability. Research efforts include developing lightweight and corrosion-resistant materials, novel composite materials, and additive manufacturing techniques to reduce costs and extend turbine lifespan.

5. Predictive Modeling and Control Strategies:

Researchers are developing predictive modeling tools and control strategies to optimize the operation of tidal energy systems in real-time. Machine learning algorithms and data-driven approaches are being used to predict tidal currents, optimize turbine operation, and maximize energy production while ensuring grid stability.

6. Hybrid Tidal Energy Systems:

Hybrid tidal energy systems that integrate tidal power with other renewable energy sources, such as wind and solar, are being explored to enhance overall system reliability and performance. Research is focusing on optimal system integration, energy storage solutions, and grid management strategies for hybrid renewable energy systems.

7. Floating Tidal Energy Platforms:

Floating tidal energy platforms offer the potential to access deeper waters with stronger tidal currents, expanding the geographic range for tidal energy deployment. Recent research has investigated innovative floating platform designs, mooring systems, and dynamic positioning technologies to enable cost-effective and scalable tidal energy projects.

8. Community Engagement and Stakeholder Consultation:

Recognizing the importance of community engagement and stakeholder consultation in tidal energy project development, researchers are conducting social science studies to understand public perceptions, values, and concerns related to tidal energy. Strategies for effective community engagement, stakeholder participation, and collaborative decision-making are being explored to promote sustainable project development and social acceptance.

These research findings and advancements demonstrate the ongoing efforts to advance tidal power energy technology, improve project feasibility and sustainability, and address challenges associated with environmental, social, and economic considerations. Continued research, innovation, and collaboration are essential to realize the full potential of tidal energy as a clean and renewable energy source.

CHAPTER THIRTEEN: Thermal Pollution

Thermal pollution refers to the degradation of water quality caused by the release of heated water into natural water bodies, such as rivers, lakes, and oceans. It is primarily associated with industrial processes, power generation, and urbanization, leading to elevated water temperatures that can have harmful effects on aquatic ecosystems and human health.

Thermal pollution poses significant challenges to the health and sustainability of aquatic ecosystems and human communities, highlighting the need for effective regulation, technological innovation, and collaborative efforts to mitigate its adverse effects and promote environmental stewardship.

Sources of Thermal Pollution

- I. Industrial Facilities: Manufacturing plants, refineries, and chemical processing plants often use water for cooling purposes, resulting in the discharge of heated effluent into nearby water bodies.
- II. Power Plants: Thermal power plants, such as coal-fired and nuclear power plants, rely on water for cooling purposes, releasing heated water back into rivers or coastal areas.
- III. Urbanization: Urban runoff, stormwater drainage, and wastewater discharges from urban areas can contribute to thermal pollution, especially in densely populated regions with limited water circulation.

Effects of Thermal Pollution

- i. Aquatic Ecosystems: Elevated water temperatures can disrupt aquatic ecosystems by reducing oxygen levels, altering habitat suitability, and disrupting ecological balance. Heat-sensitive species, such as fish, invertebrates, and aquatic plants, may be particularly vulnerable to thermal stress.
- ii. Biodiversity Loss: Thermal pollution can lead to declines in biodiversity, with sensitive species facing habitat loss, reduced reproductive success, and increased susceptibility to diseases and predation.
- iii. Water Quality Degradation: High water temperatures can promote the growth of algae and other aquatic organisms, leading to eutrophication, algal blooms, and the depletion of dissolved oxygen levels, further exacerbating water quality degradation.

Human Health Impacts

- a) Recreational Activities: Elevated water temperatures can pose health risks to swimmers, boaters, and other recreational users by increasing the likelihood of bacterial contamination, waterborne illnesses, and skin irritations.
- b) Drinking Water Supplies: Thermal pollution can affect drinking water sources by altering water quality parameters, such as taste, odor, and turbidity, and compromising the effectiveness of water treatment processes.

Regulatory Framework and Mitigation Measures

- > Water Quality Standards: Regulatory agencies establish water quality standards and thermal pollution criteria to protect aquatic ecosystems and human health from the adverse effects of thermal pollution.
- > Cooling Technologies: Industries and power plants employ various cooling technologies, such as cooling towers, heat exchangers, and closed-loop systems, to minimize thermal discharges and reduce environmental impacts.
- > Habitat Restoration: Restoration efforts, such as riparian vegetation planting, streambank stabilization, and wetland creation, can help mitigate the effects of thermal pollution by providing shade, reducing erosion, and improving habitat quality for aquatic species.
- > Pollution Prevention: Pollution prevention measures, including source reduction,

wastewater treatment, and pollution control technologies, are essential for minimizing thermal pollution and promoting sustainable water management practices.

Case Studies and Examples

■ Fukushima Daiichi Nuclear Disaster: The release of heated water from the Fukushima nuclear power plant into the Pacific Ocean following the 2011 tsunami resulted in elevated water temperatures, radioactive contamination, and ecological impacts on marine ecosystems.

■ The Hudson River: Discharges from industrial facilities and power plants along the Hudson River in the United States have led to thermal pollution, affecting aquatic biodiversity and recreational activities in the region.

Future Outlook and Challenges

I. Climate Change: The impacts of climate change, including rising global temperatures and altered precipitation patterns, are expected to exacerbate thermal pollution and its associated environmental and human health impacts.

II. Sustainable Development: Addressing thermal pollution requires integrated approaches that balance the needs of economic development, environmental conservation, and social equity, emphasizing the importance of sustainable water management practices and stakeholder engagement.